

هندسة تكنولوجيا الأمن السيبراني
المرحلة الثانية – الفصل الأول
أسس الالكترونيك



وزارة التعليم العالي والبحث العلمي
الجامعة التقنية الوسطى
الكلية التقنية الهندسية الكهربائية

Lecture 1

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What is Electronics Fundamentals ?

Why we study this course in Cyber Security Technology
Engineering Department ?

TYPES OF MATERIALS

❑ Materials are divided into three types according to their conductivity:

- Conductors – Copper
- Insulators – Wood
- Semiconductors – Silicon (Si), Germanium (Ge)

❑ The term conductor is applied to any material that will support a generous flow of charge when a voltage source of limited magnitude is applied across its terminals.

❑ An insulator is a material that offers a very low level of conductivity under pressure from an applied voltage source.

❑ A semiconductor is a material that has a conductivity level some-where between the extremes of an insulator and a conductor.

- ❑ The property that determined the type of the material is called **Resistivity (ρ)**. Where the Resistivity (ρ) measures the resistance of a material towards the flow of charges.

$$\rho = \frac{RA}{L} = \frac{\Omega(cm^2)}{cm} = \Omega - cm$$

- ❑ The resistivity of conductors $< 10^{-6}$
- ❑ The resistivity of insulators $> 10^{12}$
- ❑ The resistivity of Silicon $= 50 \times 10^3$
- ❑ The resistivity of Germanium $= 50$

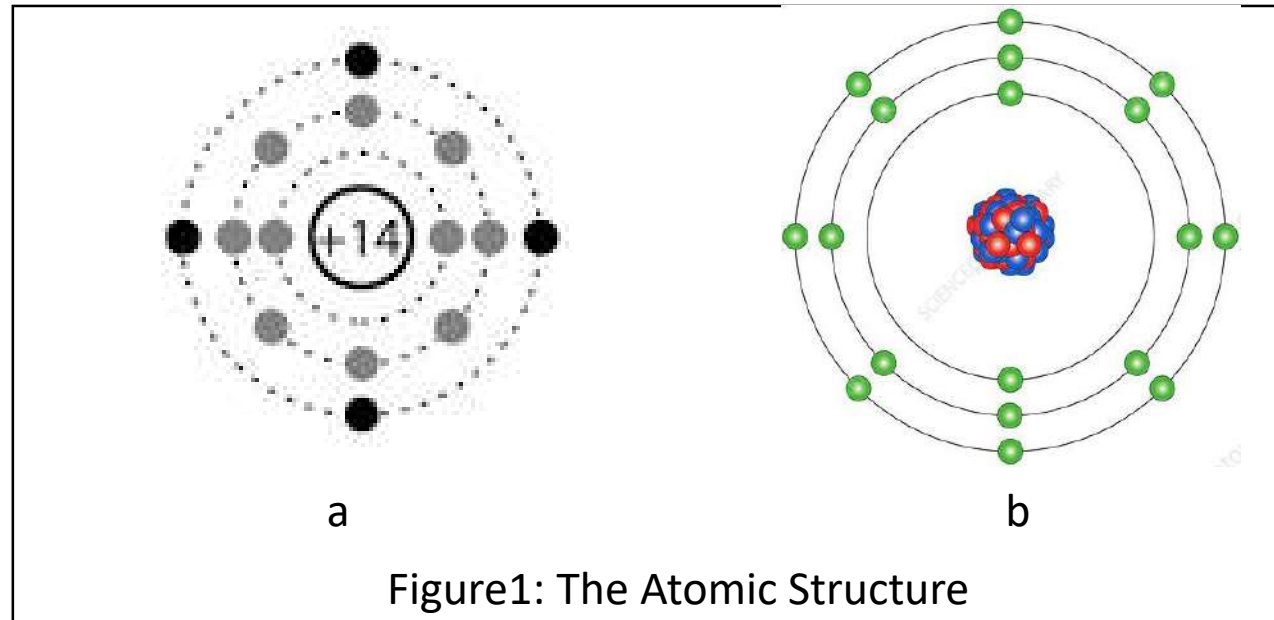
INTRINSIC MATERIALS

- ❑ Semiconductors are divided into three types according to their purity: Intrinsic materials and Extrinsic materials
- ❑ **Intrinsic materials** are those semiconductors that have been carefully refined to reduce the impurities to a very low level—essentially as pure as can be made available through modern technology.
- ❑ The most widely used semiconductors are Silicon (Si) and Germanium (Ge) because they can be manufactured to a very high purity level.
- ❑ The free electrons in the material due only to natural causes are referred to as intrinsic carriers.

- ❑ At the room temperature, intrinsic germanium material will have approximately 2.5×10^{13} free carriers per cubic centimeter.
- ❑ At the room temperature, intrinsic silicon material will have approximately 2.5×10^{10} free carriers per cubic centimeter.
- ❑ According to the number of free carriers per cubic centimeter, the germanium is a better conductor at room temperature.

Stability in Intrinsic semiconductors

Semiconductors (elements of Group 4 in the periodic table) try to reform their structure to be like the structure of noble gases in order to be more stable, where the most outer orbit of the noble gases contains 8 electrons. Figure 1,a shows the atomic structure of silicon, and Figure 1,b shows the atomic structure of Argon (a noble gas).



In order to copying the structure of noble gas, atoms of a semiconductor will share their four electrons (electrons that lies in the most outer orbits) with each others to obtain eight electrons in their most outer orbit as shown in figure 2

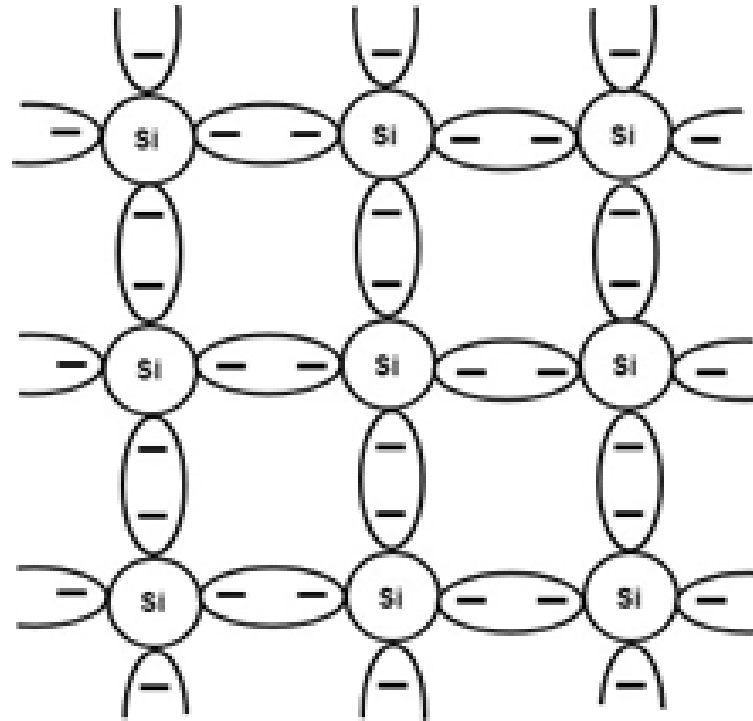


Figure 2: Crystal Lattice of Silicon

Energy Levels

- ❑ Figure 2,a shows the atomic structure of The Silicon and Figure 2,b Shows the atomic structure of The Germanium.

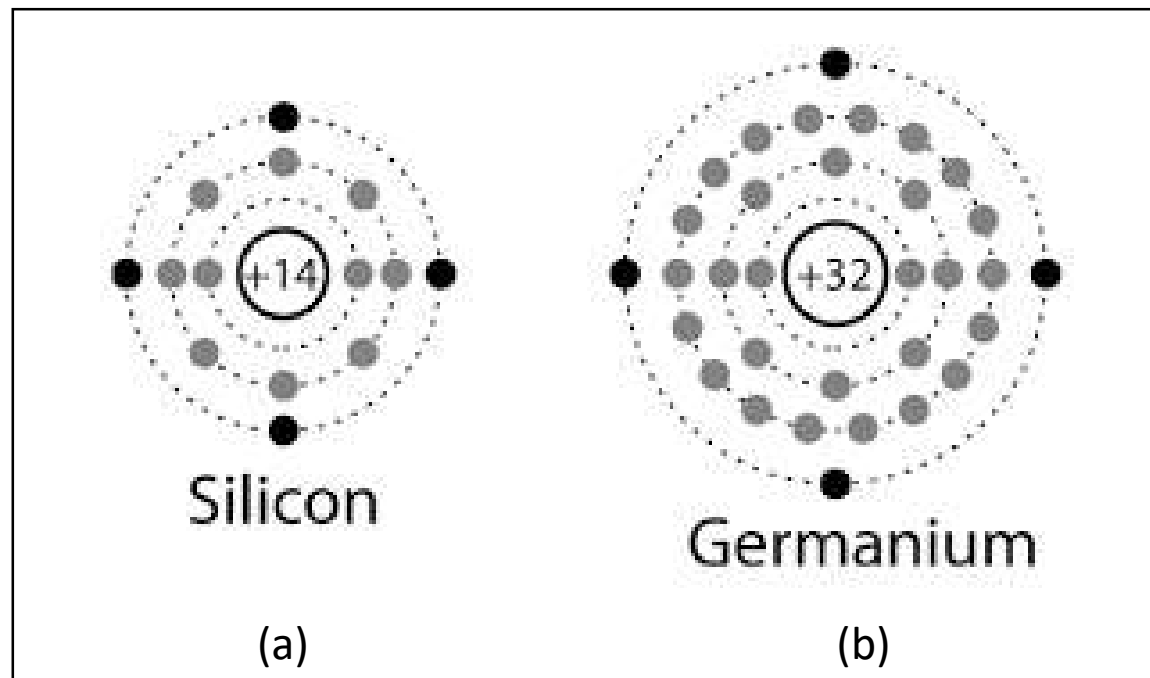


Figure (3)

- ❑ We will focus on the most outer orbit of the atom because the position of the electron will decide whether this material will allow electric current to pass through or not.
- ❑ The most outer orbit consists of two regions: conduction band and the valence band.

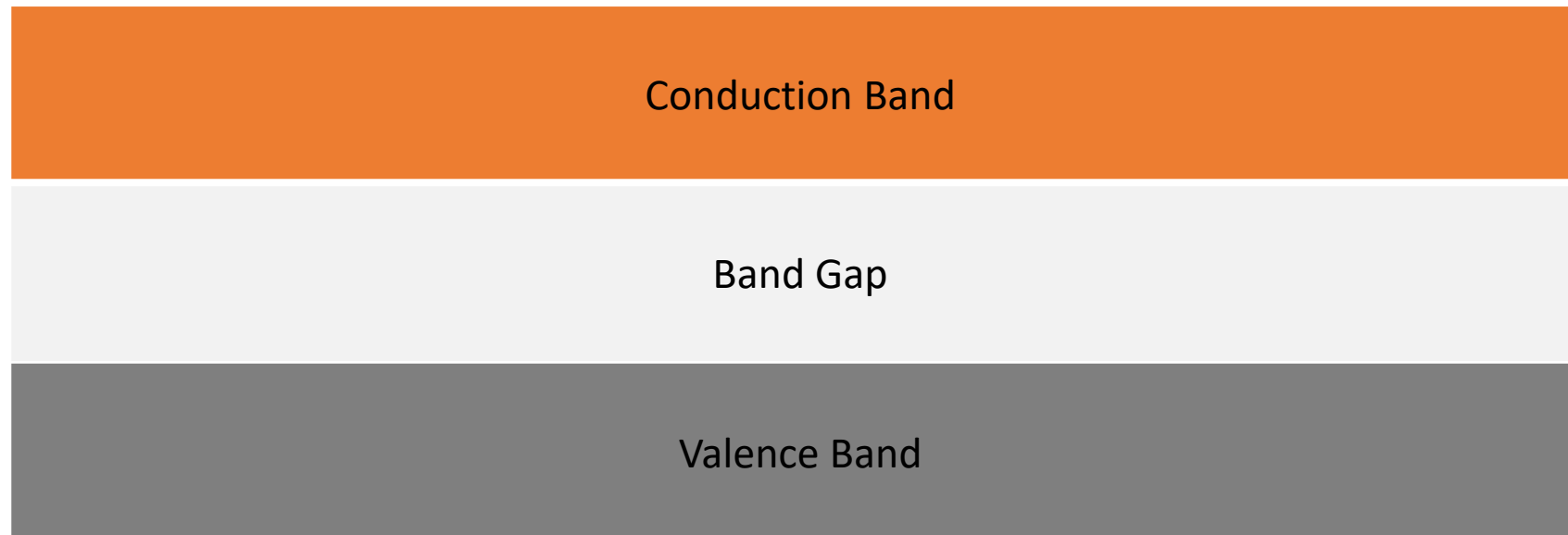


Figure 4

❑ Figure 5 shows the bands of most outer orbit for conductor, insulator, and semiconductor

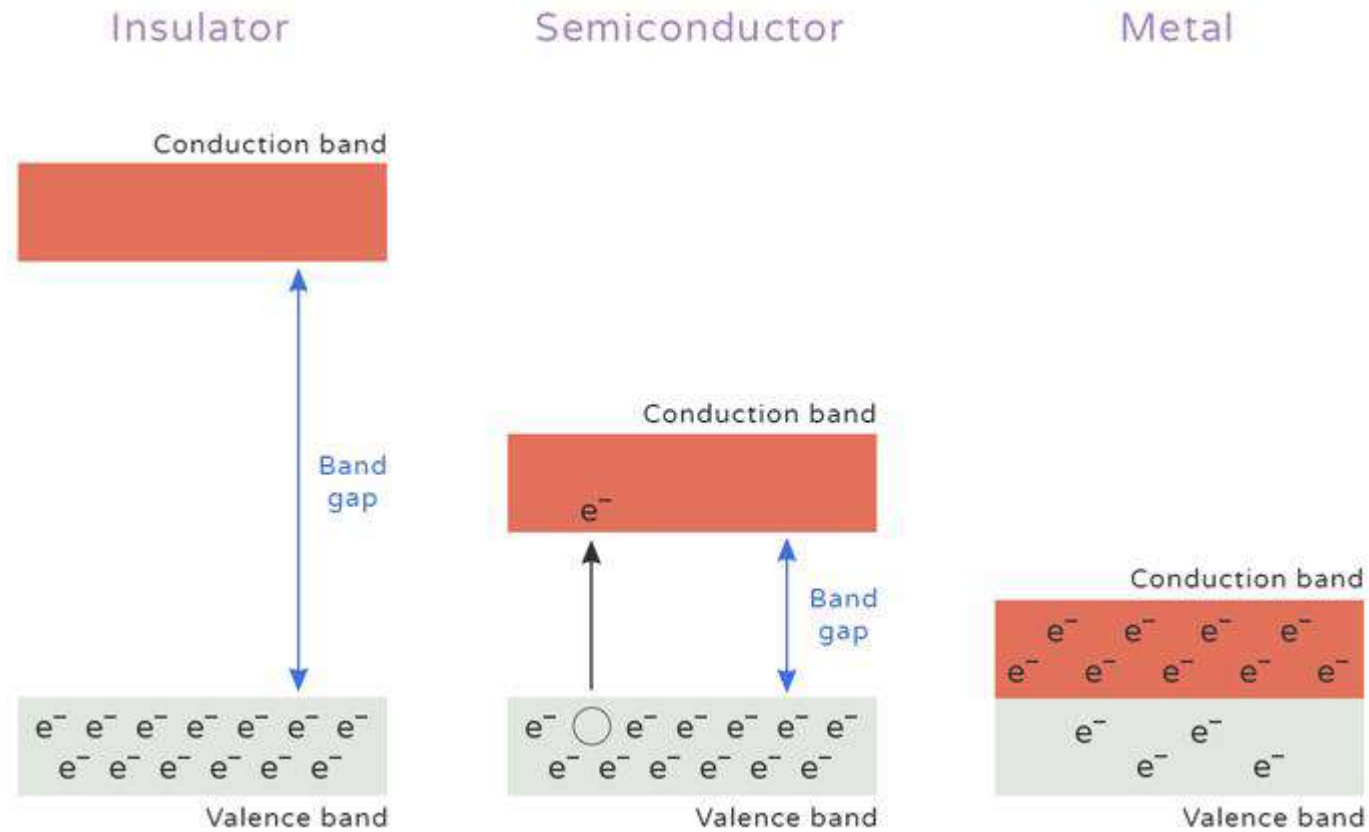


Figure 5

- ❑ Normally and except for conductors, the electrons lie in the valence band, and when they absorb power (from heat, light, electricity, pressure,) they will travel to conduction band.
- ❑ The amount of the energy that is required for the electrons to travel from valence band to conduction band is called Energy gap (E_g)
- ❑ The Energy gap for insulators > 5 eV
- ❑ The Energy gap for semiconductors ≈ 1 eV
- ❑ The valence band and the conduction band for conductors are overlapped, so the Energy gap for it is zero

□ The equation of the Energy gap:

$$W = QV \quad eV$$

$$W = QV = (1.6 \times 10^{-19} \text{ C})(1V)$$

$$1eV = 1.6 \times 10^{-19} \text{ J}$$

EXTRINSIC SEMICONDUCTORS

- ❑ In order to make the semiconductor act as conductor all the time, an atom of an element (called impurity atom) add to each 10 million atoms of a semiconductor. This process called **DOPING**. The resultant semiconductor of this process called **EXTRINSIC SEMICONDUCTORS**.
- ❑ There are two types of extrinsic material; n-type and p-type.
- ❑ N-type extrinsic semiconductor is created by adding those impurity elements that have five valence electrons (pentavalent), such as antimony, arsenic, and phosphorus.
- ❑ Diffused impurities with five valence electrons are called donor atoms.

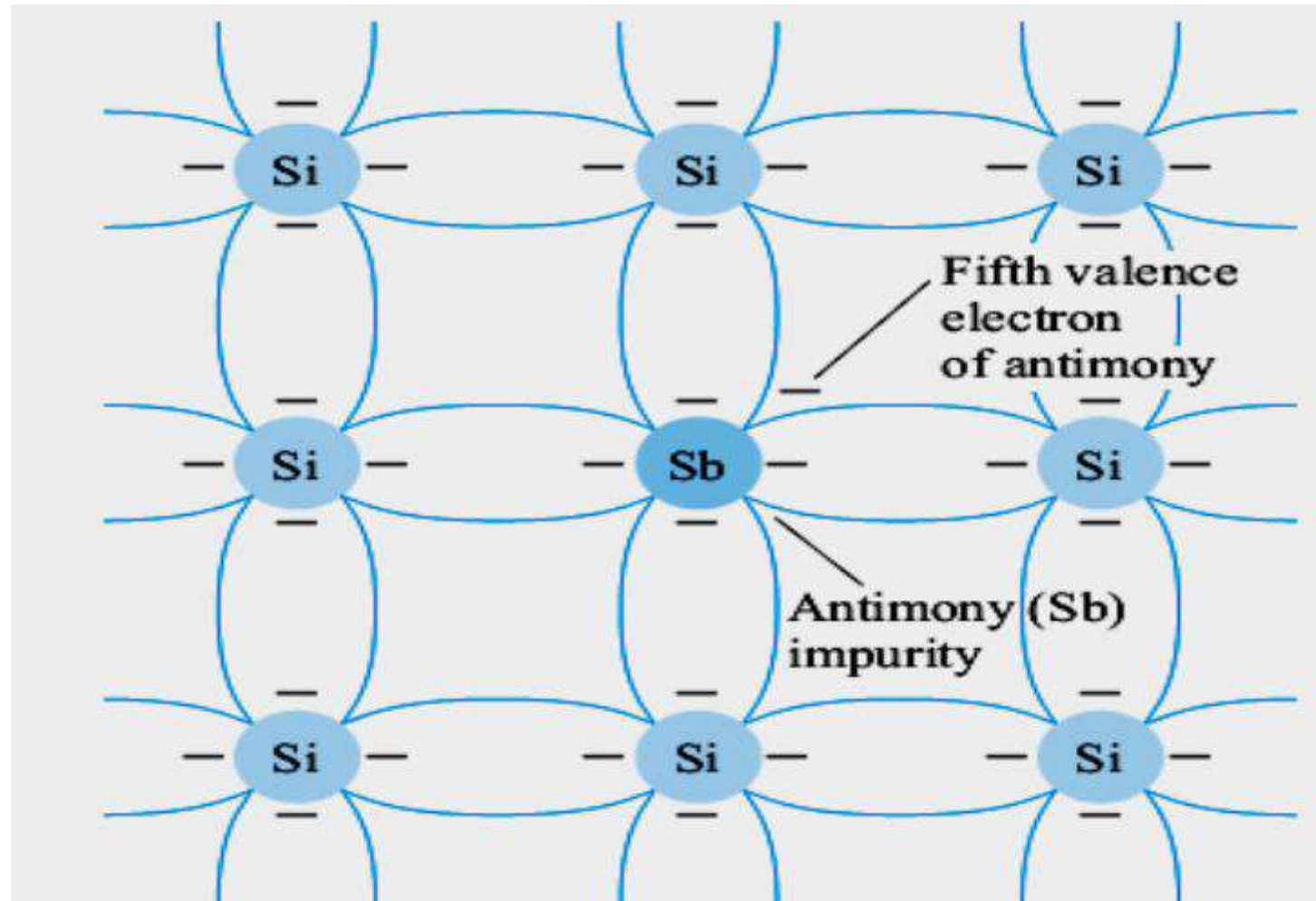


Figure 6: Crystal Lattice of silicon doped with Antimony

- ❑ The p-type material is formed by doping a pure germanium or silicon crystal with impurity atoms having three valence electrons.
- ❑ The diffused impurities with three valence electrons are called acceptor atoms.
- ❑ In an n-type material the electron is called the majority carrier and the hole the minority carrier. In a p-type material the hole is the majority carrier and the electron is the minority carrier.

- ❑ In the intrinsic state, the number of free electrons in Ge or Si is due only to those few electrons in the valence band that have acquired sufficient energy from thermal or light sources to break the covalent bond or to the few impurities that could not be removed.
- ❑ The vacancies left behind in the covalent bonding structure represent our very limited supply of holes. In an n-type material, the number of holes has not changed significantly from this intrinsic level. The net result, therefore, is that the number of electrons far outweighs the number of holes.

- ❑ In an n-type material, the electron is called the majority carrier and the hole the minority carrier.
- ❑ For the p-type material the number of holes far outweighs the number of electrons.
- ❑ In a p-type material the hole is the majority carrier and the electron is the minority carrier.
- ❑ When the fifth electron of a donor atom leaves the parent atom, the atom remaining acquires a net positive charge: hence the plus sign in the donor-ion representation. For similar reasons, the minus sign appears in the acceptor ion.

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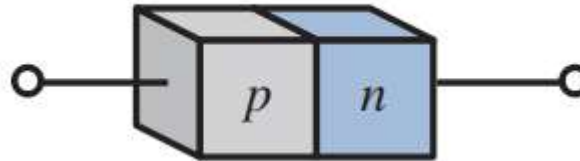
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Lecture 2 + 3

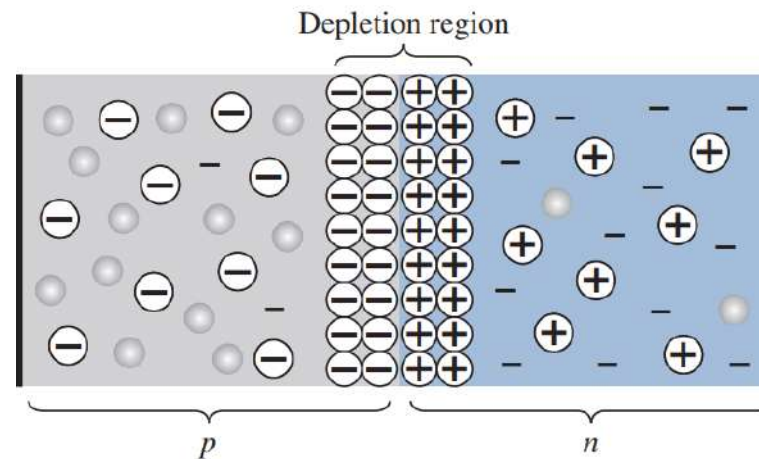
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Semiconductor diode

- ❑ The semiconductor diode is created by simply joining an n-type and a p-type material together.
- ❑ In another word, The semiconductor diode is created by joining of one material with a majority carrier of electrons to one with a majority carrier of holes.



- ❑ At the instant the two materials are “joined” the electrons and the holes in the region of the junction will combine, resulting in a lack of free carriers in the region near the junction, as shown in the figure below.
- ❑ The only particles displayed in this region are the positive and the negative ions remaining once the free carriers have been absorbed.



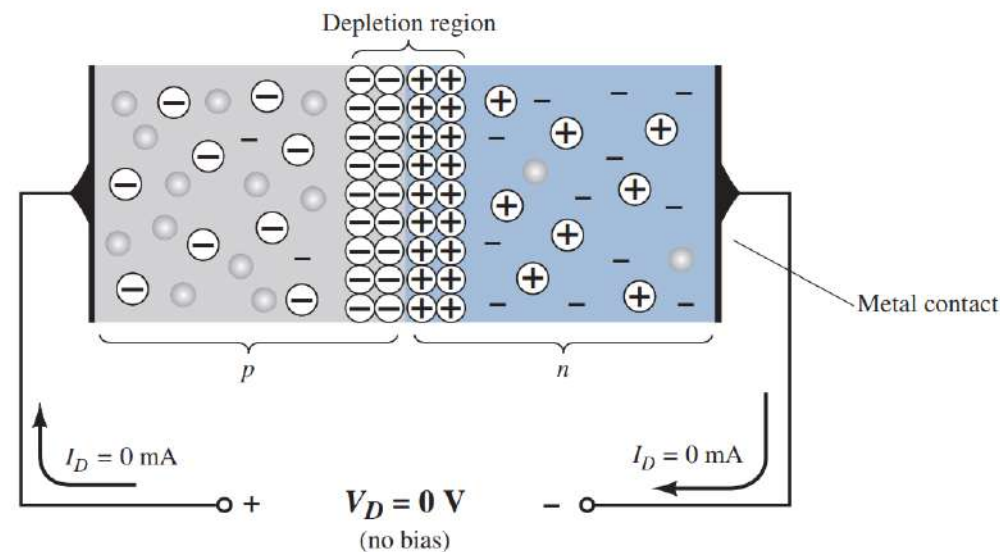
- ❑ This region of uncovered positive and negative ions is called the depletion region due to the “depletion” of free carriers in the region.

Biasing in Semiconductor diode

- ❑ The term **bias** refers to the application of an external voltage across the two terminals of the device to extract a response.
- ❑ In this course we will study three states of biasing:
 1. No – Biasing
 2. Forward – Biasing
 3. Reverse – biasing

No – Biasing ($V_D = 0$)

- At no – biasing condition, the diode terminals are isolated from the polarity of the source as shown in the figure below

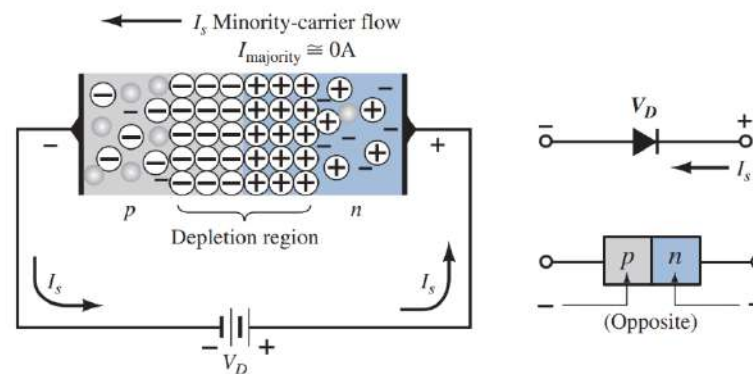


- ❑ Under no-bias conditions, any minority carriers (holes) in the n-type material that find themselves within the depletion region for any reason will pass quickly into the p-type material. The closer the minority carrier is to the junction, the greater is the attraction for the layer of negative ions and the less is the opposition offered by the positive ions in the depletion region of the n-type material.
- ❑ And also, any minority carriers (electrons) in the p-type material that find themselves within the depletion region for any reason will pass quickly into the n-type material. The closer the minority carrier is to the junction, the greater is the attraction for the layer of Positive ions and the less is the opposition offered by the negative ions in the depletion region of the p-type material.

- ❑ The majority carriers (electrons) of the n-type material must overcome the attractive forces of the layer of positive ions in the n-type material and the shield of negative ions in the p-type material to migrate into the area beyond the depletion region of the p-type material. However, the number of majority carriers is so large in the n-type material that there will invariably be a small number of majority carriers with sufficient kinetic energy to pass through the depletion region into the p-type material.
- ❑ Again, the same type of discussion can be applied to the majority carriers (holes) of the p-type material.

- ❑ As the magnitude of the both minority charges is the same, but in opposite direction, then the resultant of the net flow of the minority charges is zero
- ❑ In the absence of an applied bias across a semiconductor diode, the net flow of charge in one direction is zero. In other words, the current under no-bias conditions is zero.

- ❑ If an external potential of V is applied across the p - n junction such that the positive terminal is connected to the n -type material and the negative terminal is connected to the p -type material as shown in Figure below, the number of uncovered positive ions in the depletion region of the n -type material will increase due to the large number of free electrons drawn to the positive potential of the applied voltage. For similar reasons, the number of uncovered negative ions will increase in the p -type material.

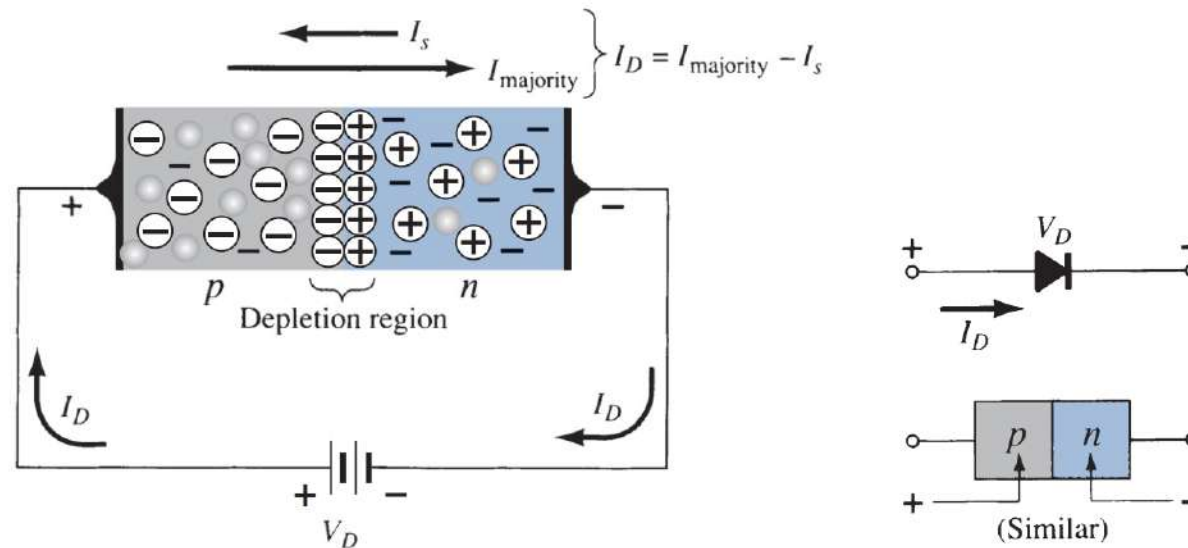


- ❑ This widening of the depletion region will establish too great a barrier for the majority carriers to overcome, effectively reducing the majority carrier flow to zero.
- ❑ The number of minority carriers, however, entering the depletion region will not change, resulting in minority-carrier flow vectors of the same magnitude indicated with no applied voltage.
- ❑ The current that exists under reverse-bias conditions is called the reverse saturation current and is represented by I_s .

- ❑ The reverse saturation current is seldom more than a few microamperes and typically in nA, except for high-power devices.
- ❑ The term saturation comes from the fact that it reaches its maximum level quickly and does not change significantly with increases in the reverse-bias potential.
- ❑ Note, in particular, that the direction of I_s is against the arrow of the symbol. Note also that the negative side of the applied voltage is connected to the p-type material and the positive side to the n-type material, the difference in underlined letters for each region revealing a reverse-bias condition.

Forward-bias Condition

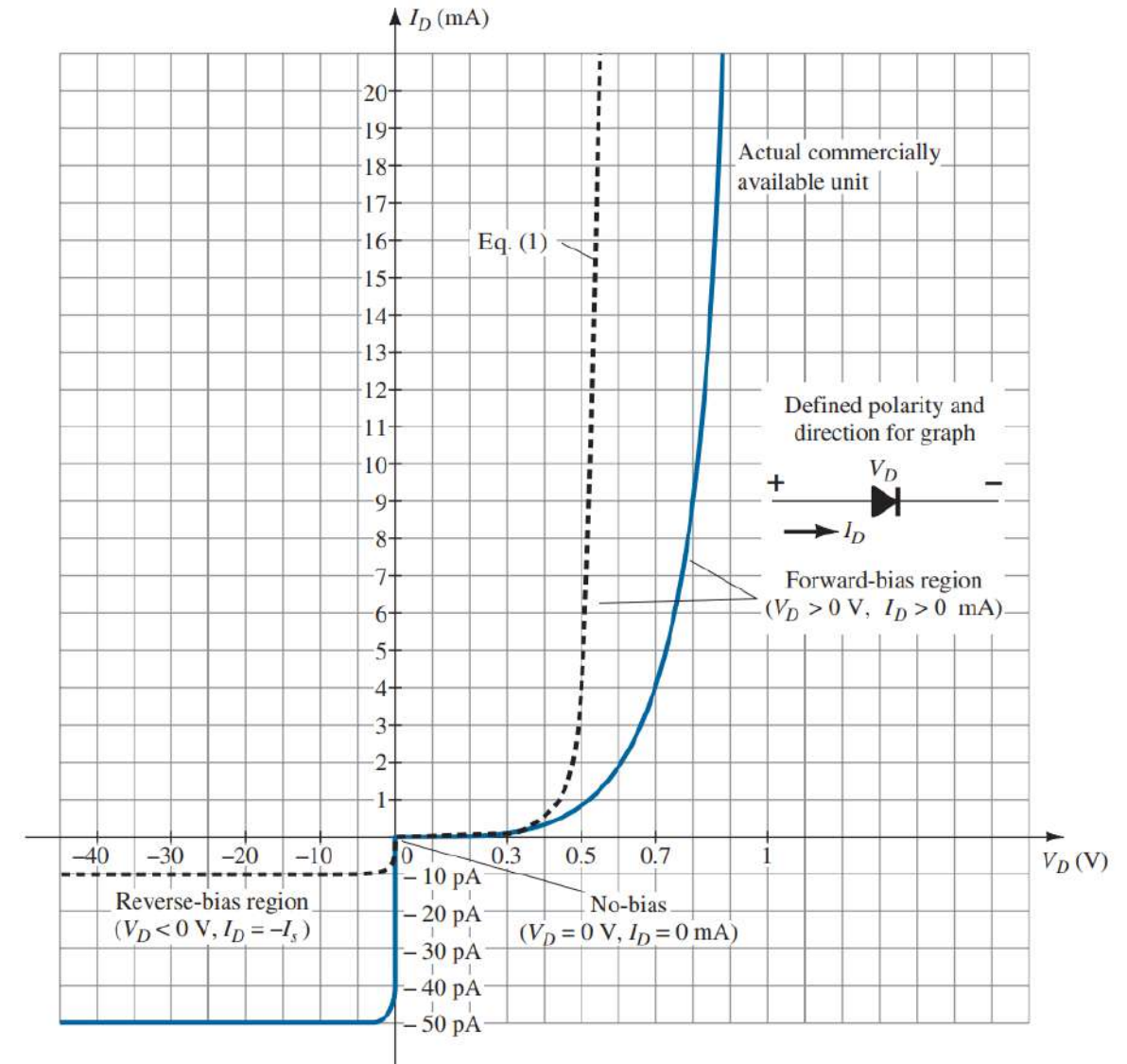
- ❑ A forward-bias or “on” condition is established by applying the positive potential to the p-type material and the negative potential to the n-type material as shown in figure below



- ❑ The application of a forward-bias potential V_D will “pressure” electrons in the n-type material and holes in the p-type material to recombine with the ions near the boundary and reduce the width of the depletion region.
- ❑ The resulting minority-carrier flow of electrons from the p-type material to the n-type material (and of holes from the n-type Semiconductor diode material to the p-type material) has not changed in magnitude (since the conduction level is controlled primarily by the limited number of impurities in the material), but the reduction in the width of the depletion region has resulted in a heavy majority flow across the junction.

- ❑ An electron of the n -type material now “sees” a reduced barrier at the junction due to the reduced depletion region and a strong attraction for the positive potential applied to the p -type material.
- ❑ As the applied bias increases in magnitude, the depletion region will continue to decrease in width until a flood of electrons can pass through the junction, resulting in an exponential rise in current.

- ❑ The sharp change in direction of the curve at $V_D = 0$ V is simply due to the change in current scales from above the axis to below the axis.
- ❑ Note that above the axis the scale is in milliamperes (mA), whereas below the axis it is in picoamperes (pA).
- ❑ For the voltage V_D there is also a measurable change in scale between the right-hand region of the graph and the left-hand region.
- ❑ For positive values of V_D the scale is in tenths of volts, and for the negative region it is in tens of volts



Silicon semiconductor diode characteristics

Diode Current

□ The diode current I_D can be expressed using Shockley's equation:

$$I_D = I_S \left(e^{\frac{V_D}{n V_T}} - 1 \right)$$

I_S , is the reverse saturation current

V_D , is the applied forward-bias voltage across the diode

n , is an ideality factor ($n=1$ will be assumed throughout this text unless otherwise noted).

V_T , is the thermal voltage

❑ The thermal voltage can be determined using the following equation:

$$V_T = \frac{k T_k}{q}$$

❑ k , is Boltzmann's constant = 1.38×10^{-23}

❑ T_k , is the absolute temperature in kelvins = $273 +$ the temperature in $^{\circ}\text{C}$

❑ q , is the magnitude of electronic charge 1.602×10^{-19}

□ If we expand the equation of diode current, we will get:

$$I_D = I_s e^{V_D/nV_T} - I_s$$

□ For positive values of V_D , the first term of the above equation will grow very quickly and totally overpower the effect of the second term. The result is the following equation, which only has positive values and takes on the exponential format

$$I_D \cong I_s e^{V_D/nV_T}$$

- ❑ For negative values of V_D , the exponential term drops very quickly below the level of I_s ,
Semiconductor diode and the resulting equation for I_D is simply:

$$I_D \cong -I_s$$

Example: Find diode current if the applied voltage is 5v and the saturation current is 10 pA at room temperature.

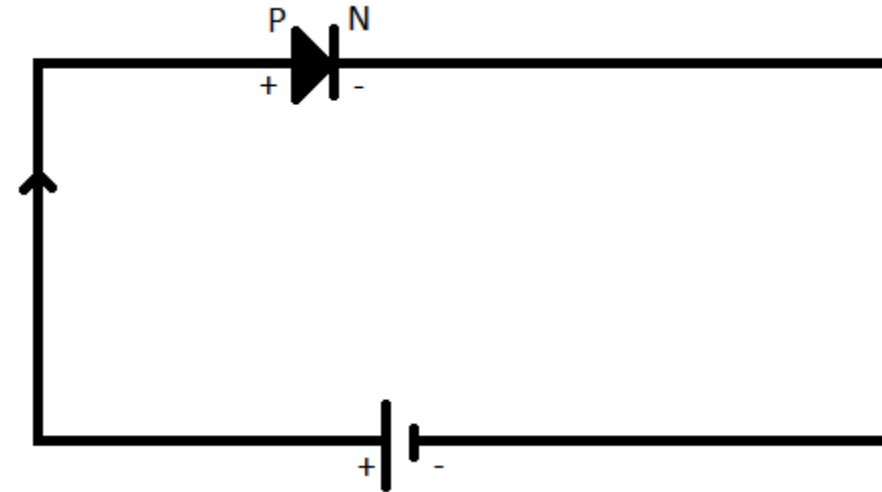
Solution:

$$I_D = I_s \left(e^{\frac{V_D}{n V_T}} - 1 \right)$$

$$V_T = \frac{k T_k}{q}$$

$$T_k = 300 + 27 = 300 \text{ kelvin}$$

$$V_T = \frac{1.38 \times 10^{-23} \times 300}{1.602 \times 10^{-19}} = 25.875 \text{ mV}$$



$$I_D = 10 \times 10^{-12} \left(e^{\left(\frac{5 \text{ V}}{25.875 \text{ mV}} \right)} - 1 \right)$$

$$I_D =$$

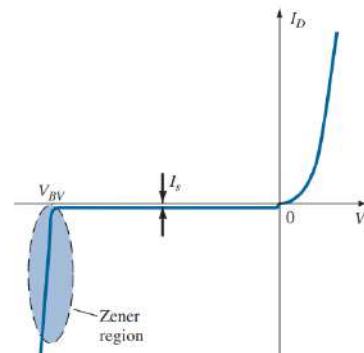
Example: Find diode current if the applied voltage is 5v and the saturation current is 10 pA at room temperature.

Solution:

$$I_D = -I_s$$

BREAKDOWN REGION

- ❑ In reverse biasing, there is a point where the application of large voltage with the reverse polarity will result in a sharp change in the characteristics, as shown in the figure below.
- ❑ The current increases at a very rapid rate in a direction opposite to that of the positive voltage region.
- ❑ The reverse-bias potential that results in this dramatic change in characteristics is called the *breakdown potential* and is given the label V_{BV} .
- ❑ This sharp change in the characteristic at any level is called *the Zener region*.



In the previous section we found that a p - n junction will permit a generous flow of charge when forward-biased and a very small level of current when reverse-biased. Both conditions are reviewed in Fig. 21, with the heavy current vector in Fig. 21a matching the direction of the arrow in the diode symbol and the significantly smaller vector in the opposite direction in Fig. 21b representing the reverse saturation current.

An analogy often used to describe the behavior of a semiconductor diode is a mechanical switch. In Fig. 21a the diode is acting like a closed switch permitting a generous flow of charge in the direction indicated. In Fig. 21b the level of current is so small in most cases that it can be approximated as 0 A and represented by an open switch.

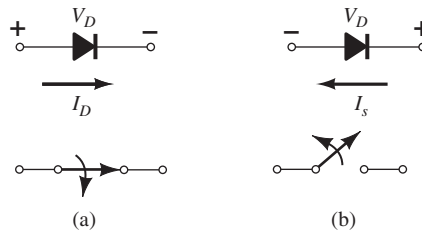


FIG. 21

Ideal semiconductor diode: (a) forward-biased; (b) reverse-biased.

In other words:

The semiconductor diode behaves in a manner similar to a mechanical switch in that it can control whether current will flow between its two terminals.

However, it is important to also be aware that:

The semiconductor diode is different from a mechanical switch in the sense that when the switch is closed it will only permit current to flow in one direction.

Ideally, if the semiconductor diode is to behave like a closed switch in the forward-bias region, the resistance of the diode should be $0\ \Omega$. In the reverse-bias region its resistance should be $\infty\ \Omega$ to represent the open-circuit equivalent. Such levels of resistance in the forward- and reverse-bias regions result in the characteristics of Fig. 22.

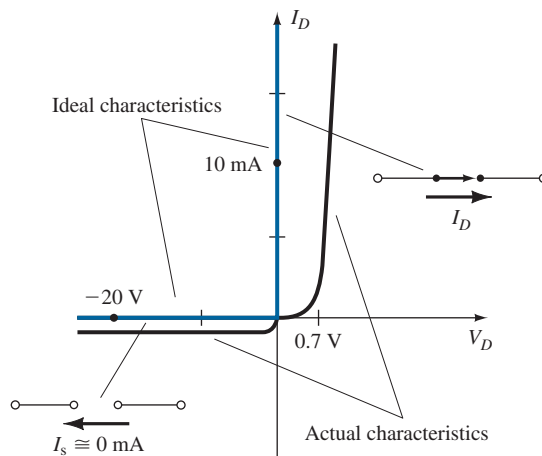


FIG. 22

Ideal versus actual semiconductor characteristics.

The characteristics have been superimposed to compare the ideal Si diode to a real-world Si diode. First impressions might suggest that the commercial unit is a poor impression of the ideal switch. However, when one considers that the only major difference is that the commercial diode rises at a level of 0.7 V rather than 0 V, there are a number of similarities between the two plots.

When a switch is closed the resistance between the contacts is assumed to be 0 Ω . At the plot point chosen on the vertical axis the diode current is 5 mA and the voltage across the diode is 0 V. Substituting into Ohm's law results in

$$R_F = \frac{V_D}{I_D} = \frac{0 \text{ V}}{5 \text{ mA}} = 0 \Omega \quad (\text{short-circuit equivalent})$$

In fact:

At any current level on the vertical line, the voltage across the ideal diode is 0 V and the resistance is 0 Ω .

For the horizontal section, if we again apply Ohm's law, we find

$$R_R = \frac{V_D}{I_D} = \frac{20 \text{ V}}{0 \text{ mA}} \cong \infty \Omega \quad (\text{open-circuit equivalent})$$

Again:

Because the current is 0 mA anywhere on the horizontal line, the resistance is considered to be infinite ohms (an open-circuit) at any point on the axis.

Due to the shape and the location of the curve for the commercial unit in the forward-bias region there will be a resistance associated with the diode that is greater than 0 Ω . However, if that resistance is small enough compared to other resistors of the network in series with the diode, it is often a good approximation to simply assume the resistance of the commercial unit is 0 Ω . In the reverse-bias region, if we assume the reverse saturation current is so small it can be approximated as 0 mA, we have the same open-circuit equivalence provided by the open switch.

The result, therefore, is that there are sufficient similarities between the ideal switch and the semiconductor diode to make it an effective electronic device. In the next section the various resistance levels of importance are determined for use in the chapter "Diode Applications", where the response of diodes in an actual network is examined.

8 RESISTANCE LEVELS

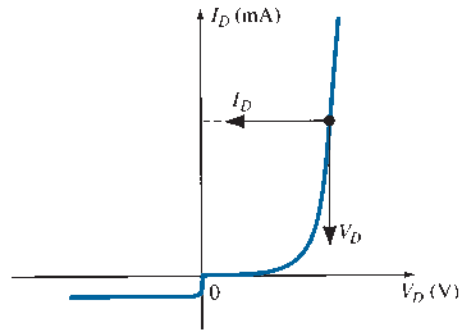
As the operating point of a diode moves from one region to another the resistance of the diode will also change due to the nonlinear shape of the characteristic curve. It will be demonstrated in the next few paragraphs that the type of applied voltage or signal will define the resistance level of interest. Three different levels will be introduced in this section, which will appear again as we examine other devices. It is therefore paramount that their determination be clearly understood.

DC or Static Resistance

The application of a dc voltage to a circuit containing a semiconductor diode will result in an operating point on the characteristic curve that will not change with time. The resistance of the diode at the operating point can be found simply by finding the corresponding levels of V_D and I_D as shown in Fig. 23 and applying the following equation:

$$R_D = \frac{V_D}{I_D} \quad (4)$$

The dc resistance levels at the knee and below will be greater than the resistance levels obtained for the vertical rise section of the characteristics. The resistance levels in the reverse-bias region will naturally be quite high. Since ohmmeters typically employ a relatively constant-current source, the resistance determined will be at a preset current level (typically, a few milliamperes).


FIG. 23

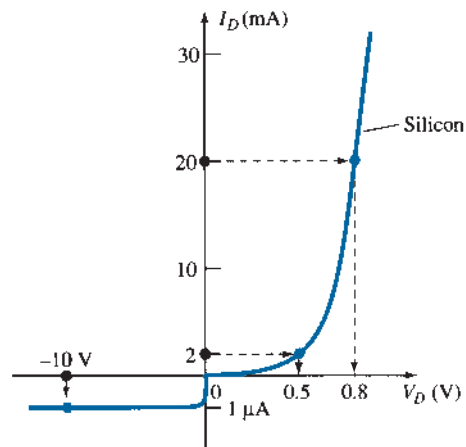
Determining the dc resistance of a diode at a particular operating point.

In general, therefore, the higher the current through a diode, the lower is the dc resistance level.

Typically, the dc resistance of a diode in the active (most utilized) will range from about $10\ \Omega$ to $80\ \Omega$.

EXAMPLE 3 Determine the dc resistance levels for the diode of Fig. 24 at

- $I_D = 2\text{ mA}$ (low level)
- $I_D = 20\text{ mA}$ (high level)
- $V_D = -10\text{ V}$ (reverse-biased)


FIG. 24

Example 3.

Solution:

- At $I_D = 2\text{ mA}$, $V_D = 0.5\text{ V}$ (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.5\text{ V}}{2\text{ mA}} = 250\ \Omega$$

- At $I_D = 20\text{ mA}$, $V_D = 0.8\text{ V}$ (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.8\text{ V}}{20\text{ mA}} = 40\ \Omega$$

c. At $V_D = -10\text{ V}$, $I_D = -I_s = -1\text{ }\mu\text{A}$ (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{10\text{ V}}{1\text{ }\mu\text{A}} = \mathbf{10\text{ M}\Omega}$$

clearly supporting some of the earlier comments regarding the dc resistance levels of a diode.

AC or Dynamic Resistance

Eq. (4) and Example 3 reveal that

the dc resistance of a diode is independent of the shape of the characteristic in the region surrounding the point of interest.

If a sinusoidal rather than a dc input is applied, the situation will change completely. The varying input will move the instantaneous operating point up and down a region of the characteristics and thus defines a specific change in current and voltage as shown in Fig. 25. With no applied varying signal, the point of operation would be the Q -point appearing on Fig. 25, determined by the applied dc levels. The designation Q -point is derived from the word *quiescent*, which means “still or unvarying.”

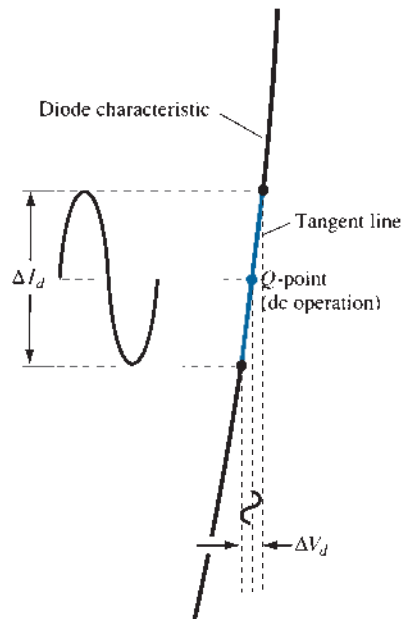


FIG. 25

Defining the dynamic or ac resistance.

A straight line drawn tangent to the curve through the Q -point as shown in Fig. 26 will define a particular change in voltage and current that can be used to determine the *ac* or *dynamic* resistance for this region of the diode characteristics. An effort should be made to keep the change in voltage and current as small as possible and equidistant to either side of the Q -point. In equation form,

$$r_d = \frac{\Delta V_d}{\Delta I_d} \quad (5)$$

where Δ signifies a finite change in the quantity.

The steeper the slope, the lower is the value of ΔV_d for the same change in ΔI_d and the lower is the resistance. The *ac* resistance in the vertical-rise region of the characteristic is therefore quite small, whereas the *ac* resistance is much higher at low current levels.

*In general, therefore, the lower the Q -point of operation (smaller current or lower voltage), the higher is the *ac* resistance.*

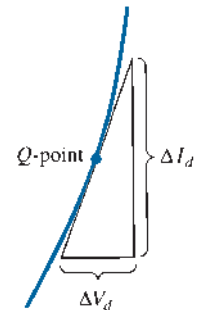
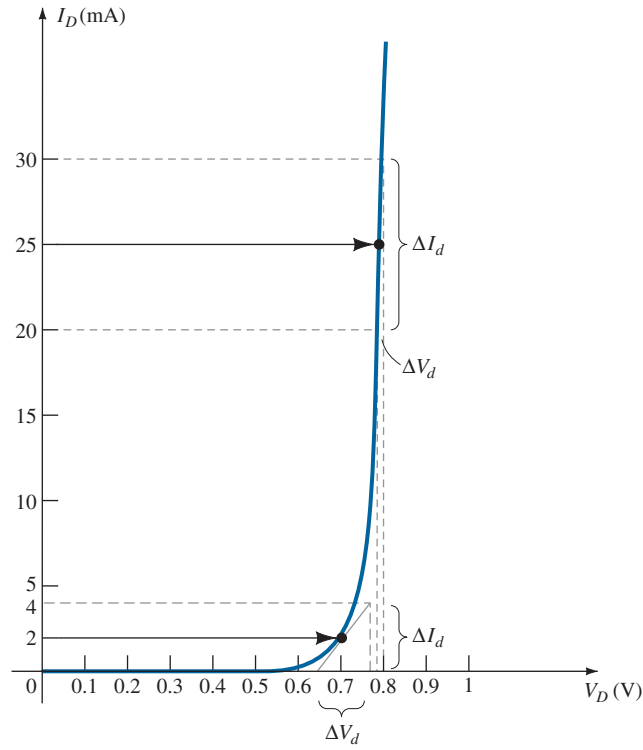


FIG. 26

*Determining the *ac* resistance at a Q -point.*

EXAMPLE 4 For the characteristics of Fig. 27:

- Determine the ac resistance at $I_D = 2$ mA.
- Determine the ac resistance at $I_D = 25$ mA.
- Compare the results of parts (a) and (b) to the dc resistances at each current level.

**FIG. 27**

Example 4.

Solution:

- For $I_D = 2$ mA, the tangent line at $I_D = 2$ mA was drawn as shown in Fig. 27 and a swing of 2 mA above and below the specified diode current was chosen. At $I_D = 4$ mA, $V_D = 0.76$ V, and at $I_D = 0$ mA, $V_D = 0.65$ V. The resulting changes in current and voltage are, respectively,

$$\Delta I_d = 4 \text{ mA} - 0 \text{ mA} = 4 \text{ mA}$$

and

$$\Delta V_d = 0.76 \text{ V} - 0.65 \text{ V} = 0.11 \text{ V}$$

and the ac resistance is

$$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.11 \text{ V}}{4 \text{ mA}} = \mathbf{27.5 \Omega}$$

- For $I_D = 25$ mA, the tangent line at $I_D = 25$ mA was drawn as shown in Fig. 27 and a swing of 5 mA above and below the specified diode current was chosen. At $I_D = 30$ mA, $V_D = 0.8$ V, and at $I_D = 20$ mA, $V_D = 0.78$ V. The resulting changes in current and voltage are, respectively,

$$\Delta I_d = 30 \text{ mA} - 20 \text{ mA} = 10 \text{ mA}$$

and

$$\Delta V_d = 0.8 \text{ V} - 0.78 \text{ V} = 0.02 \text{ V}$$

and the ac resistance is

$$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.02 \text{ V}}{10 \text{ mA}} = \mathbf{2 \Omega}$$

- For $I_D = 2$ mA, $V_D = 0.7$ V and

$$R_D = \frac{V_D}{I_D} = \frac{0.7 \text{ V}}{2 \text{ mA}} = \mathbf{350 \Omega}$$

which far exceeds the r_d of 27.5 Ω .

For $I_D = 25 \text{ mA}$, $V_D = 0.79 \text{ V}$ and

$$R_D = \frac{V_D}{I_D} = \frac{0.79 \text{ V}}{25 \text{ mA}} = \mathbf{31.62 \text{ } \Omega}$$

which far exceeds the r_d of $2 \text{ } \Omega$.

We have found the dynamic resistance graphically, but there is a basic definition in differential calculus that states:

The derivative of a function at a point is equal to the slope of the tangent line drawn at that point.

Equation (5), as defined by Fig. 26, is, therefore, essentially finding the derivative of the function at the Q -point of operation. If we find the derivative of the general equation (2) for the semiconductor diode with respect to the applied forward bias and then invert the result, we will have an equation for the dynamic or ac resistance in that region. That is, taking the derivative of Eq. (2) with respect to the applied bias will result in

$$\frac{d}{dV_D}(I_D) = \frac{d}{dV_D}[I_s(e^{V_D/nV_T} - 1)]$$

and

$$\frac{dI_D}{dV_D} = \frac{1}{nV_T}(I_D + I_s)$$

after we apply differential calculus. In general, $I_D \gg I_s$ in the vertical-slope section of the characteristics and

$$\frac{dI_D}{dV_D} \cong \frac{I_D}{nV_T}$$

Flipping the result to define a resistance ratio ($R = V/I$) gives

$$\frac{dV_D}{dI_D} = r_d = \frac{nV_T}{I_D}$$

Substituting $n = 1$ and $V_T \cong 26 \text{ mV}$ from Example 1 results in

$$\boxed{r_d = \frac{26 \text{ mV}}{I_D}} \quad (6)$$

The significance of Eq. (6) must be clearly understood. It implies that

the dynamic resistance can be found simply by substituting the quiescent value of the diode current into the equation.

There is no need to have the characteristics available or to worry about sketching tangent lines as defined by Eq. (5). It is important to keep in mind, however, that Eq. (6) is accurate only for values of I_D in the vertical-rise section of the curve. For lesser values of I_D , $n = 2$ (silicon) and the value of r_d obtained must be multiplied by a factor of 2. For small values of I_D below the knee of the curve, Eq. (6) becomes inappropriate.

All the resistance levels determined thus far have been defined by the p - n junction and do not include the resistance of the semiconductor material itself (called *body* resistance) and the resistance introduced by the connection between the semiconductor material and the external metallic conductor (called *contact* resistance). These additional resistance levels can be included in Eq. (6) by adding a resistance denoted r_B :

$$\boxed{r'_d = \frac{26 \text{ mV}}{I_D} + r_B} \quad \text{ohms} \quad (7)$$

The resistance r'_d , therefore, includes the dynamic resistance defined by Eq. (6) and the resistance r_B just introduced. The factor r_B can range from typically $0.1 \text{ } \Omega$ for high-power devices to $2 \text{ } \Omega$ for some low-power, general-purpose diodes. For Example 4 the ac resistance at 25 mA was calculated to be $2 \text{ } \Omega$. Using Eq. (6), we have

$$r_d = \frac{26 \text{ mV}}{I_D} = \frac{26 \text{ mV}}{25 \text{ mA}} = \mathbf{1.04 \text{ } \Omega}$$

The difference of about $1\ \Omega$ could be treated as the contribution of r_B .

For Example 4 the ac resistance at 2 mA was calculated to be $27.5\ \Omega$. Using Eq. (6) but multiplying by a factor of 2 for this region (in the knee of the curve $n = 2$),

$$r_d = 2 \left(\frac{26\ \text{mV}}{I_D} \right) = 2 \left(\frac{26\ \text{mV}}{2\ \text{mA}} \right) = 2(13\ \Omega) = 26\ \Omega$$

The difference of $1.5\ \Omega$ could be treated as the contribution due to r_B .

In reality, determining r_d to a high degree of accuracy from a characteristic curve using Eq. (5) is a difficult process at best and the results have to be treated with skepticism. At low levels of diode current the factor r_B is normally small enough compared to r_d to permit ignoring its impact on the ac diode resistance. At high levels of current the level of r_B may approach that of r_d , but since there will frequently be other resistive elements of a much larger magnitude in series with the diode, we will assume in this text that the ac resistance is determined solely by r_d , and the impact of r_B will be ignored unless otherwise noted. Technological improvements of recent years suggest that the level of r_B will continue to decrease in magnitude and eventually become a factor that can certainly be ignored in comparison to r_d .

The discussion above centered solely on the forward-bias region. In the reverse-bias region we will assume that the change in current along the I_s line is nil from 0 V to the Zener region and the resulting ac resistance using Eq. (5) is sufficiently high to permit the open-circuit approximation.

Typically, the ac resistance of a diode in the active region will range from about $1\ \Omega$ to $100\ \Omega$.

Average AC Resistance

If the input signal is sufficiently large to produce a broad swing such as indicated in Fig. 28, the resistance associated with the device for this region is called the *average ac resistance*. The average ac resistance is, by definition, the resistance determined by a straight line drawn between the two intersections established by the maximum and minimum values of input voltage. In equation form (note Fig. 28),

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \Big|_{\text{pt. to pt.}} \quad (8)$$

For the situation indicated by Fig. 28,

$$\Delta I_d = 17\ \text{mA} - 2\ \text{mA} = 15\ \text{mA}$$

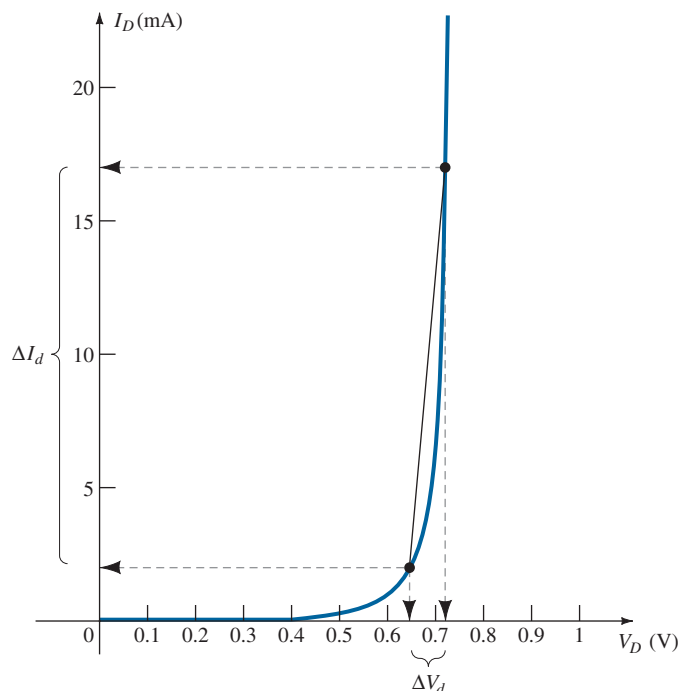


FIG. 28

Determining the average ac resistance between indicated limits.

and

$$\Delta V_d = 0.725 \text{ V} - 0.65 \text{ V} = 0.075 \text{ V}$$

with

$$r_{\text{av}} = \frac{\Delta V_d}{\Delta I_d} = \frac{0.075 \text{ V}}{15 \text{ mA}} = 5 \, \Omega$$

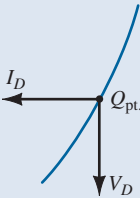
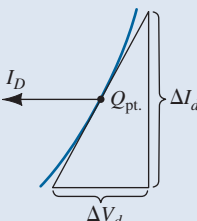
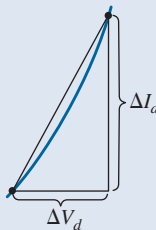
If the ac resistance (r_d) were determined at $I_D = 2 \text{ mA}$, its value would be more than $5 \, \Omega$, and if determined at 17 mA , it would be less. In between, the ac resistance would make the transition from the high value at 2 mA to the lower value at 17 mA . Equation (7) defines a value that is considered the average of the ac values from 2 mA to 17 mA . The fact that one resistance level can be used for such a wide range of the characteristics will prove quite useful in the definition of equivalent circuits for a diode in a later section.

As with the dc and ac resistance levels, the lower the level of currents used to determine the average resistance, the higher is the resistance level.

Summary Table

Table 6 was developed to reinforce the important conclusions of the last few pages and to emphasize the differences among the various resistance levels. As indicated earlier, the content of this section is the foundation for a number of resistance calculations to be performed in later sections.

TABLE 6
Resistance Levels

Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26 \text{ mV}}{I_D}$	Defined by a tangent line at the Q -point	
Average ac	$r_{\text{av}} = \left. \frac{\Delta V_d}{\Delta I_d} \right _{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

9 DIODE EQUIVALENT CIRCUITS

An equivalent circuit is a combination of elements properly chosen to best represent the actual terminal characteristics of a device or system in a particular operating region.

In other words, once the equivalent circuit is defined, the device symbol can be removed from a schematic and the equivalent circuit inserted in its place without severely affecting the actual behavior of the system. The result is often a network that can be solved using traditional circuit analysis techniques.

Piecewise-Linear Equivalent Circuit

One technique for obtaining an equivalent circuit for a diode is to approximate the characteristics of the device by straight-line segments, as shown in Fig. 29. The resulting equivalent circuit is called a *piecewise-linear equivalent circuit*. It should be obvious from Fig. 29 that the straight-line segments do not result in an exact duplication of the actual characteristics, especially in the knee region. However, the resulting segments are sufficiently close to the actual curve to establish an equivalent circuit that will provide an excellent first approximation to the actual behavior of the device. For the sloping section of the equivalence the average ac resistance as introduced in Section 8 is the resistance level appearing in the equivalent circuit of Fig. 28 next to the actual device. In essence, it defines the resistance level of the device when it is in the “on” state. The ideal diode is included to establish that there is only one direction of conduction through the device, and a reverse-bias condition will result in the open-circuit state for the device. Since a silicon semiconductor diode does not reach the conduction state until V_D reaches 0.7 V with a forward bias (as shown in Fig. 29), a battery V_K opposing the conduction direction must appear in the equivalent circuit as shown in Fig. 30. The battery simply specifies that the voltage across the device must be greater than the threshold battery voltage before conduction through the device in the direction dictated by the ideal diode can be established. When conduction is established the resistance of the diode will be the specified value of r_{av} .

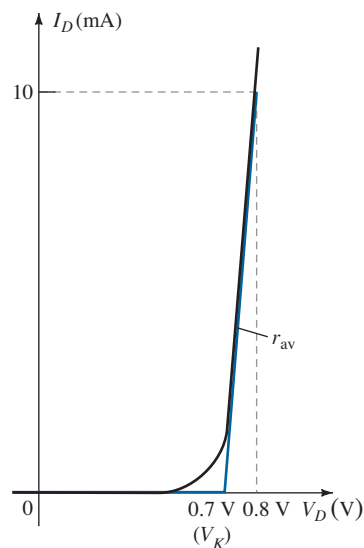


FIG. 29

Defining the piecewise-linear equivalent circuit using straight-line segments to approximate the characteristic curve.

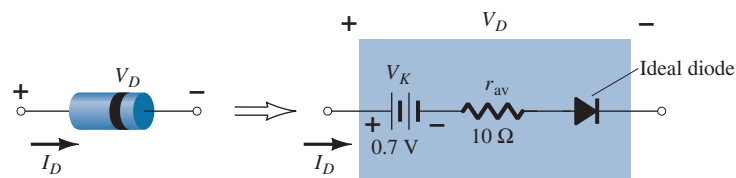


FIG. 30

Components of the piecewise-linear equivalent circuit.

Keep in mind, however, that V_K in the equivalent circuit is not an independent voltage source. If a voltmeter is placed across an isolated diode on the top of a laboratory bench, a reading of 0.7 V will not be obtained. The battery simply represents the horizontal offset of the characteristics that must be exceeded to establish conduction.

The approximate level of r_{av} can usually be determined from a specified operating point on the specification sheet (to be discussed in Section 10). For instance, for a silicon semiconductor diode, if $I_F = 10$ mA (a forward conduction current for the diode) at

$V_D = 0.8 \text{ V}$, we know that for silicon a shift of 0.7 V is required before the characteristics rise, and we obtain

$$r_{av} = \left. \frac{\Delta V_d}{\Delta I_d} \right|_{\text{pt. to pt.}} = \frac{0.8 \text{ V} - 0.7 \text{ V}}{10 \text{ mA} - 0 \text{ mA}} = \frac{0.1 \text{ V}}{10 \text{ mA}} = \mathbf{10 \, \Omega}$$

as obtained for Fig. 29.

If the characteristics or specification sheet for a diode is not available the resistance r_{av} can be approximated by the ac resistance r_d .

Simplified Equivalent Circuit

For most applications, the resistance r_{av} is sufficiently small to be ignored in comparison to the other elements of the network. Removing r_{av} from the equivalent circuit is the same as implying that the characteristics of the diode appear as shown in Fig. 31. Indeed, this approximation is frequently employed in semiconductor circuit analysis. The reduced equivalent circuit appears in the same figure. It states that a forward-biased silicon diode in an electronic system under dc conditions has a drop of 0.7 V across it in the conduction state at any level of diode current (within rated values, of course).

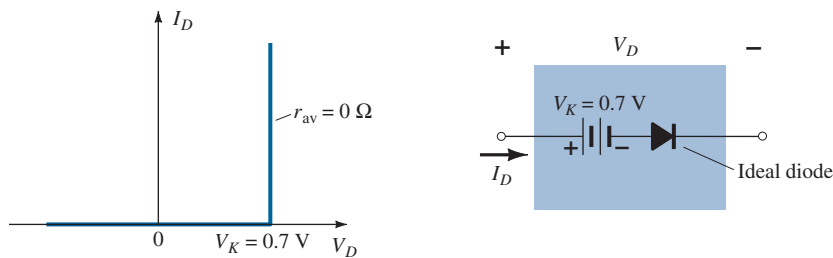


FIG. 31

Simplified equivalent circuit for the silicon semiconductor diode.

Ideal Equivalent Circuit

Now that r_{av} has been removed from the equivalent circuit, let us take the analysis a step further and establish that a 0.7-V level can often be ignored in comparison to the applied voltage level. In this case the equivalent circuit will be reduced to that of an ideal diode as shown in Fig. 32 with its characteristics.

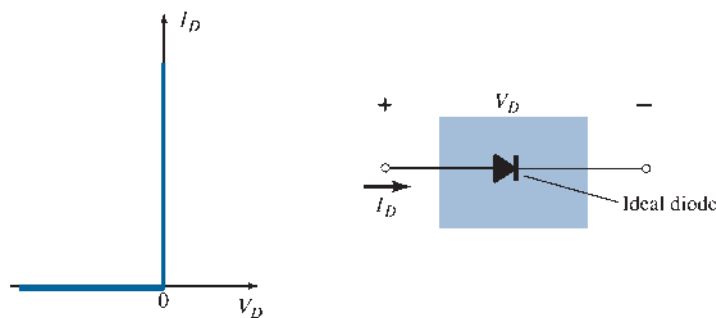


FIG. 32

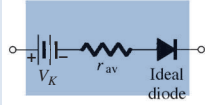
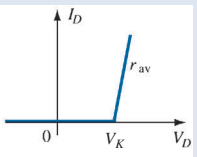
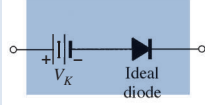
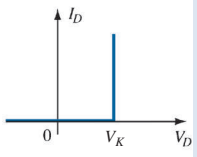
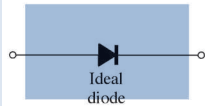
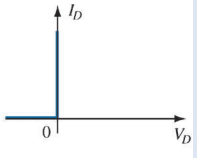
Ideal diode and its characteristics.

In industry a popular substitution for the phrase “diode equivalent circuit” is diode *model*—a model by definition being a representation of an existing device, object, system, and so on.

Summary Table

For clarity, the diode models employed for the range of circuit parameters and applications are provided in Table 7 with their piecewise-linear characteristics. There are always exceptions to the general rule, but it is fairly safe to say that the simplified equivalent model will

TABLE 7
Diode Equivalent Circuits (Models)

Type	Conditions	Model	Characteristics
Piecewise-linear model			
Simplified model	$R_{\text{network}} \gg r_{\text{av}}$		
Ideal device	$R_{\text{network}} \gg r_{\text{av}}$ $E_{\text{network}} \gg V_K$		

be employed most frequently in the analysis of electronic systems, whereas the ideal diode is frequently applied in the analysis of power supply systems where larger voltages are encountered.

10 TRANSITION AND DIFFUSION CAPACITANCE

It is important to realize that:

Every electronic or electrical device is frequency sensitive.

That is, the terminal characteristics of any device will change with frequency. Even the resistance of a basic resistor, as of any construction, will be sensitive to the applied frequency. At low to mid-frequencies most resistors can be considered fixed in value. However, as we approach high frequencies, stray capacitive and inductive effects start to play a role and will affect the total impedance level of the element.

For the diode it is the stray capacitance levels that have the greatest effect. At low frequencies and relatively small levels of capacitance the reactance of a capacitor, determined by $X_C = 1/2\pi fC$, is usually so high it can be considered infinite in magnitude, represented by an open circuit, and ignored. At high frequencies, however, the level of X_C can drop to the point where it will introduce a low-reactance “shorting” path. If this shorting path is across the diode, it can essentially keep the diode from affecting the response of the network.

In the p – n semiconductor diode, there are two capacitive effects to be considered. Both types of capacitance are present in the forward- and reverse-bias regions, but one so outweighs the other in each region that we consider the effects of only one in each region.

Recall that the basic equation for the capacitance of a parallel-plate capacitor is defined by $C = \epsilon A/d$, where ϵ is the permittivity of the dielectric (insulator) between the plates of area A separated by a distance d . In a diode the depletion region (free of carriers) behaves essentially like an insulator between the layers of opposite charge. Since the depletion width (d) will increase with increased reverse-bias potential, the resulting transition capacitance will decrease, as shown in Fig. 33. The fact that the capacitance is dependent on the applied reverse-bias potential has application in a number of electronic systems.

This capacitance, called the transition (C_T), barriers, or depletion region capacitance, is determined by

$$C_T = \frac{C(0)}{(1 + |V_R/V_K|)^n} \quad (9)$$

having to resort to lengthy mathematical procedures. This is usually accomplished through the approximation process, which can develop into an art itself. Although the results obtained using the actual characteristics may be slightly different from those obtained using a series of approximations, keep in mind that the characteristics obtained from a specification sheet may be slightly different from those of the device in actual use. In other words, for example, the characteristics of a 1N4001 semiconductor diode may vary from one element to the next in the same lot. The variation may be slight, but it will often be sufficient to justify the approximations employed in the analysis. Also consider the other elements of the network: Is the resistor labeled $100\ \Omega$ exactly $100\ \Omega$? Is the applied voltage exactly $10\ \text{V}$ or perhaps $10.08\ \text{V}$? All these tolerances contribute to the general belief that a response determined through an appropriate set of approximations can often be “as accurate” as one that employs the full characteristics. In this text the emphasis is toward developing a working knowledge of a device through the use of appropriate approximations, thereby avoiding an unnecessary level of mathematical complexity. Sufficient detail will normally be provided, however, to permit a detailed mathematical analysis if desired.

2 LOAD-LINE ANALYSIS

The circuit of Fig. 1 is the simplest of diode configurations. It will be used to describe the analysis of a diode circuit using its actual characteristics. In the next section we will replace the characteristics by an approximate model for the diode and compare solutions. Solving the circuit of Fig. 1 is all about finding the current and voltage levels that will satisfy both the characteristics of the diode and the chosen network parameters at the same time.

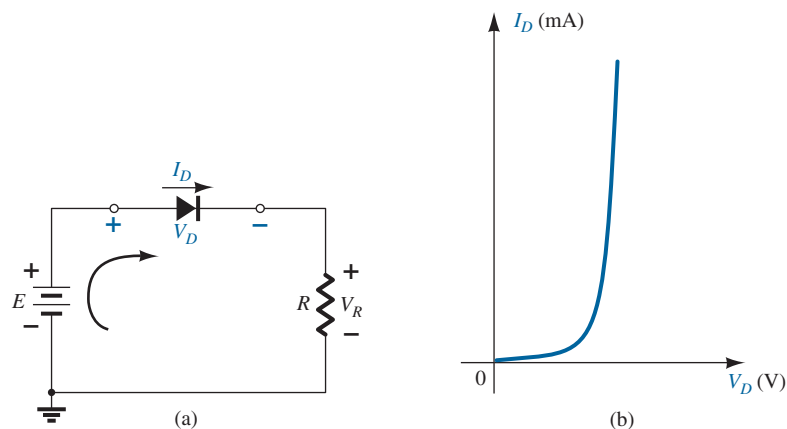
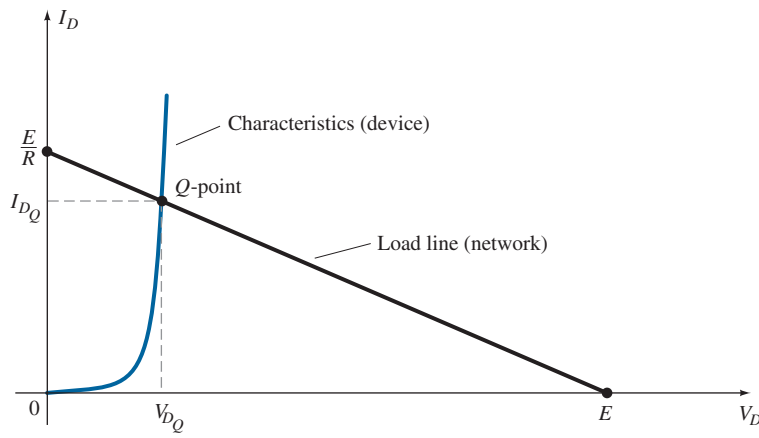


FIG. 1

Series diode configuration: (a) circuit; (b) characteristics.

In Fig. 2 the diode characteristics are placed on the same set of axes as a straight line defined by the parameters of the network. The straight line is called a *load line* because the intersection on the vertical axis is defined by the applied load R . The analysis to follow is therefore called *load-line analysis*. The intersection of the two curves will define the solution for the network and define the current and voltage levels for the network.

Before reviewing the details of drawing the load line on the characteristics, we need to determine the expected response of the simple circuit of Fig. 1. Note in Fig. 1 that the effect of the “pressure” established by the dc supply is to establish a conventional current in the direction indicated by the clockwise arrow. The fact that the direction of this current has the same direction as the arrow in the diode symbol reveals that the diode is in the “on” state and will conduct a high level of current. The polarity of the applied voltage has resulted in a forward-bias situation. With the current direction established, the polarities for the voltage across the diode and resistor can be superimposed. The polarity of V_D and the direction of I_D clearly reveal that the diode is indeed in the forward-bias state, resulting in a voltage across the diode in the neighborhood of $0.7\ \text{V}$ and a current on the order of $10\ \text{mA}$ or more.


FIG. 2

Drawing the load line and finding the point of operation.

The intersections of the load line on the characteristics of Fig. 2 can be determined by first applying Kirchhoff's voltage law in the clockwise direction, which results in

$$+E - V_D - V_R = 0$$

or

$$E = V_D + I_D R \quad (1)$$

The two variables of Eq. (1), V_D and I_D , are the same as the diode axis variables of Fig. 2. This similarity permits plotting Eq. (1) on the same characteristics of Fig. 2.

The intersections of the load line on the characteristics can easily be determined if one simply employs the fact that anywhere on the horizontal axis $I_D = 0$ A and anywhere on the vertical axis $V_D = 0$ V.

If we set $V_D = 0$ V in Eq. (1) and solve for I_D , we have the magnitude of I_D on the vertical axis. Therefore, with $V_D = 0$ V, Eq. (1) becomes

$$\begin{aligned} E &= V_D + I_D R \\ &= 0 \text{ V} + I_D R \end{aligned}$$

and

$$I_D = \frac{E}{R} \bigg|_{V_D=0 \text{ V}} \quad (2)$$

as shown in Fig. 2. If we set $I_D = 0$ A in Eq. (1) and solve for V_D , we have the magnitude of V_D on the horizontal axis. Therefore, with $I_D = 0$ A, Eq. (1) becomes

$$\begin{aligned} E &= V_D + I_D R \\ &= V_D + (0 \text{ A})R \end{aligned}$$

and

$$V_D = E \bigg|_{I_D=0 \text{ A}} \quad (3)$$

as shown in Fig. 2. A straight line drawn between the two points will define the load line as depicted in Fig. 2. Change the level of R (the load) and the intersection on the vertical axis will change. The result will be a change in the slope of the load line and a different point of intersection between the load line and the device characteristics.

We now have a load line defined by the network and a characteristic curve defined by the device. The point of intersection between the two is the point of operation for this circuit. By simply drawing a line down to the horizontal axis, we can determine the diode voltage V_{DQ} , whereas a horizontal line from the point of intersection to the vertical axis will provide the level of I_{DQ} . The current I_D is actually the current through the entire series configuration of Fig. 1a. The point of operation is usually called the *quiescent point* (abbreviated “Q-point”) to reflect its “still, unmoving” qualities as defined by a dc network.

The solution obtained at the intersection of the two curves is the same as would be obtained by a simultaneous mathematical solution of

$$I_D = \frac{E}{R} - \frac{V_D}{R} \quad [\text{derived from Eq. (1)}]$$

and

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

Since the curve for a diode has nonlinear characteristics, the mathematics involved would require the use of nonlinear techniques that are beyond the needs and scope of this text. The load-line analysis described above provides a solution with a minimum of effort and a “pictorial” description of why the levels of solution for V_{D_Q} and I_{D_Q} were obtained. The next example demonstrates the techniques introduced above and reveals the relative ease with which the load line can be drawn using Eqs. (2) and (3).

EXAMPLE 1 For the series diode configuration of Fig. 3a, employing the diode characteristics of Fig. 3b, determine:

- V_{D_Q} and I_{D_Q} .
- V_R .

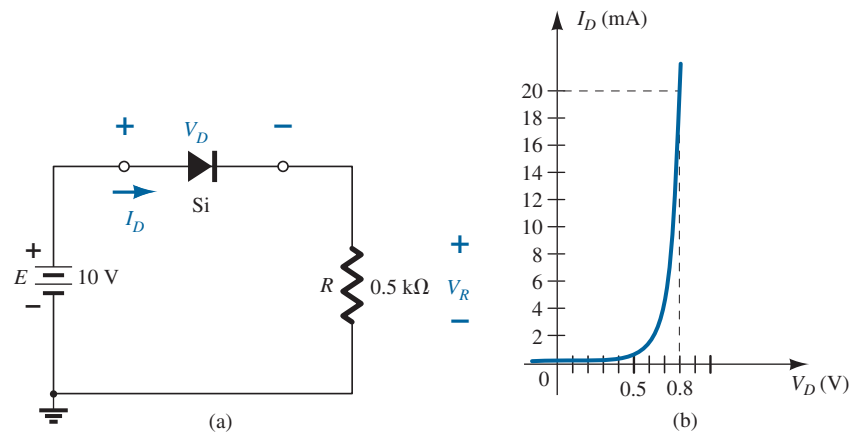


FIG. 3

(a) Circuit; (b) characteristics.

Solution:

$$\text{a. Eq. (2):} \quad I_D = \frac{E}{R} \bigg|_{V_D=0\text{ V}} = \frac{10\text{ V}}{0.5\text{ k}\Omega} = 20\text{ mA}$$

$$\text{Eq. (3):} \quad V_D = E \big|_{I_D=0\text{ A}} = 10\text{ V}$$

The resulting load line appears in Fig. 4. The intersection between the load line and the characteristic curve defines the Q -point as

$$V_{D_Q} \cong 0.78\text{ V}$$

$$I_{D_Q} \cong 18.5\text{ mA}$$

The level of V_D is certainly an estimate, and the accuracy of I_D is limited by the chosen scale. A higher degree of accuracy would require a plot that would be much larger and perhaps unwieldy.

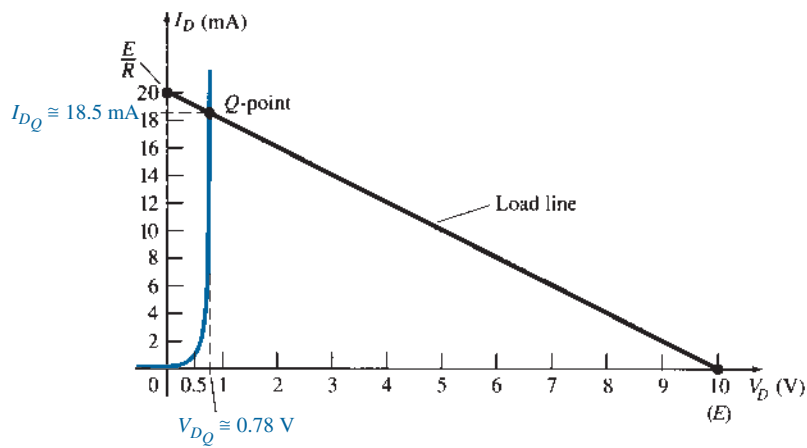
$$\text{b. } V_R = E - V_D = 10\text{ V} - 0.78\text{ V} = 9.22\text{ V}$$

As noted in the example above,

the load line is determined solely by the applied network, whereas the characteristics are defined by the chosen device.

Changing the model we use for the diode will not disturb the network so the load line to be drawn will be exactly the same as appearing in the example above.

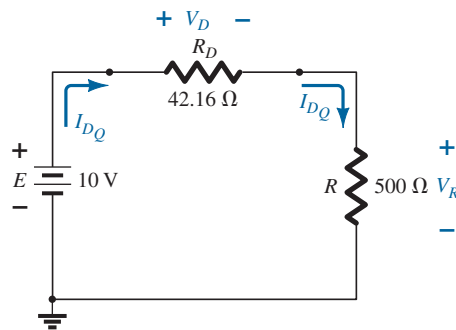
Since the network of Example 1 is a dc network the Q -point of Fig. 4 will remain fixed with $V_{D_Q} = 0.78\text{ V}$ and $I_{D_Q} = 18.5\text{ mA}$.

**FIG. 4***Solution to Example 1.*

Using the Q -point values, the dc resistance for Example 1 is

$$R_D = \frac{V_{DQ}}{I_{DQ}} = \frac{0.78 \text{ V}}{18.5 \text{ mA}} = 42.16 \, \Omega$$

An equivalent network (for these operating conditions only) can then be drawn as shown in Fig. 5.

**FIG. 5***Network equivalent to Fig. 4.*

The current

$$I_D = \frac{E}{R_D + R} = \frac{10 \text{ V}}{42.16 \, \Omega + 500 \, \Omega} = \frac{10 \text{ V}}{542.16 \, \Omega} \cong 18.5 \text{ mA}$$

and
$$V_R = \frac{RE}{R_D + R} = \frac{(500 \, \Omega)(10 \text{ V})}{42.16 \, \Omega + 500 \, \Omega} = 9.22 \text{ V}$$

matching the results of Example 1.

In essence, therefore, once a dc Q -point has been determined the diode can be replaced by its dc resistance equivalent. This concept of replacing a characteristic by an equivalent model is an important one. Let us now see what effect different equivalent models for the diode will have on the response in Example 1.

EXAMPLE 2 Repeat Example 1 using the approximate equivalent model for the silicon semiconductor diode.

Solution: The load line is redrawn as shown in Fig. 6 with the same intersections as defined in Example 1. The characteristics of the approximate equivalent circuit for the diode have also been sketched on the same graph. The resulting Q -point is

$$\begin{aligned} V_{DQ} &= 0.7 \text{ V} \\ I_{DQ} &= 18.5 \text{ mA} \end{aligned}$$

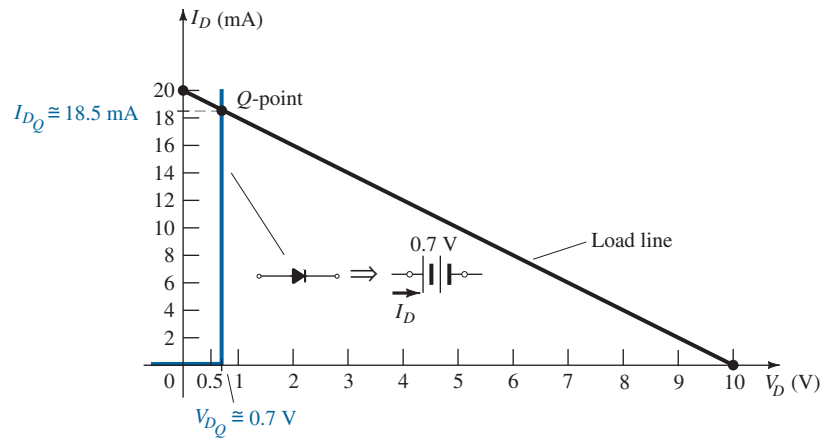


FIG. 6

Solution to Example 1 using the diode approximate model.

The results obtained in Example 2 are quite interesting. The level of I_{D_Q} is exactly the same as obtained in Example 1 using a characteristic curve that is a great deal easier to draw than that appearing in Fig. 4. The $V_D = 0.7 \text{ V}$ here and the 0.78 V from Example 1 are of a different magnitude to the hundredths place, but they are certainly in the same neighborhood if we compare their magnitudes to the magnitudes of the other voltages of the network.

For this situation the dc resistance of the Q-point is

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0.7 \text{ V}}{18.5 \text{ mA}} = 37.84 \Omega$$

which is still relatively close to that obtained for the full characteristics.

In the next example we go a step further and substitute the ideal model. The results will reveal the conditions that must be satisfied to apply the ideal equivalent properly.

EXAMPLE 3 Repeat Example 1 using the ideal diode model.

Solution: As shown in Fig. 7, the load line is the same, but the ideal characteristics now intersect the load line on the vertical axis. The Q-point is therefore defined by

$$\begin{aligned} V_{D_Q} &= 0 \text{ V} \\ I_{D_Q} &= 20 \text{ mA} \end{aligned}$$

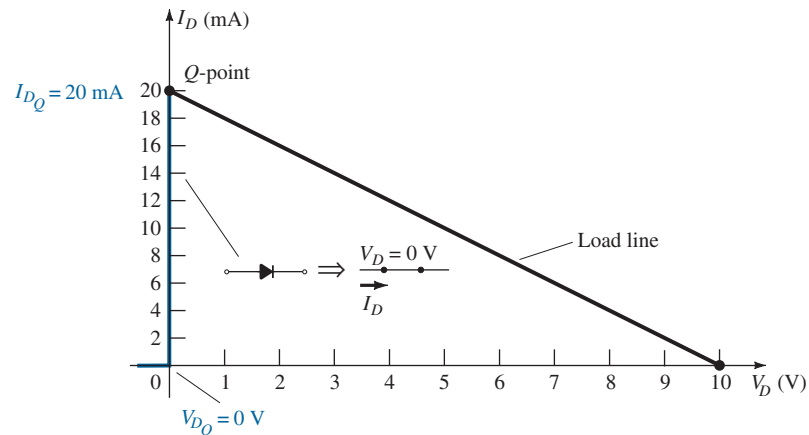


FIG. 7

Solution to Example 1 using the ideal diode model.

The results are sufficiently different from the solutions of Example 1 to cause some concern about their accuracy. Certainly, they do provide some indication of the level of voltage and current to be expected relative to the other voltage levels of the network, but the additional effort of simply including the 0.7-V offset suggests that the approach of Example 2 is more appropriate.

Use of the ideal diode model therefore should be reserved for those occasions when the role of a diode is more important than voltage levels that differ by tenths of a volt and in those situations where the applied voltages are considerably larger than the threshold voltage V_K . In the next few sections the approximate model will be employed exclusively since the voltage levels obtained will be sensitive to variations that approach V_K . In later sections the ideal model will be employed more frequently since the applied voltages will frequently be quite a bit larger than V_K and the authors want to ensure that the role of the diode is correctly and clearly understood.

In this case,

$$R_D = \frac{V_{D_Q}}{I_{D_Q}} = \frac{0 \text{ V}}{20 \text{ mA}} = 0 \Omega \text{ (or a short-circuit equivalent)}$$

3 SERIES DIODE CONFIGURATIONS

In the last section we found that the results obtained using the approximate piecewise-linear equivalent model were quite close, if not equal, to the response obtained using the full characteristics. In fact, if one considers all the variations possible due to tolerances, temperature, and so on, one could certainly consider one solution to be “as accurate” as the other. Since the use of the approximate model normally results in a reduced expenditure of time and effort to obtain the desired results, it is the approach that will be employed in this text unless otherwise specified. Recall the following:

The primary purpose of this text is to develop a general knowledge of the behavior, capabilities, and possible areas of application of a device in a manner that will minimize the need for extensive mathematical developments.

For all the analysis to follow in this chapter it is assumed that

The forward resistance of the diode is usually so small compared to the other series elements of the network that it can be ignored.

This is a valid approximation for the vast majority of applications that employ diodes. Using this fact will result in the approximate equivalents for a silicon diode and an ideal diode that appear in Table 1. For the conduction region the only difference between the silicon diode and the ideal diode is the vertical shift in the characteristics, which is accounted for in the equivalent model by a dc supply of 0.7 V opposing the direction of forward current through the device. For voltages less than 0.7 V for a silicon diode and 0 V for the ideal diode the resistance is so high compared to other elements of the network that its equivalent is the open circuit.

For a Ge diode the offset voltage is 0.3 V and for a GaAs diode it is 1.2 V. Otherwise the equivalent networks are the same. For each diode the label Si, Ge, or GaAs will appear along with the diode symbol. For networks with ideal diodes the diode symbol will appear as shown in Table 1 without any labels.

The approximate models will now be used to investigate a number of series diode configurations with dc inputs. This will establish a foundation in diode analysis that will carry over into the sections to follow. The procedure described can, in fact, be applied to networks with any number of diodes in a variety of configurations.

For each configuration the state of each diode must first be determined. Which diodes are “on” and which are “off”? Once determined, the appropriate equivalent can be substituted and the remaining parameters of the network determined.

In general, a diode is in the “on” state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol, and $V_D \geq 0.7 \text{ V}$ for silicon, $V_D \geq 0.3 \text{ V}$ for germanium, and $V_D \geq 1.2 \text{ V}$ for gallium arsenide.

For each configuration, mentally replace the diodes with resistive elements and note the resulting current direction as established by the applied voltages (“pressure”). If the resulting

TABLE 1

Approximate and Ideal Semiconductor Diode Models.

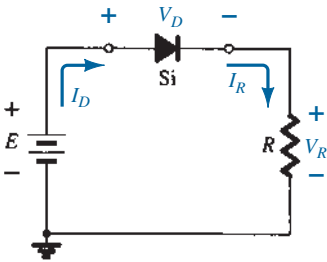
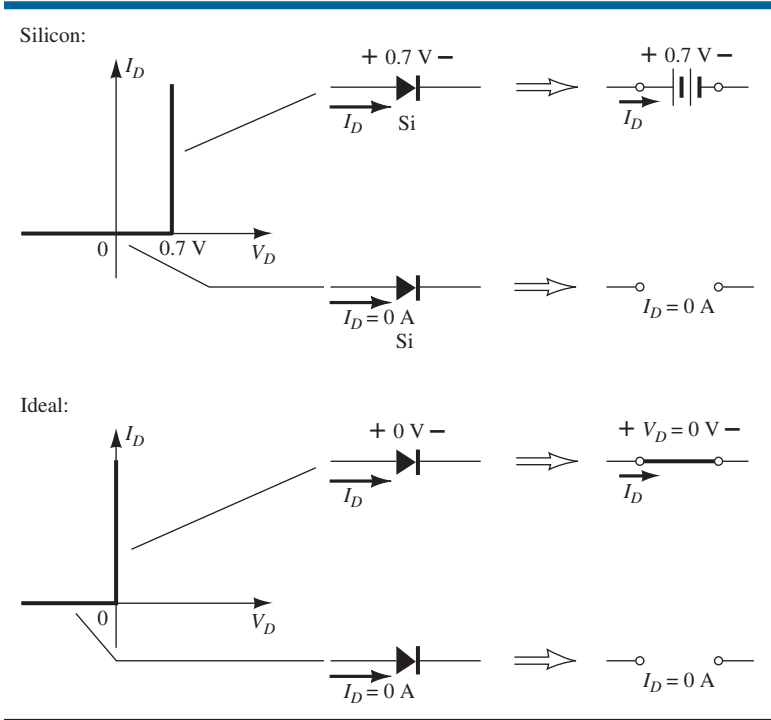


FIG. 8
Series diode configuration.

direction is a “match” with the arrow in the diode symbol, conduction through the diode will occur and the device is in the “on” state. The description above is, of course, contingent on the supply having a voltage greater than the “turn-on” voltage (V_K) of each diode.

If a diode is in the “on” state, one can either place a 0.7-V drop across the element or redraw the network with the V_K equivalent circuit as defined in Table 1. In time the preference will probably simply be to include the 0.7-V drop across each “on” diode and to draw a diagonal line through each diode in the “off” or open state. Initially, however, the substitution method will be used to ensure that the proper voltage and current levels are determined.

The series circuit of Fig. 8 described in some detail in Section 2 will be used to demonstrate the approach described in the above paragraphs. The state of the diode is first determined by mentally replacing the diode with a resistive element as shown in Fig. 9a. The resulting direction of I is a match with the arrow in the diode symbol, and since $E > V_K$, the diode is in the “on” state. The network is then redrawn as shown in Fig. 9b with the appropriate equivalent model for the forward-biased silicon diode. Note for future reference that the polarity of V_D is the same as would result if in fact the diode were a resistive element. The resulting voltage and current levels are the following:

$V_D = V_K$

(4)

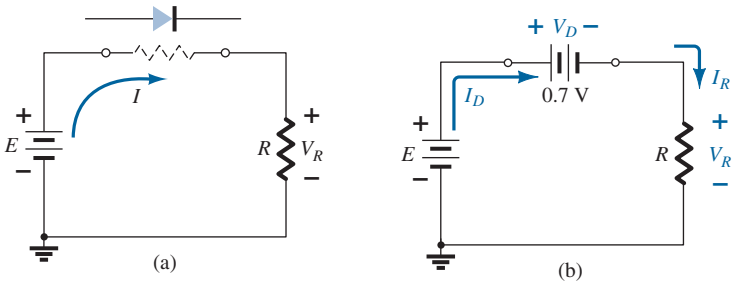


FIG. 9
(a) Determining the state of the diode of Fig. 8; (b) substituting the equivalent model for the “on” diode of Fig. 9a.

$$V_R = E - V_K \quad (5)$$

$$I_D = I_R = \frac{V_R}{R} \quad (6)$$

In Fig. 10 the diode of Fig. 7 has been reversed. Mentally replacing the diode with a resistive element as shown in Fig. 11 will reveal that the resulting current direction does not match the arrow in the diode symbol. The diode is in the “off” state, resulting in the equivalent circuit of Fig. 12. Due to the open circuit, the diode current is 0 A and the voltage across the resistor R is the following:

$$V_R = I_R R = I_D R = (0 \text{ A})R = 0 \text{ V}$$

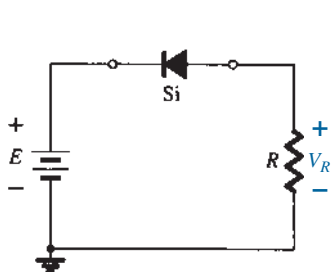


FIG. 10

Reversing the diode of Fig. 8.

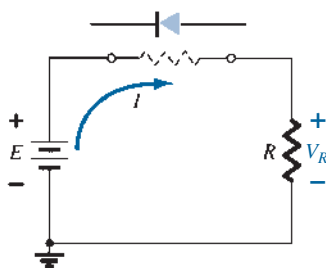


FIG. 11

Determining the state of the diode of Fig. 10.

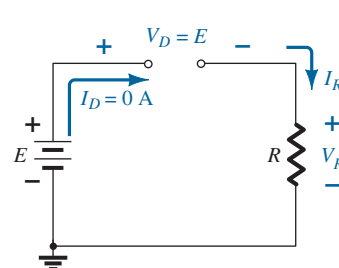


FIG. 12

Substituting the equivalent model for the “off” diode of Fig. 10.

The fact that $V_R = 0 \text{ V}$ will establish E volts across the open circuit as defined by Kirchhoff's voltage law. Always keep in mind that under any circumstances—dc, ac instantaneous values, pulses, and so on—Kirchhoff's voltage law must be satisfied!

EXAMPLE 4 For the series diode configuration of Fig. 13, determine V_D , V_R , and I_D .

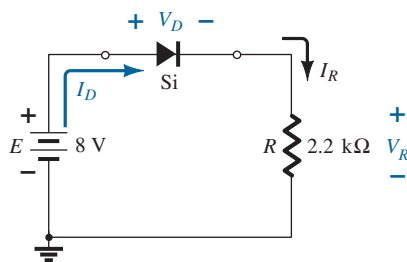


FIG. 13

Circuit for Example 4.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = 0.7 \text{ V}$$

$$V_R = E - V_D = 8 \text{ V} - 0.7 \text{ V} = 7.3 \text{ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3 \text{ V}}{2.2 \text{ k}\Omega} \cong 3.32 \text{ mA}$$

EXAMPLE 5 Repeat Example 4 with the diode reversed.

Solution: Removing the diode, we find that the direction of I is opposite to the arrow in the diode symbol and the diode equivalent is the open circuit no matter which model is employed. The result is the network of Fig. 14, where $I_D = 0$ A due to the open circuit. Since $V_R = I_R R$, we have $V_R = (0)R = 0$ V. Applying Kirchhoff's voltage law around the closed loop yields

$$E - V_D - V_R = 0$$

and

$$V_D = E - V_R = E - 0 = E = 8 \text{ V}$$

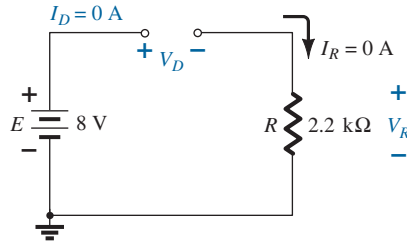


FIG. 14

Determining the unknown quantities for Example 5.

In particular, note in Example 5 the high voltage across the diode even though it is an “off” state. The current is zero, but the voltage is significant. For review purposes, keep the following in mind for the analysis to follow:

An open circuit can have any voltage across its terminals, but the current is always 0 A. A short circuit has a 0-V drop across its terminals, but the current is limited only by the surrounding network.

In the next example the notation of Fig. 15 will be employed for the applied voltage. It is a common industry notation and one with which the reader should become very familiar.

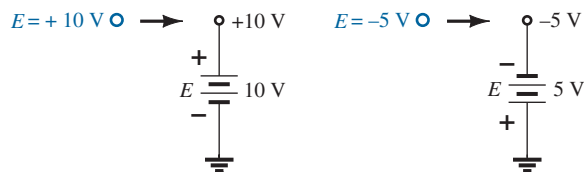


FIG. 15

Source notation.

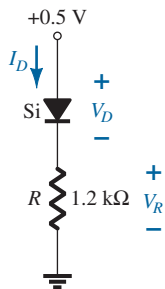


FIG. 16

Series diode circuit for Example 6.

EXAMPLE 6 For the series diode configuration of Fig. 16, determine V_D , V_R , and I_D .

Solution: Although the “pressure” establishes a current with the same direction as the arrow symbol, the level of applied voltage is insufficient to turn the silicon diode “on.” The point of operation on the characteristics is shown in Fig. 17, establishing the open-circuit equivalent as the appropriate approximation, as shown in Fig. 18. The resulting voltage and current levels are therefore the following:

$$I_D = 0 \text{ A}$$

$$V_R = I_R R = I_D R = (0 \text{ A}) 1.2 \text{ k}\Omega = 0 \text{ V}$$

and

$$V_D = E = 0.5 \text{ V}$$

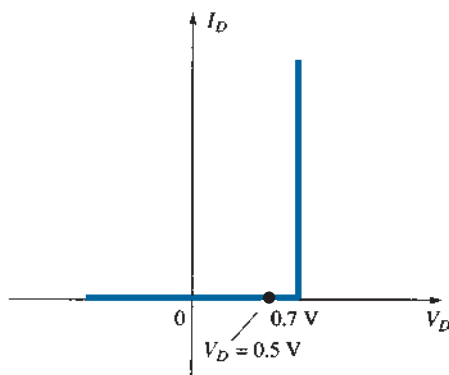


FIG. 17

Operating point with $E = 0.5 \text{ V}$.

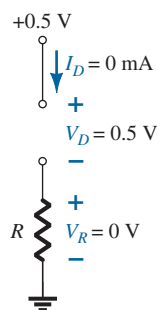


FIG. 18

Determining I_D , V_R , and V_D for the circuit of Fig. 16.

EXAMPLE 7 Determine V_o and I_D for the series circuit of Fig. 19.

Solution: An attack similar to that applied in Example 4 will reveal that the resulting current has the same direction as the arrowheads of the symbols of both diodes, and the network of Fig. 20 results because $E = 12 \text{ V} > (0.7 \text{ V} + 1.8 \text{ V}) = 2.5 \text{ V}$. Note the redrawn supply of 12 V and the polarity of V_o across the $680\text{-}\Omega$ resistor. The resulting voltage is

$$V_o = E - V_{K_1} - V_{K_2} = 12 \text{ V} - 2.5 \text{ V} = 9.5 \text{ V}$$

and

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{9.5 \text{ V}}{680 \Omega} = 13.97 \text{ mA}$$

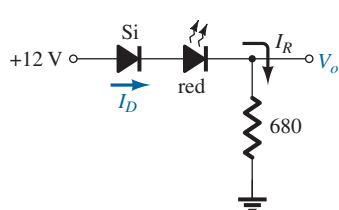


FIG. 19

Circuit for Example 7.

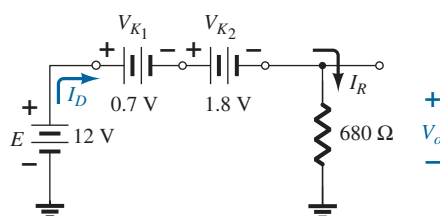


FIG. 20

Determining the unknown quantities for Example 7.

EXAMPLE 8 Determine I_D , V_{D_2} , and V_o for the circuit of Fig. 21.

Solution: Removing the diodes and determining the direction of the resulting current I result in the circuit of Fig. 22. There is a match in current direction for one silicon diode but not for the other silicon diode. The combination of a short circuit in series with an open circuit always results in an open circuit and $I_D = 0 \text{ A}$, as shown in Fig. 23.

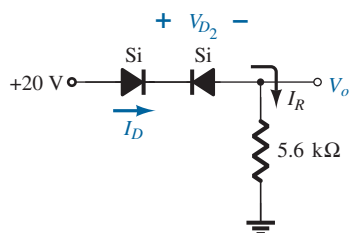


FIG. 21

Circuit for Example 8.

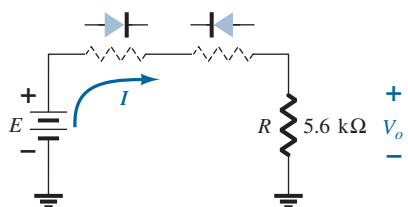


FIG. 22

Determining the state of the diodes of Fig. 21.

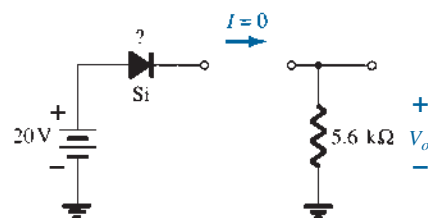


FIG. 23

Substituting the equivalent state for the open diode.

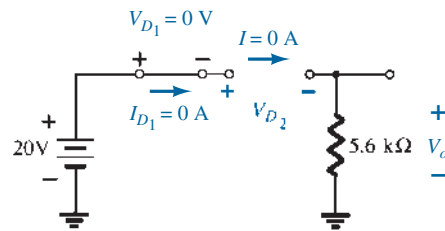


FIG. 24

Determining the unknown quantities for the circuit of Example 8.

The question remains as to what to substitute for the silicon diode. For the analysis to follow in this chapter, simply recall for the actual practical diode that when $I_D = 0$ A, $V_D = 0$ V (and vice versa). The conditions described by $I_D = 0$ A and $V_{D1} = 0$ V are indicated in Fig. 24. We have

$$V_o = I_R R = I_D R = (0 \text{ A}) R = 0 \text{ V}$$

and

$$V_{D2} = V_{\text{open circuit}} = E = 20 \text{ V}$$

Applying Kirchhoff's voltage law in a clockwise direction gives

$$E - V_{D1} - V_{D2} - V_o = 0$$

and

$$\begin{aligned} V_{D2} &= E - V_{D1} - V_o = 20 \text{ V} - 0 - 0 \\ &= 20 \text{ V} \end{aligned}$$

with

$$V_o = 0 \text{ V}$$

EXAMPLE 9 Determine I , V_1 , V_2 , and V_o for the series dc configuration of Fig. 25.

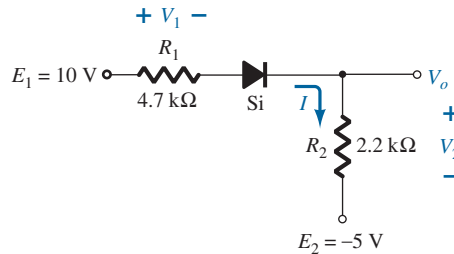


FIG. 25

Circuit for Example 9.

Solution: The sources are drawn and the current direction indicated as shown in Fig. 26. The diode is in the “on” state and the notation appearing in Fig. 27 is included to indicate this state. Note that the “on” state is noted simply by the additional $V_D = 0.7$ V on the figure. This eliminates the need to redraw the network and avoids any confusion that may

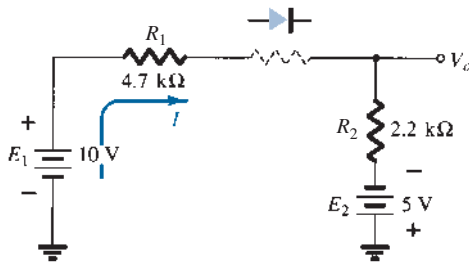


FIG. 26

Determining the state of the diode for the network of Fig. 25.

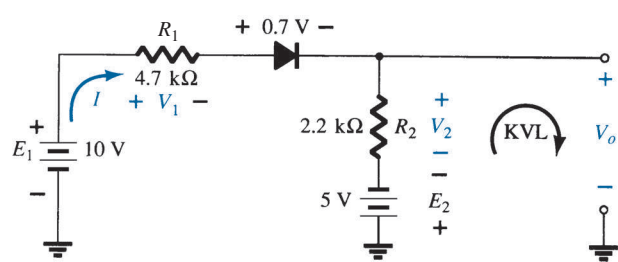


FIG. 27

Determining the unknown quantities for the network of Fig. 25. KVL, Kirchhoff voltage loop.

result from the appearance of another source. As indicated in the introduction to this section, this is probably the path and notation that one will take when a level of confidence has been established in the analysis of diode configurations. In time the entire analysis will be performed simply by referring to the original network. Recall that a reverse-biased diode can simply be indicated by a line through the device.

The resulting current through the circuit is

$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10 \text{ V} + 5 \text{ V} - 0.7 \text{ V}}{4.7 \text{ k}\Omega + 2.2 \text{ k}\Omega} = \frac{14.3 \text{ V}}{6.9 \text{ k}\Omega} \\ \cong \mathbf{2.07 \text{ mA}}$$

and the voltages are

$$V_1 = IR_1 = (2.07 \text{ mA})(4.7 \text{ k}\Omega) = \mathbf{9.73 \text{ V}}$$

$$V_2 = IR_2 = (2.07 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{4.55 \text{ V}}$$

Applying Kirchhoff's voltage law to the output section in the clockwise direction results in

$$-E_2 + V_2 - V_o = 0$$

and

$$V_o = V_2 - E_2 = 4.55 \text{ V} - 5 \text{ V} = \mathbf{-0.45 \text{ V}}$$

The minus sign indicates that V_o has a polarity opposite to that appearing in Fig. 25.

4 PARALLEL AND SERIES-PARALLEL CONFIGURATIONS

The methods applied in Section 3 can be extended to the analysis of parallel and series-parallel configurations. For each area of application, simply match the sequential series of steps applied to series diode configurations.

EXAMPLE 10 Determine V_o , I_1 , I_{D1} , and I_{D2} for the parallel diode configuration of Fig. 28.

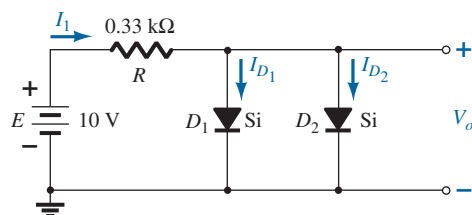


FIG. 28

Network for Example 10.

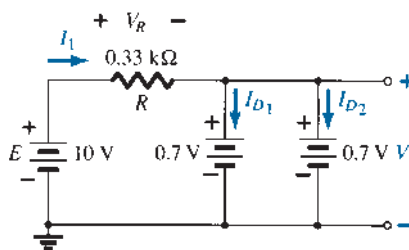


FIG. 29

Determining the unknown quantities for the network of Example 10.

Solution: For the applied voltage the “pressure” of the source acts to establish a current through each diode in the same direction as shown in Fig. 29. Since the resulting current direction matches that of the arrow in each diode symbol and the applied voltage is greater than 0.7 V, both diodes are in the “on” state. The voltage across parallel elements is always the same and

$$V_o = \mathbf{0.7 \text{ V}}$$

The current is

$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10 \text{ V} - 0.7 \text{ V}}{0.33 \text{ k}\Omega} = \mathbf{28.18 \text{ mA}}$$

Assuming diodes of similar characteristics, we have

$$I_{D1} = I_{D2} = \frac{I_1}{2} = \frac{28.18 \text{ mA}}{2} = \mathbf{14.09 \text{ mA}}$$

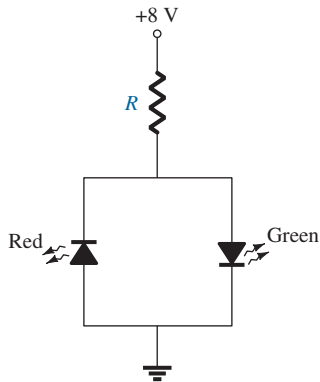


FIG. 30

Network for Example 11.

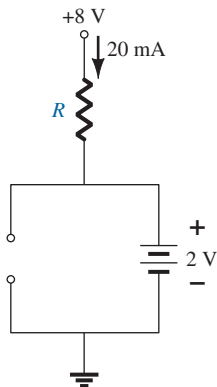


FIG. 31

Operating conditions for the network of Fig. 30.

This example demonstrates one reason for placing diodes in parallel. If the current rating of the diodes of Fig. 28 is only 20 mA, a current of 28.18 mA would damage the device if it appeared alone in Fig. 28. By placing two in parallel, we limit the current to a safe value of 14.09 mA with the same terminal voltage.

EXAMPLE 11 In this example there are two LEDs that can be used as a polarity detector. Apply a positive source voltage and a green light results. Negative supplies result in a red light. Packages of such combinations are commercially available.

Find the resistor R to ensure a current of 20 mA through the “on” diode for the configuration of Fig. 30. Both diodes have a reverse breakdown voltage of 3 V and an average turn-on voltage of 2 V.

Solution: The application of a positive supply voltage results in a conventional current that matches the arrow of the green diode and turns it on.

The polarity of the voltage across the green diode is such that it reverse biases the red diode by the same amount. The result is the equivalent network of Fig. 31.

Applying Ohm’s law, we obtain

$$I = 20 \text{ mA} = \frac{E - V_{\text{LED}}}{R} = \frac{8 \text{ V} - 2 \text{ V}}{R}$$

and

$$R = \frac{6 \text{ V}}{20 \text{ mA}} = 300 \, \Omega$$

Note that the reverse breakdown voltage across the red diode is 2 V, which is fine for an LED with a reverse breakdown voltage of 3 V.

However, if the green diode were to be replaced by a blue diode, problems would develop, as shown in Fig. 32. Recall that the forward bias required to turn on a blue diode is about 5 V. The result would appear to require a smaller resistor R to establish the current of 20 mA. However, note that the reverse bias voltage of the red LED is 5 V, but the reverse breakdown voltage of the diode is only 3 V. The result is the voltage across the red LED would lock in at 3 V as shown in Fig. 33. The voltage across R would be 5 V and the current limited to 20 mA with a 250 Ω resistor but neither LED would be on.

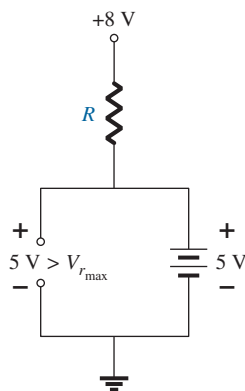


FIG. 32

Network of Fig. 31 with a blue diode.

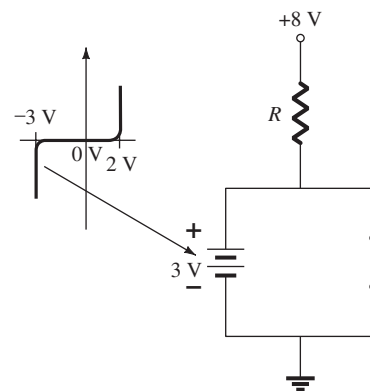
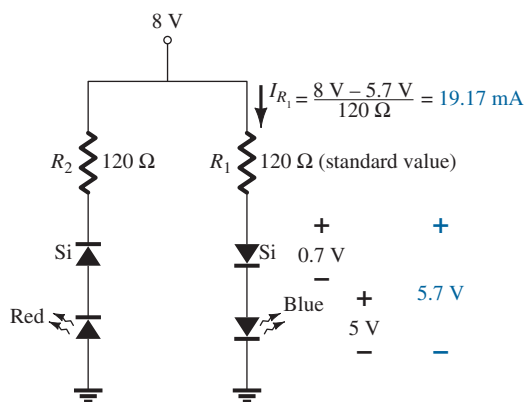


FIG. 33

Demonstrating damage to the red LED if the reverse breakdown voltage is exceeded.

A simple solution to the above is to add the appropriate resistance level in series with each diode to establish the desired 20 mA and to include another diode to add to the reverse-bias total reverse breakdown voltage rating, as shown in Fig. 34. When the blue LED is on, the diode in series with the blue LED will also be on, causing a total voltage drop of 5.7 V across the two series diodes and a voltage of 2.3 V across the resistor R_1 , establishing a high emission current of 19.17 mA. At the same time the red LED diode and

**FIG. 34**

Protective measure for the red LED of Fig. 33.

its series diode will also be reverse biased, but now the standard diode with a reverse breakdown voltage of 20 V will prevent the full reverse-bias voltage of 8 V from appearing across the red LED. When forward biased, the resistor R_2 will establish a current of 19.63 mA to ensure a high level of intensity for the red LED.

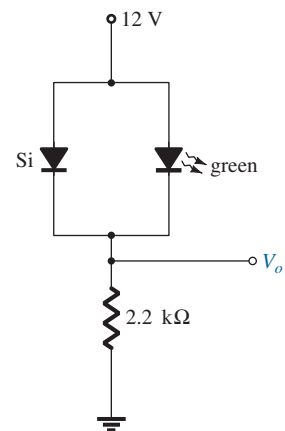
EXAMPLE 12 Determine the voltage V_o for the network of Fig. 35.

Solution: Initially, it might appear that the applied voltage will turn both diodes “on” because the applied voltage (“pressure”) is trying to establish a conventional current through each diode that would suggest the “on” state. However, if both were on, there would be more than one voltage across the parallel diodes, violating one of the basic rules of network analysis: The voltage must be the same across parallel elements.

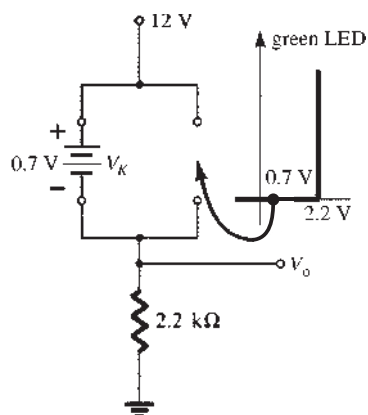
The resulting action can best be explained by remembering that there is a period of build-up of the supply voltage from 0 V to 12 V even though it may take milliseconds or microseconds. At the instant the increasing supply voltage reaches 0.7 V the silicon diode will turn “on” and maintain the level of 0.7 V since the characteristic is vertical at this voltage—the current of the silicon diode will simply rise to the defined level. The result is that the voltage across the green LED will never rise above 0.7 V and will remain in the equivalent open-circuit state as shown in Fig. 36.

The result is

$$V_o = 12 \text{ V} - 0.7 \text{ V} = 11.3 \text{ V}$$

**FIG. 35**

Network for Example 12.

**FIG. 36**

Determining V_o for the network of Fig. 35.

EXAMPLE 13 Determine the currents I_1 , I_2 , and I_{D_2} for the network of Fig. 37.

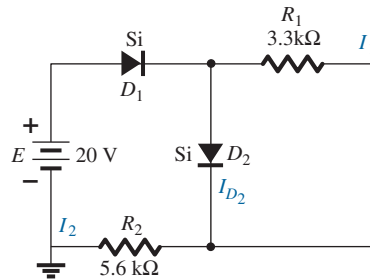


FIG. 37

Network for Example 13.

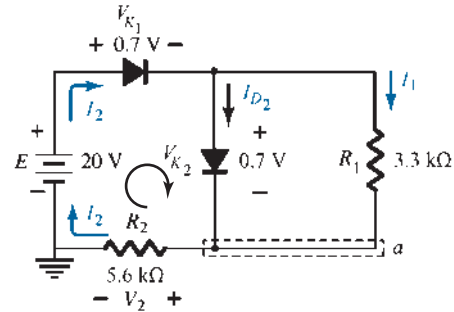


FIG. 38

Determining the unknown quantities for Example 13.

Solution: The applied voltage (pressure) is such as to turn both diodes on, as indicated by the resulting current directions in the network of Fig. 38. Note the use of the abbreviated notation for “on” diodes and that the solution is obtained through an application of techniques applied to dc series–parallel networks. We have

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = \mathbf{0.212 \text{ mA}}$$

Applying Kirchhoff’s voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

$$\text{and} \quad V_2 = E - V_{K_1} - V_{K_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = \mathbf{18.6 \text{ V}}$$

$$\text{with} \quad I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = \mathbf{3.32 \text{ mA}}$$

At the bottom node a ,

$$I_{D_2} + I_1 = I_2$$

$$\text{and} \quad I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} \cong \mathbf{3.11 \text{ mA}}$$

5 AND/OR GATES

The tools of analysis are now at our disposal, and the opportunity to investigate a computer configuration is one that will demonstrate the range of applications of this relatively simple device. Our analysis will be limited to determining the voltage levels and will not include a detailed discussion of Boolean algebra or positive and negative logic.

The network to be analyzed in Example 14 is an OR gate for positive logic. That is, the 10-V level of Fig. 39 is assigned a “1” for Boolean algebra and the 0-V input is assigned a “0.” An OR gate is such that the output voltage level will be a 1 if either *or* both inputs are a 1. The output is a 0 if both inputs are at the 0 level.

The analysis of AND/OR gates is made easier by using the approximate equivalent for a diode rather than the ideal because we can stipulate that the voltage across the diode must be 0.7 V positive for the silicon diode to switch to the “on” state.

In general, the best approach is simply to establish a “gut” feeling for the state of the diodes by noting the direction and the “pressure” established by the applied potentials. The analysis will then verify or negate your initial assumptions.

EXAMPLE 14 Determine V_o for the network of Fig. 39.

Solution: First note that there is only one applied potential; 10 V at terminal 1. Terminal 2 with a 0-V input is essentially at ground potential, as shown in the redrawn network of

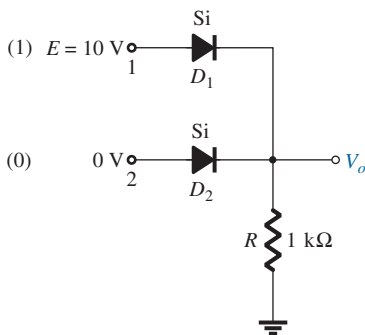


FIG. 39

Positive logic OR gate.

However, recall that we mentioned in the introduction to this section that the use of the approximate model will be an aid to the analysis. For D_1 , where will the 0.7 V come from if the input and source voltages are at the same level and creating opposing “pressures”? D_2 is assumed to be in the “on” state due to the low voltage at the cathode side and the availability of the 10-V source through the 1-k Ω resistor.

For the network of Fig. 43 the voltage at V_o is 0.7 V due to the forward-biased diode D_2 . With 0.7 V at the anode of D_1 and 10 V at the cathode, D_1 is definitely in the “off” state. The current I will have the direction indicated in Fig. 43 and a magnitude equal to

$$I = \frac{E - V_K}{R} = \frac{10 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = \mathbf{9.3 \text{ mA}}$$

The state of the diodes is therefore confirmed and our earlier analysis was correct. Although not 0 V as earlier defined for the 0 level, the output voltage is sufficiently small to be considered a 0 level. For the AND gate, therefore, a single input will result in a 0-level output. The remaining states of the diodes for the possibilities of two inputs and no inputs will be examined in the problems at the end of the chapter.

6 SINUSOIDAL INPUTS; HALF-WAVE RECTIFICATION

The diode analysis will now be expanded to include time-varying functions such as the sinusoidal waveform and the square wave. There is no question that the degree of difficulty will increase, but once a few fundamental maneuvers are understood, the analysis will be fairly direct and follow a common thread.

The simplest of networks to examine with a time-varying signal appears in Fig. 44. For the moment we will use the ideal model (note the absence of the Si, Ge, or GaAs label) to ensure that the approach is not clouded by additional mathematical complexity.

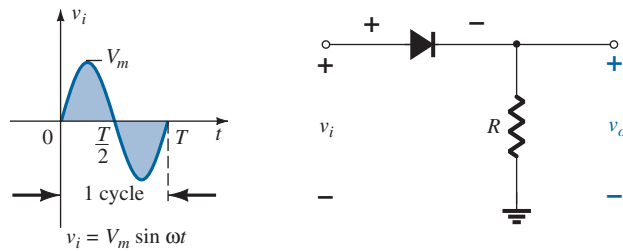
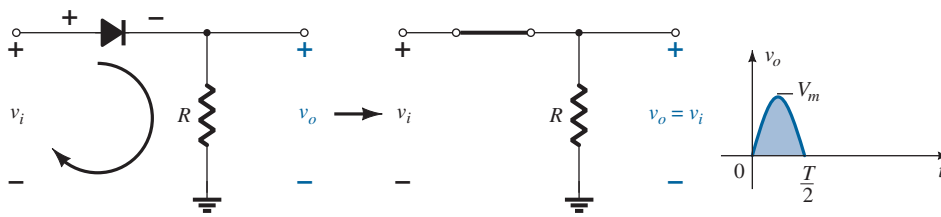


FIG. 44
Half-wave rectifier.

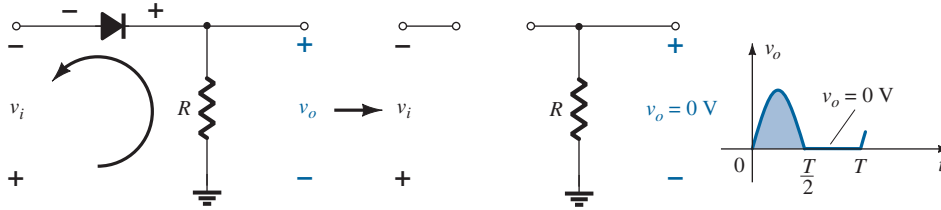
Over one full cycle, defined by the period T of Fig. 44, the average value (the algebraic sum of the areas above and below the axis) is zero. The circuit of Fig. 44, called a *half-wave rectifier*, will generate a waveform v_o that will have an average value of particular use in the ac-to-dc conversion process. When employed in the rectification process, a diode is typically referred to as a *rectifier*. Its power and current ratings are typically much higher than those of diodes employed in other applications, such as computers and communication systems.

During the interval $t = 0 \rightarrow T/2$ in Fig. 44 the polarity of the applied voltage v_i is such as to establish “pressure” in the direction indicated and turn on the diode with the polarity appearing above the diode. Substituting the short-circuit equivalence for the ideal diode will result in the equivalent circuit of Fig. 45, where it is fairly obvious that the output signal is an exact replica of the applied signal. The two terminals defining the output voltage are connected directly to the applied signal via the short-circuit equivalence of the diode.

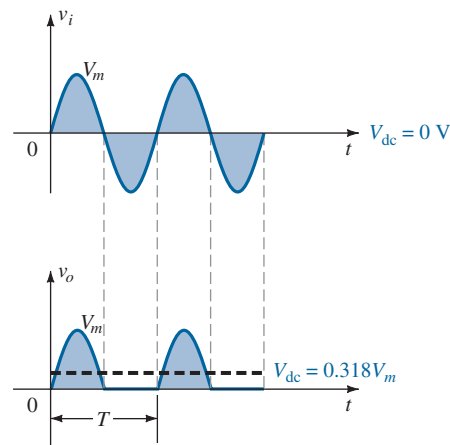
For the period $T/2 \rightarrow T$, the polarity of the input v_i is as shown in Fig. 46, and the resulting polarity across the ideal diode produces an “off” state with an open-circuit equivalent. The result is the absence of a path for charge to flow, and $v_o = iR = (0)R = 0 \text{ V}$ for the period $T/2 \rightarrow T$. The input v_i and the output v_o are sketched together in Fig. 47 for comparison purposes. The output signal v_o now has a net positive area above the axis over


FIG. 45

Conduction region ($0 \rightarrow T/2$).


FIG. 46

Nonconduction region ($T/2 \rightarrow T$).


FIG. 47

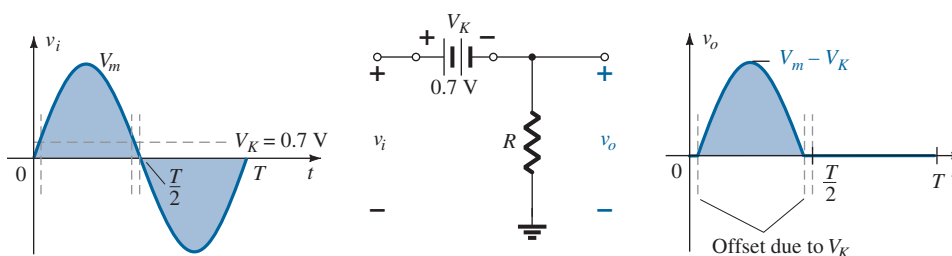
Half-wave rectified signal.

a full period and an average value determined by

$$V_{dc} = 0.318 V_m \quad \text{half-wave} \quad (7)$$

The process of removing one-half the input signal to establish a dc level is called *half-wave rectification*.

The effect of using a silicon diode with $V_K = 0.7 \text{ V}$ is demonstrated in Fig. 48 for the forward-bias region. The applied signal must now be at least 0.7 V before the diode can turn “on.” For levels of v_i less than 0.7 V , the diode is still in an open-circuit state and $v_o = 0 \text{ V}$, as shown in the same figure. When conducting, the difference between v_o and v_i is a fixed


FIG. 48

Effect of V_K on half-wave rectified signal.

level of $V_K = 0.7 \text{ V}$ and $v_o = v_i - V_K$, as shown in the figure. The net effect is a reduction in area above the axis, which reduces the resulting dc voltage level. For situations where $V_m \gg V_K$, the following equation can be applied to determine the average value with a relatively high level of accuracy.

$$V_{dc} \cong 0.318(V_m - V_K) \quad (8)$$

In fact, if V_m is sufficiently greater than V_K , Eq. (7) is often applied as a first approximation for V_{dc} .

EXAMPLE 16

- Sketch the output v_o and determine the dc level of the output for the network of Fig. 49.
- Repeat part (a) if the ideal diode is replaced by a silicon diode.
- Repeat parts (a) and (b) if V_m is increased to 200 V, and compare solutions using Eqs. (7) and (8).

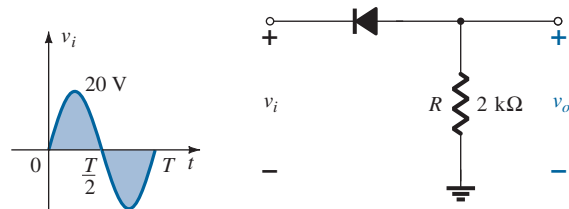


FIG. 49

Network for Example 16.

Solution:

- In this situation the diode will conduct during the negative part of the input as shown in Fig. 50, and v_o will appear as shown in the same figure. For the full period, the dc level is

$$V_{dc} = -0.318V_m = -0.318(20 \text{ V}) = -6.36 \text{ V}$$

The negative sign indicates that the polarity of the output is opposite to the defined polarity of Fig. 49.

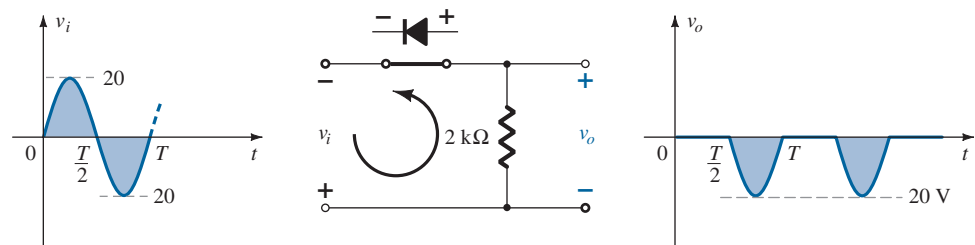


FIG. 50

Resulting v_o for the circuit of Example 16.

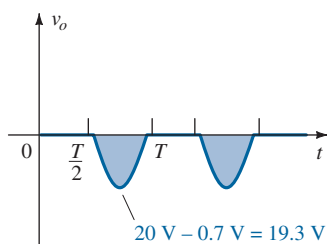


FIG. 51

Effect of V_K on output of Fig. 50.

- For a silicon diode, the output has the appearance of Fig. 51, and

$$V_{dc} \cong -0.318(V_m - 0.7 \text{ V}) = -0.318(19.3 \text{ V}) \cong -6.14 \text{ V}$$

The resulting drop in dc level is 0.22 V, or about 3.5%.

$$\text{c. Eq. (7): } V_{dc} = -0.318 V_m = -0.318(200 \text{ V}) = -63.6 \text{ V}$$

$$\begin{aligned} \text{Eq. (8): } V_{dc} &= -0.318(V_m - V_K) = -0.318(200 \text{ V} - 0.7 \text{ V}) \\ &= -(0.318)(199.3 \text{ V}) = -63.38 \text{ V} \end{aligned}$$

which is a difference that can certainly be ignored for most applications. For part (c) the offset and drop in amplitude due to V_K would not be discernible on a typical oscilloscope if the full pattern is displayed.

The peak inverse voltage (PIV) [or PRV (peak reverse voltage)] rating of the diode is of primary importance in the design of rectification systems. Recall that it is the voltage rating that must not be exceeded in the reverse-bias region or the diode will enter the Zener avalanche region. The required PIV rating for the half-wave rectifier can be determined from Fig. 52, which displays the reverse-biased diode of Fig. 44 with maximum applied voltage. Applying Kirchhoff's voltage law, it is fairly obvious that the PIV rating of the diode must equal or exceed the peak value of the applied voltage. Therefore,

$$\text{PIV rating} \geq V_m \quad \text{half-wave rectifier} \quad (9)$$

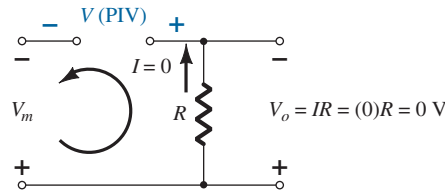


FIG. 52

Determining the required PIV rating for the half-wave rectifier.

7 FULL-WAVE RECTIFICATION

Bridge Network

The dc level obtained from a sinusoidal input can be improved 100% using a process called *full-wave rectification*. The most familiar network for performing such a function appears in Fig. 53 with its four diodes in a *bridge* configuration. During the period $t = 0$ to $T/2$ the polarity of the input is as shown in Fig. 54. The resulting polarities across the ideal diodes are also shown in Fig. 54 to reveal that D_2 and D_3 are conducting, whereas D_1 and D_4 are in the "off" state. The net result is the configuration of Fig. 55, with its indicated current and polarity across R . Since the diodes are ideal, the load voltage is $v_o = v_i$, as shown in the same figure.

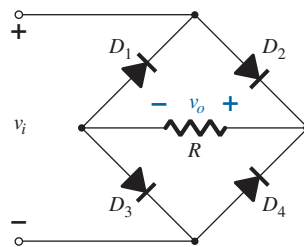
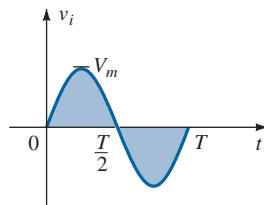


FIG. 53

Full-wave bridge rectifier.

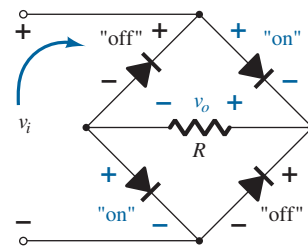


FIG. 54

Network of Fig. 53 for the period $0 \rightarrow T/2$ of the input voltage v_i .

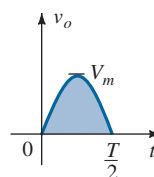
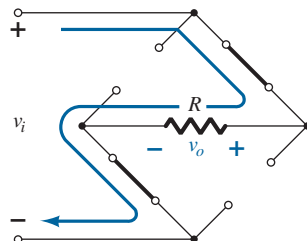
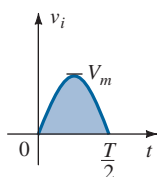


FIG. 55

Conduction path for the positive region of v_i .

For the negative region of the input the conducting diodes are D_1 and D_4 , resulting in the configuration of Fig. 56. The important result is that the polarity across the load resistor R is the same as in Fig. 54, establishing a second positive pulse, as shown in Fig. 56. Over one full cycle the input and output voltages will appear as shown in Fig. 57.

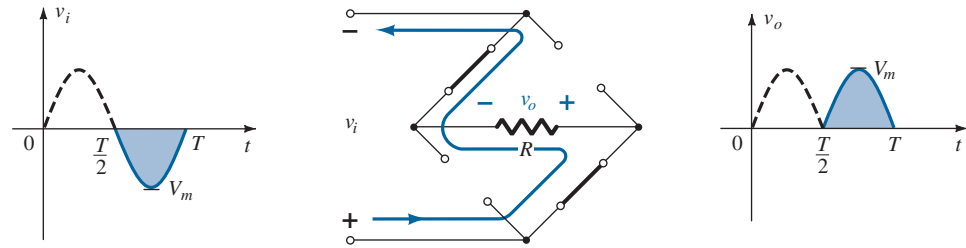


FIG. 56

Conduction path for the negative region of v_i .

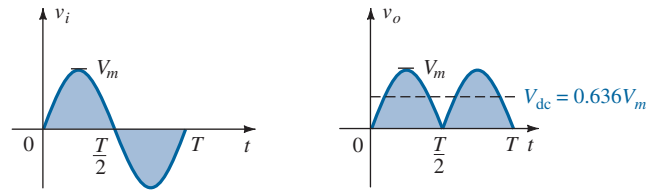


FIG. 57

Input and output waveforms for a full-wave rectifier.

Since the area above the axis for one full cycle is now twice that obtained for a half-wave system, the dc level has also been doubled and

$$V_{dc} = 2[\text{Eq. (7)}] = 2(0.318V_m)$$

or

$$V_{dc} = 0.636 V_m \quad \text{full-wave} \quad (10)$$

If silicon rather than ideal diodes are employed as shown in Fig. 58, the application of Kirchhoff's voltage law around the conduction path results in

$$v_i - V_K - v_o - V_K = 0$$

and

$$v_o = v_i - 2V_K$$

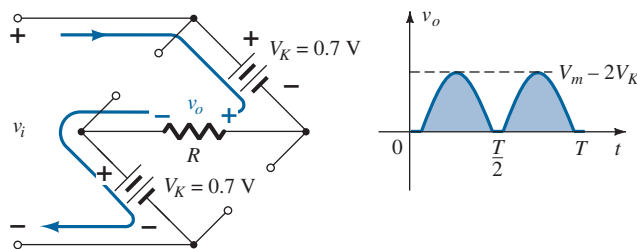


FIG. 58

Determining $V_{o_{max}}$ for silicon diodes in the bridge configuration.

The peak value of the output voltage v_o is therefore

$$V_{o_{max}} = V_m - 2V_K$$

For situations where $V_m \gg 2V_K$, the following equation can be applied for the average value with a relatively high level of accuracy:

$$V_{dc} \cong 0.636(V_m - 2V_K) \quad (11)$$

Then again, if V_m is sufficiently greater than $2V_K$, then Eq. (10) is often applied as a first approximation for V_{dc} .

PIV The required PIV of each diode (ideal) can be determined from Fig. 59 obtained at the peak of the positive region of the input signal. For the indicated loop the maximum voltage across R is V_m and the PIV rating is defined by

$$\text{PIV} \geq V_m \quad \text{full-wave bridge rectifier} \quad (12)$$

Center-Tapped Transformer

A second popular full-wave rectifier appears in Fig. 60 with only two diodes but requiring a center-tapped (CT) transformer to establish the input signal across each section of the secondary of the transformer. During the positive portion of v_i applied to the primary of the transformer, the network will appear as shown in Fig. 61 with a positive pulse across each section of the secondary coil. D_1 assumes the short-circuit equivalent and D_2 the open-circuit equivalent, as determined by the secondary voltages and the resulting current directions. The output voltage appears as shown in Fig. 61.

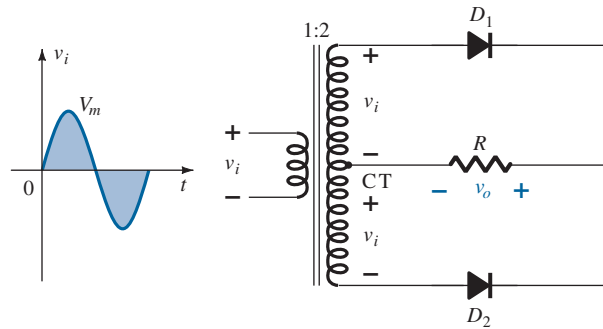


FIG. 60

Center-tapped transformer full-wave rectifier.

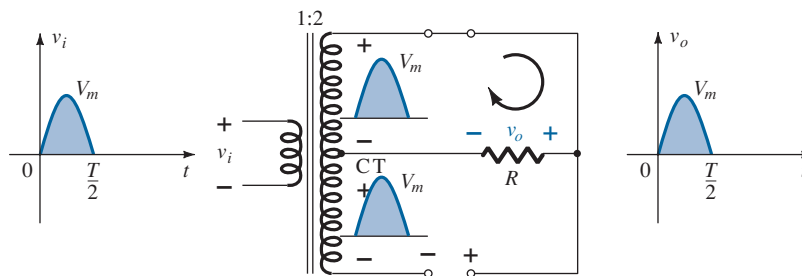


FIG. 61

Network conditions for the positive region of v_i .

During the negative portion of the input the network appears as shown in Fig. 62, reversing the roles of the diodes but maintaining the same polarity for the voltage across the load resistor R . The net effect is the same output as that appearing in Fig. 57 with the same dc levels.

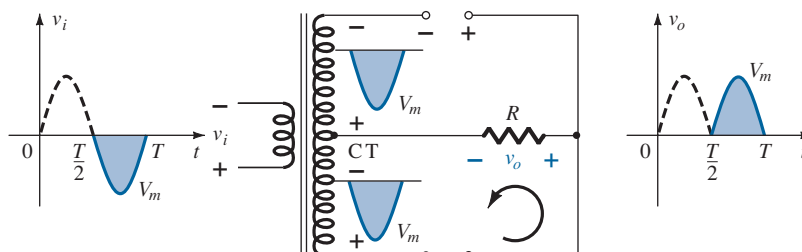


FIG. 62

Network conditions for the negative region of v_i .

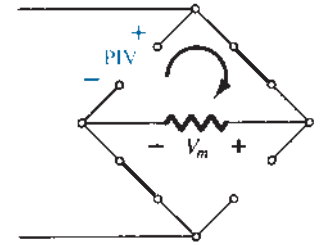


FIG. 59

Determining the required PIV for the bridge configuration.

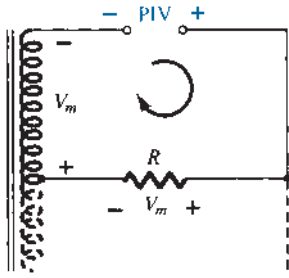


FIG. 63

Determining the PIV level for the diodes of the CT transformer full-wave rectifier.

PIV The network of Fig. 63 will help us determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and V_m as established by the adjoining loop results in

$$\begin{aligned} \text{PIV} &= V_{\text{secondary}} + V_R \\ &= V_m + V_m \end{aligned}$$

and

$$\text{PIV} \cong 2V_m$$

CT transformer, full-wave rectifier

(13)

EXAMPLE 17 Determine the output waveform for the network of Fig. 64 and calculate the output dc level and the required PIV of each diode.

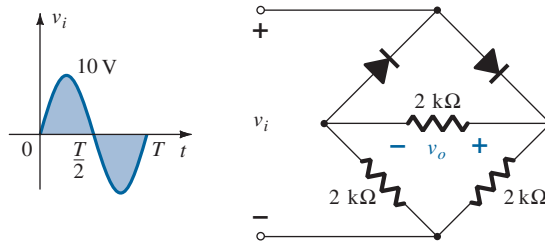


FIG. 64

Bridge network for Example 17.

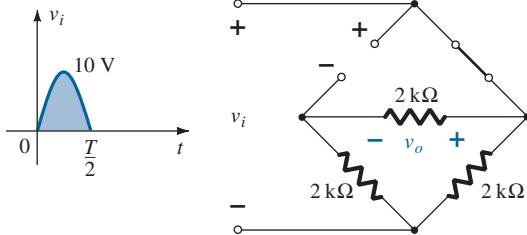


FIG. 65

Network of Fig. 64 for the positive region of v_i .

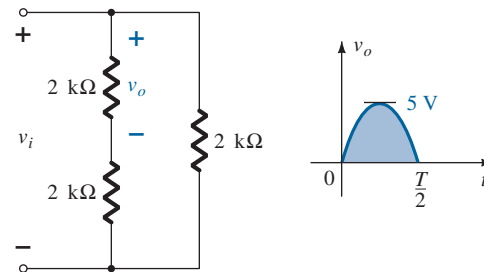


FIG. 66

Redrawn network of Fig. 65.

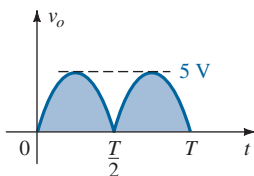


FIG. 67

Resulting output for Example 17.

Solution: The network appears as shown in Fig. 65 for the positive region of the input voltage. Redrawing the network results in the configuration of Fig. 66, where $v_o = \frac{1}{2}v_i$ or $V_{o\text{max}} = \frac{1}{2}V_{i\text{max}} = \frac{1}{2}(10 \text{ V}) = 5 \text{ V}$, as shown in Fig. 66. For the negative part of the input, the roles of the diodes are interchanged and v_o appears as shown in Fig. 67.

The effect of removing two diodes from the bridge configuration is therefore to reduce the available dc level to the following:

$$V_{\text{dc}} = 0.636(5 \text{ V}) = 3.18 \text{ V}$$

or that available from a half-wave rectifier with the same input. However, the PIV as determined from Fig. 59 is equal to the maximum voltage across R , which is 5 V, or half of that required for a half-wave rectifier with the same input.

8 CLIPPERS

The previous section on rectification gives clear evidence that diodes can be used to change the appearance of an applied waveform. This section on clippers and the next on clampers will expand on the wave-shaping abilities of diodes.

Clippers are networks that employ diodes to “clip” away a portion of an input signal without distorting the remaining part of the applied waveform.

Clippers

Clippers are diode circuit that have the ability to clip of a portion of input signal.

Series clipper

series clipper is defined as one where the diode is in series with the load.

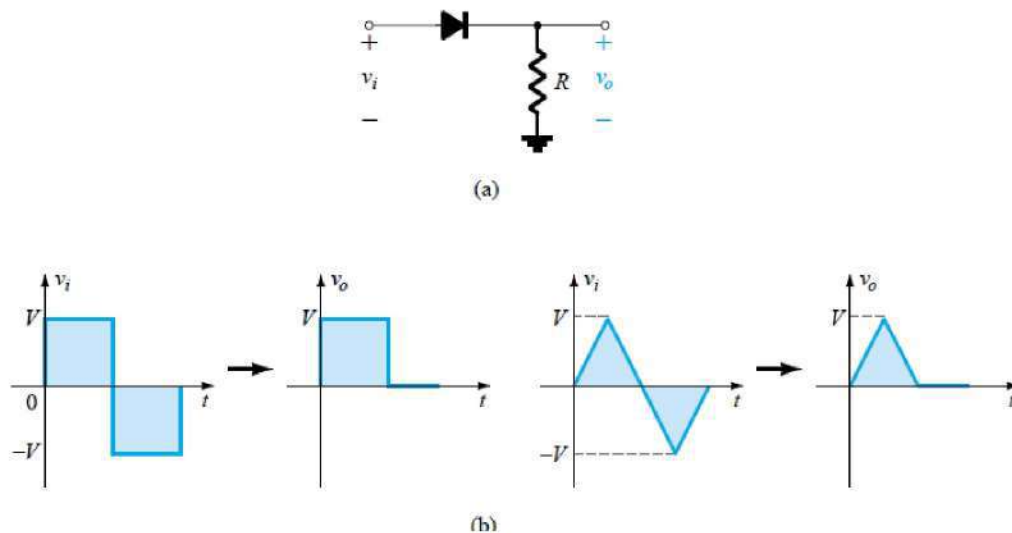


Fig. 11.1 Series clipper.

The addition of a dc supply influences the output of a clipper our initial discussion will be limited to ideal diodes.

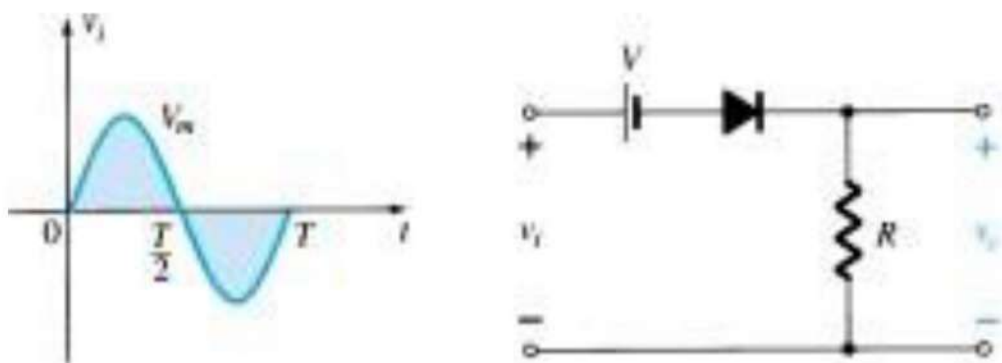


Fig. 11.2 Series clipper with dc supply

To analyzing networks as figure shown there are a few thoughts to keep in mind as you work toward a solution.

1. **Make a mental sketch of response of the network based on the direction of the diode and applied voltage levels.** The direction of the diode suggests that the signal v_i must be positive to turn it **on** and the voltage v_i must be greater than V volt to turn it **on**. The negative region of the input signal the diode is an open circuit **off**.
2. Determine the applied voltage (transition voltage) that will cause a change in state of diode.

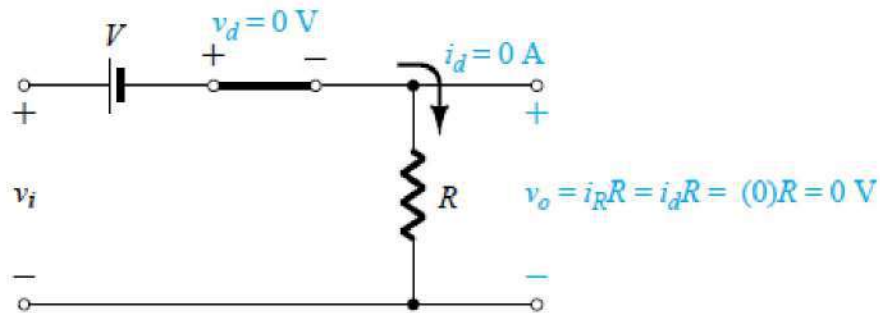


Fig. 11.3 Determining the transition level for the circuit

For the ideal diode the transition between states will occur at the point on the characteristics where

$i_d = 0 \text{ A}$ at $v_d = 0 \text{ V}$. Applying the condition

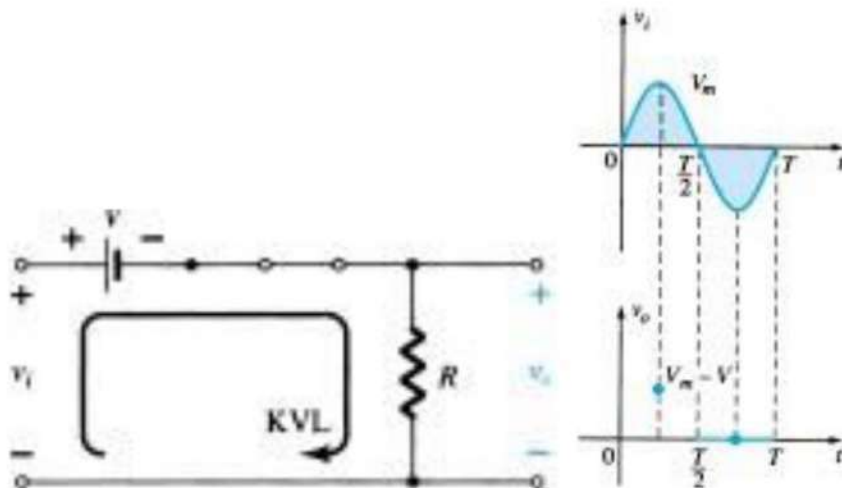


Fig. 11.4 Determining v_o .

$i_d = 0 \text{ A}$ at $v_d = 0 \text{ V}$ to the network will result.

$$v_i - V - v_o = 0$$

$$v_i - V - iR = 0$$

$$v_i - V = 0$$

$$v_i = V$$

The level of v_i that will cause the transition. For an input voltage greater than V the diode is in **on** state ($v_i > V$), while for input voltage less than V the diode is in **off** state ($v_i < V$).

- Applying Kirchhoff's voltage law to define (v_o) in each state.

-Diode on

$$v_i - V - v_o = 0$$

$$v_o = v_i - V$$

- Diode off

$$v_o = 0$$

- Sketch the input signal above the output signal and determine the output at instance values of the input.

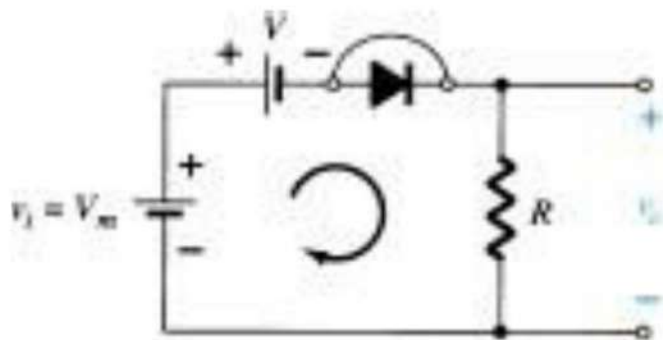
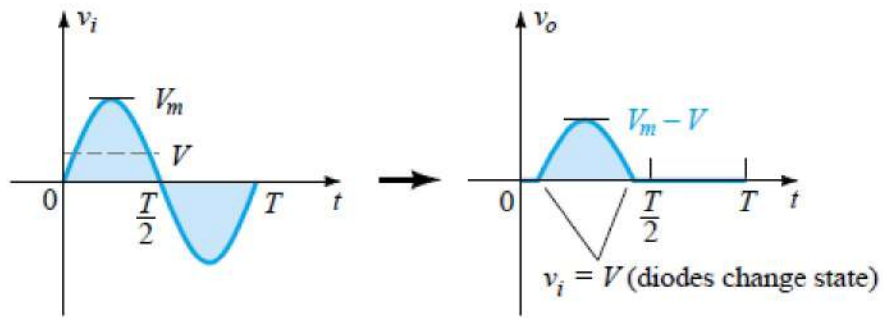


Fig.11.5 Determining v_o when $v_i = V_m$

Fig.11.6 Sketching v_o .**Example 11.1**

Determine the output waveform for the network

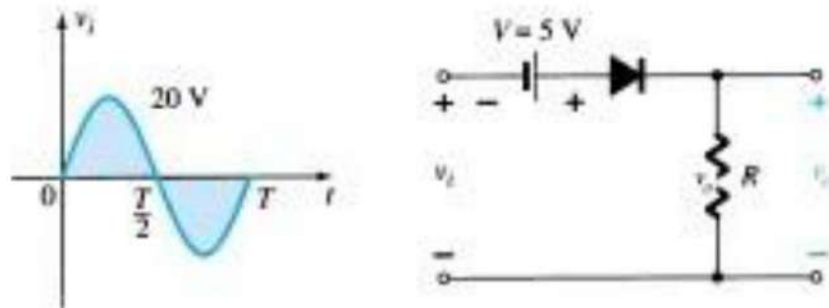
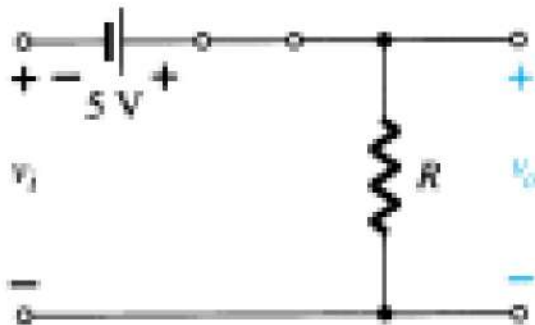


Fig.11.7 Example 11.1

SolutionFig.11.8 v_o with diode in the "on" state.

-The diode will be in the **on** state for the positive half cycle v_i

$$v_i + V - i_d R = 0$$

-Applying $i_d = 0$ and $v_d = 0$ to obtain transition level

$$v_i + V - 0 = 0$$

$$v_i = -V = -5V$$

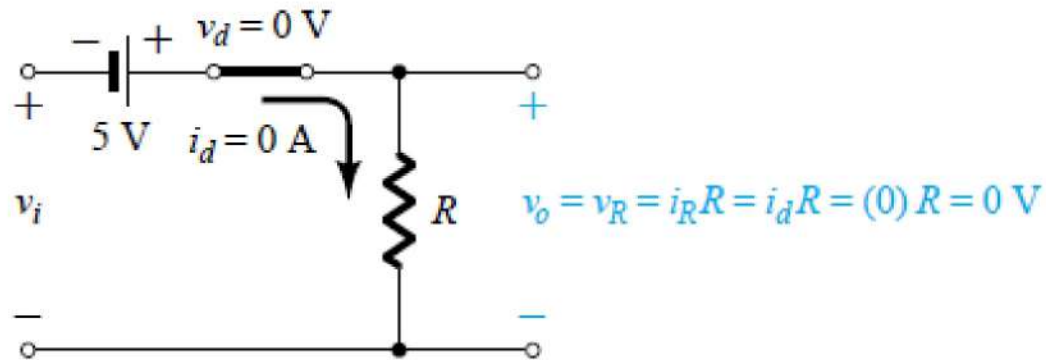


Fig.11.9 Determining the transition level for the clipper

For v_i more than -5V the diode on, for v_i less than -5V the diode off.

-When diode **on**

$$v_i + 5 - v_o = 0$$

$$v_o = v_i + 5$$

-When diode **off**

$$v_o = 0$$

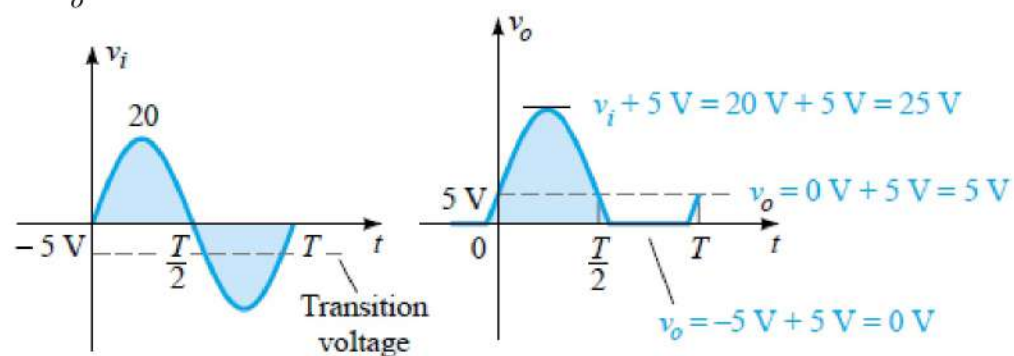


Fig.11.10 Sketching v_o

Clippers with square wave

The analysis of clippers with square wave inputs is much easier than clippers with sinusoidal input, the network can be analyzed as if it had only two dc level inputs.

Example 11.2

Repeat the example for the square wave input

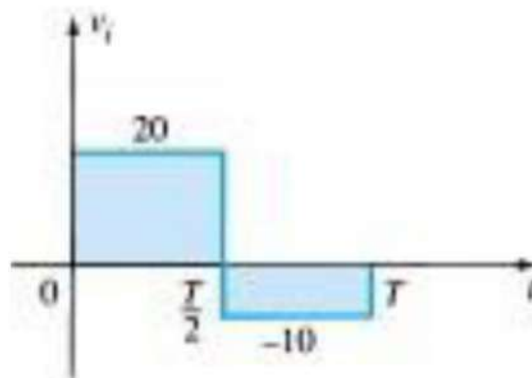


Fig 11.11 Applied signal for Example 11.2.

Solution

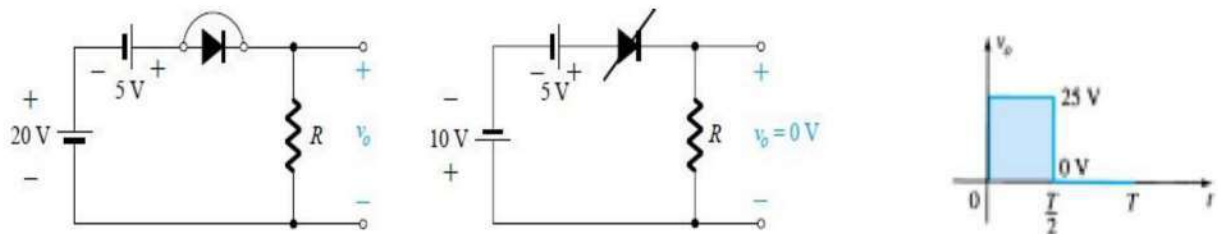


Fig 11.12 Determining v_o

-For $v_i=20V$ the diode on

$$v_o = v_i + 5 = 20 + 5 = 25V$$

-For $v_i=-10V$ the diode of

$$v_o = v_i + 5 = -10 + 5 = -5V$$

Parallel clipper:

parallel clipper is defined as one where the diode is in branch parallel with the load.

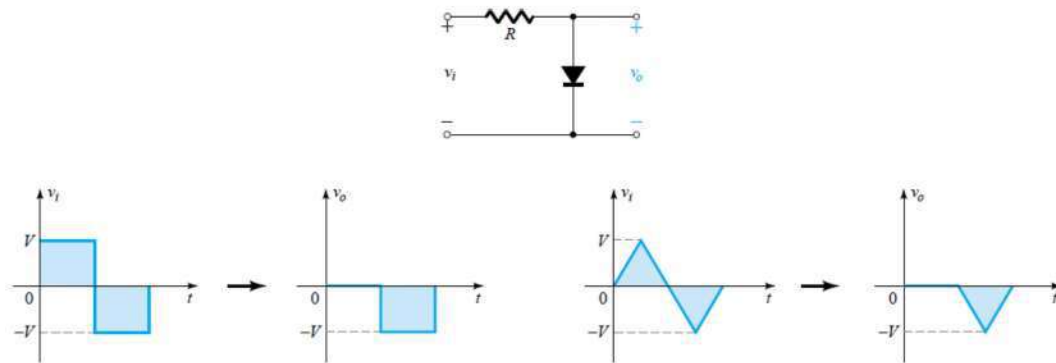


Fig 11.13 Response to a parallel clipper.

Example 11.3

Determine v_o

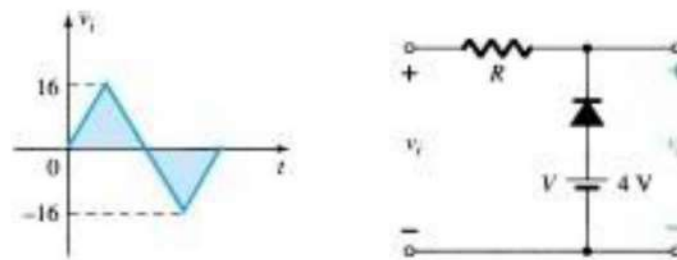


Fig 11.14 Applied signal and circuit for Example 11.2.

Solution

- The direction of the diode Strongly suggested that the diode on for the negative cycle
- The transition state $i_d=0$ and $v_d=0$

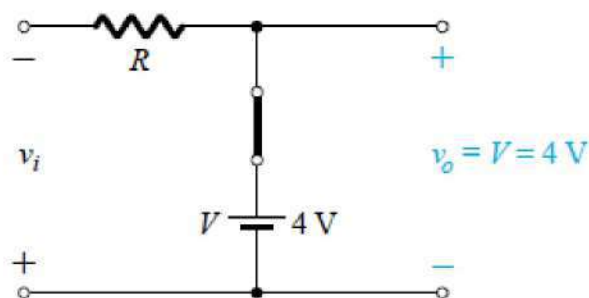
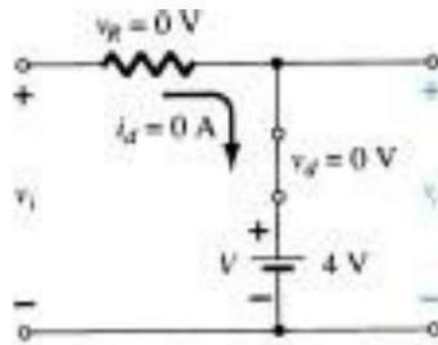


Fig 11.15 v_o for the negative region of v_i .

Fig 11.16 Determining the transition level for ($v_i < 4$) Diode on,

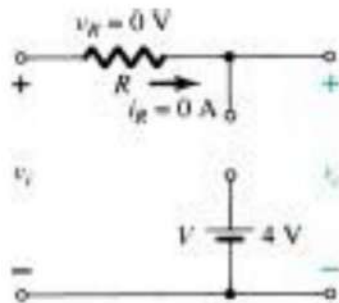
$$v_i - i_d R - 4 = 0$$

$$v_i = 4$$

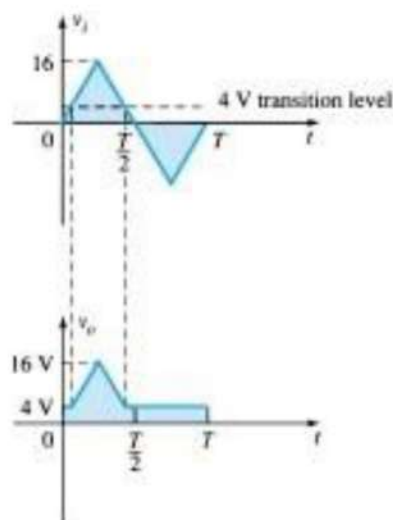
For ($v_i > 4$) Diode off,

$$v_o = v_i$$

$$v_o = 4V$$

Fig 11.17 Determining v_o for the open state of the diode.

-Sketching

Fig 11.18 Sketching v_o

Example

Repeat example using a silicon diode

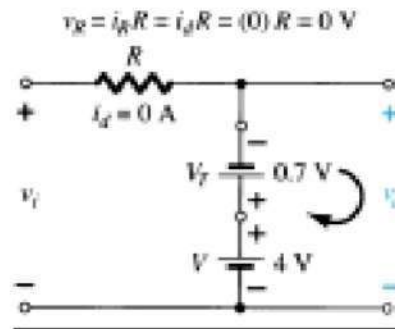


Fig 11.19 Determining the transition level

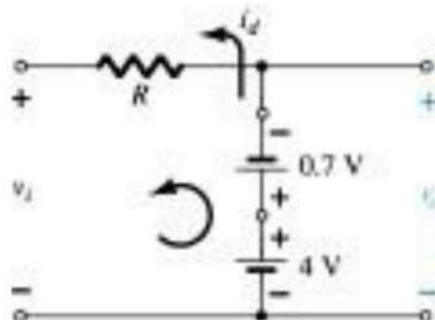
Solution

- The diode on for the negative cycle
- The transition state $i_d = 0$ and $v_d = 0.7 \text{ V}$

Applying Kirchhoff's law

$$v_i - i_d R + 0.7 - 4 = 0$$

$$v_i = 4 - 0.7 = 3.3 \text{ V}$$



For ($v_i > 3.3 \text{ V}$) Diode off, $v_o = v_i$

For ($v_i < 3.3 \text{ V}$) Diode on, $v_o = 3.3 \text{ V}$

Fig 11.20 Determining v_o

-Sketching

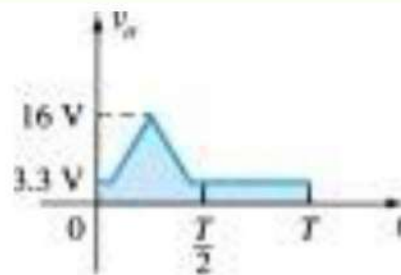


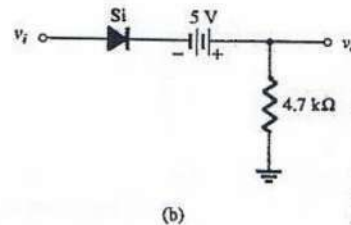
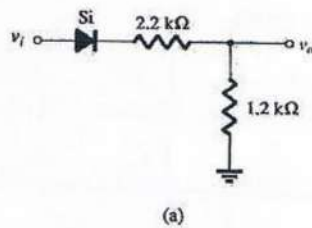
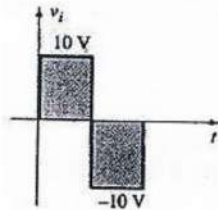
Fig 11.21 Sketching v_o

Note:-Including the effect of V_T will complicate the analysis somewhat, but once the analysis is understood with the ideal diode the procedure including the effect of V_T will not be that difficult.

Problems

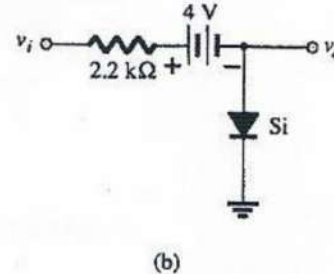
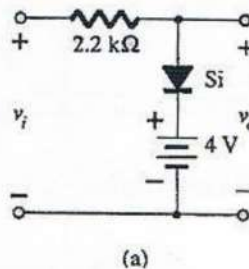
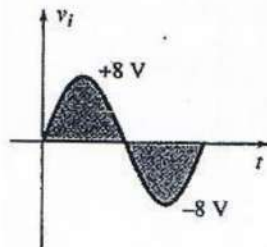
Q1: Sketch and determine v_o for configurations shown.

(Ans: positive pulse 3.28V, positive pulse 14.3V)

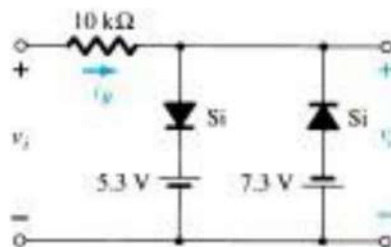
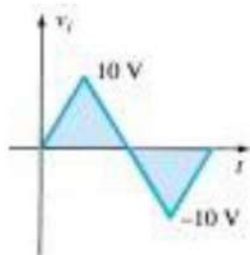


Q2: Sketch and determine v_o for configurations shown.

(Ans: clipped at 4.7V, positive clipped at 0.7V; negative -12V)



Q3: Sketch i_R and v_o for the network





وزارة التعليم العالي والبحث العلمي
الجامعة التقنية الوسطى
الكلية التقنية الهندسية الكهربائية
قسم هندسة تكنولوجيا الأمن السيبراني

أسس الإلكترونيك

المرحلة الثانية

TRANSISTOR (1)

مدرس المادة

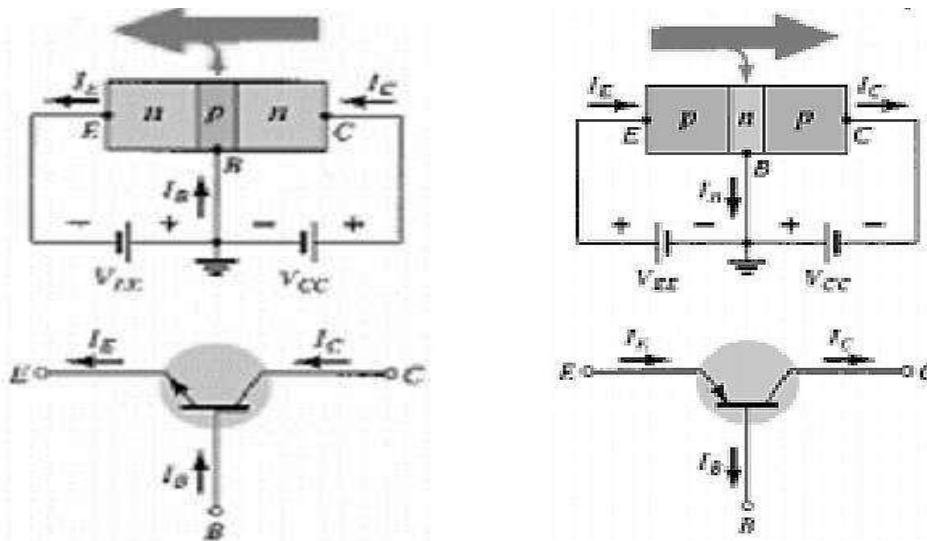
م.م. مصطفى احمد سعدي

Chapter Two

Bipolar Junction Transistors (BJT)

1- Introduction:

The transistor is a three-layer semiconductor device consisting of either two n- and one p-type layers of material or two p- and one n-type layers of material.



1- Common-Base Configuration

In common base, the base is usually the terminal closest to, or at, ground potential.

$$I_E = I_C + I_B$$

$$I_E \cong I_C$$

A transistor is in the “on” state, the base-to-emitter voltage will be assumed to be the following:

$$V_{BE} = 0.7$$

In the dc mode the levels of I_C and I_E due to the majority carriers are related by a quantity called alpha and defined by the following equation:

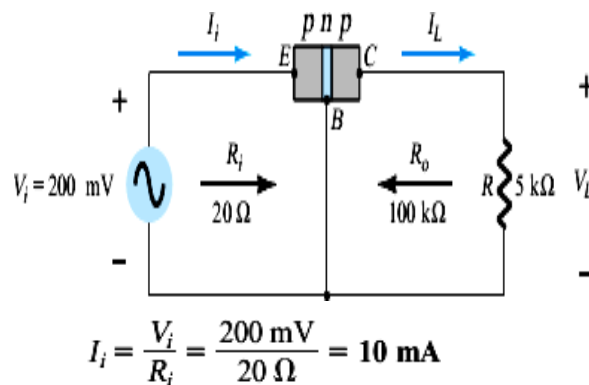
$$\alpha_{dc} = \frac{I_C}{I_E}$$

But for common base called current gain with a level of typically extends from 0.90 to 0.998.

For the common-base configuration the ac input resistance is quite small and typically varies from 10 to 100 Ω . The output resistance is quite high and typically varies from 50 k Ω to 1 M Ω (100 k Ω).

Example:

Using a common value of 20 Ω for the input resistance, we find that



If we assume for the moment that $\alpha_{ac} = 1$ ($I_c = I_e$),

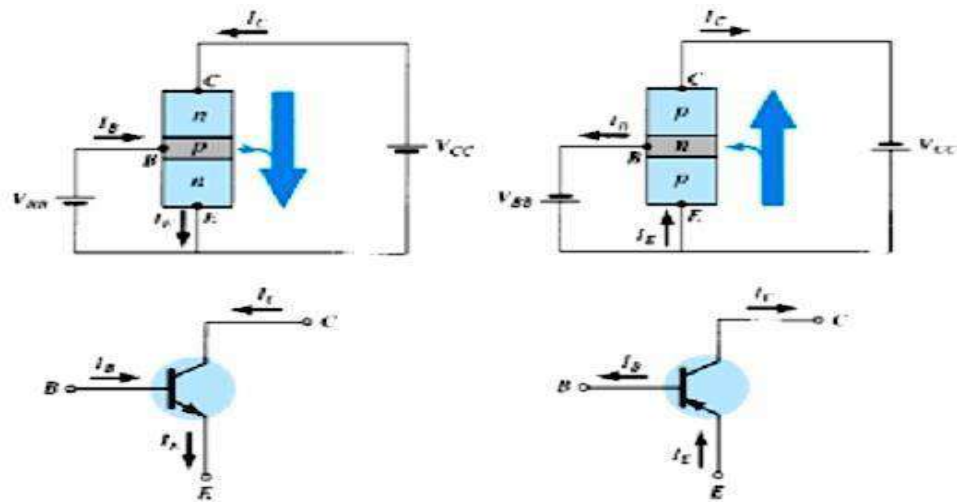
$$I_L = I_i = 10 \text{ mA}$$

and

$$\begin{aligned} V_L &= I_L R \\ &= (10 \text{ mA})(5 \text{ k}\Omega) \\ &= 50 \text{ V} \end{aligned}$$

The voltage amplification is

$$A_v = \frac{V_L}{V_i} = \frac{50 \text{ V}}{200 \text{ mV}} = 250$$



2-Common-Emitter Configuration

In the dc mode the levels of I_C and I_B are related by a quantity called beta and defined by the following equation which is the current gain of common emitter:

$$\beta_{dc} = \frac{I_C}{I_B}$$

For practical devices the level of typically ranges from about 50 to over 400, with most in the midrange.

A relationship can be developed between β and α using the basic relationships introduced thus far. Using $\beta = I_C/I_B$ we have $I_B = I_C/\beta$, and

$$I_C = \beta I_B$$

$$I_E = I_C + I_B$$

$$\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$$

and since

$$I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

we have

$$I_E = (\beta + 1)I_B$$

from $\alpha = I_C/I_E$ we have

$I_E = I_C/\alpha$. Substituting into

and dividing both sides of the equation by I_C will result in

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta}$$

or

$$\beta = \alpha\beta + \alpha = (\beta + 1)\alpha$$

so that

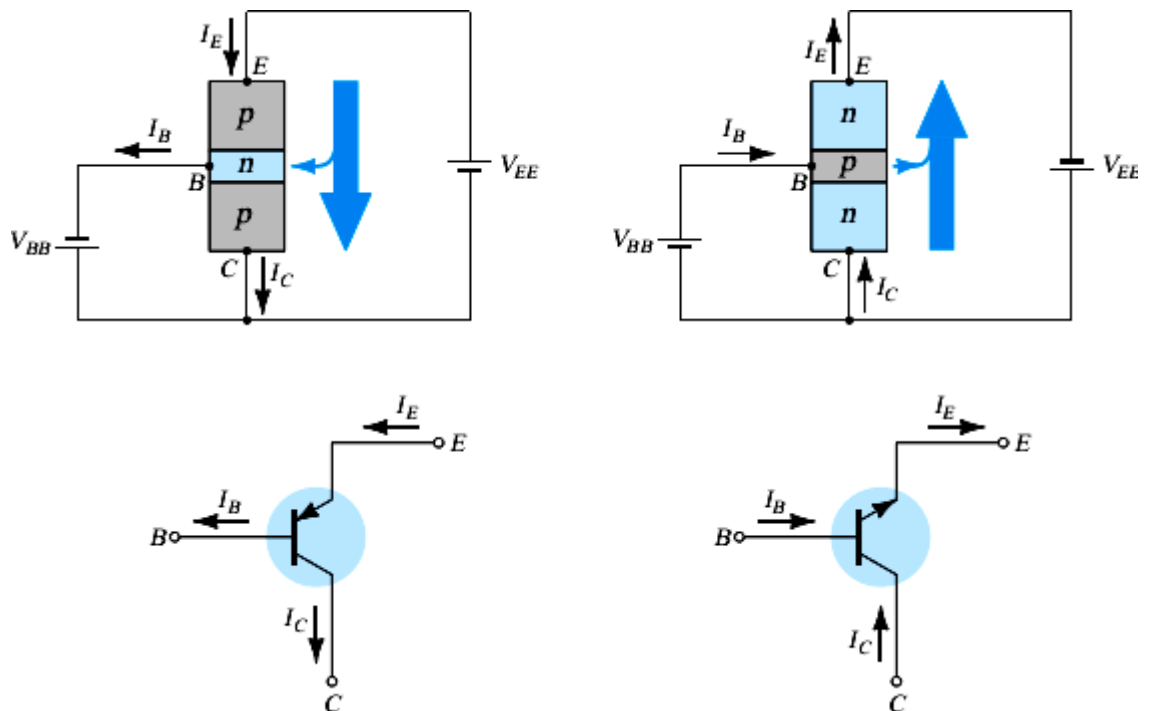
$$\alpha = \frac{\beta}{\beta + 1}$$

or

$$\beta = \frac{\alpha}{1 - \alpha}$$

3-Common-Collector Configuration

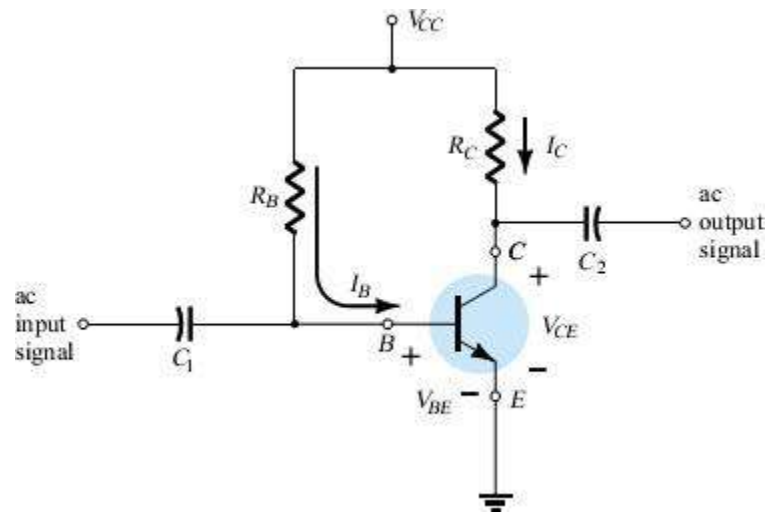
The common-collector configuration is used primarily for impedance-matching purposes since it has a high input impedance and low output impedance, opposite to that of the common-base and common-emitter configurations.



For all practical purposes, the output characteristics of the common-collector configuration are the same as for the common-emitter configuration.

DC Biasing—BJTs

1- Fixed-Bias Circuit



Writing Kirchhoff's voltage equation in the clockwise direction for the loop, we obtain

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Applying Kirchhoff's voltage law in the clockwise direction

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

As a brief review of single- and double-subscript notation recall that

$$V_{CE} = V_C - V_E$$

But in this case, since $V_E = 0$ V, we have

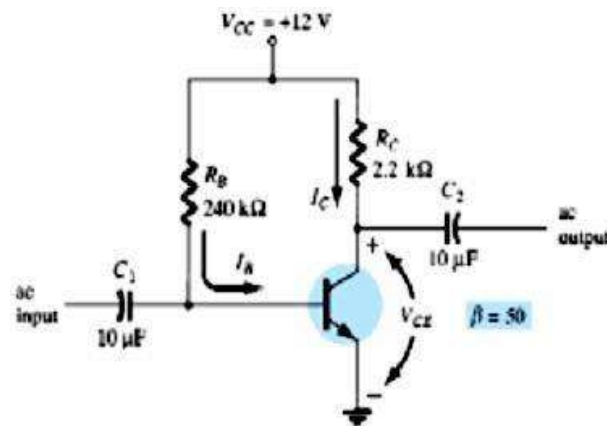
$$V_{CE} = V_C$$

In addition

$$V_{BE} = V_B - V_E \text{ then } V_{BE} = V_B$$

Example: Determine the following for the fixed-bias configuration of Fig. below.

(a) I_{BQ} and I_{CQ} (b) V_{CEQ} (c) V_B and V_C . (d) V_{BC} .



$$(a) \quad I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{240 \text{ k}\Omega} = 47.08 \text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = (50)(47.08 \text{ }\mu\text{A}) = 2.35 \text{ mA}$$

$$(b) \quad \begin{aligned} V_{CEQ} &= V_{CC} - I_C R_C \\ &= 12 \text{ V} - (2.35 \text{ mA})(2.2 \text{ k}\Omega) \\ &= 6.83 \text{ V} \end{aligned}$$

$$(c) \quad \begin{aligned} V_B &= V_{BE} = 0.7 \text{ V} \\ V_C &= V_{CE} = 6.83 \text{ V} \end{aligned}$$

(d) Using double-subscript notation yields

$$\begin{aligned} V_{BC} &= V_B - V_C = 0.7 \text{ V} - 6.83 \text{ V} \\ &= -6.13 \text{ V} \end{aligned}$$

The saturation current for the fixed-bias configuration is

$$I_{Csat} = \frac{V_{CC}}{R_C}$$

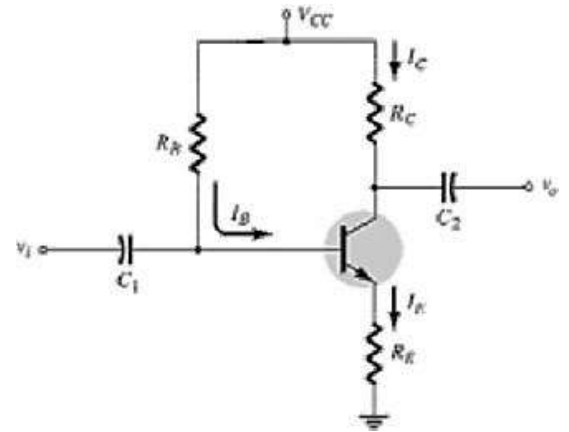
Where $V_{CE} = 0$ for this case

The addition of the emitter resistor to the dc bias of the BJT provides improved stability, that is, the dc bias currents and voltages remain closer to where they were set by the circuit when outside conditions, such as temperature, and transistor beta, change.

The base-emitter loop of the network is:

Then

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$



Note that aside from the base-to-emitter voltage V_{BE} , the resistor R_E is reflected back to the input base circuit by a factor $(\beta + 1)$.

$$R_i = (\beta + 1)R_E$$

Writing Kirchhoff's voltage law for collector-emitter loop in the clockwise direction will result in

$$-V_{CC} + I_C R_C + V_{CE} + I_E R_E = 0$$

Substituting $I_C \cong I_E$

$$-V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$$

$$V_{CE} - V_{CC} + I_C (R_C + R_E) = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_E = I_E R_E$$

while the voltage from collector to ground can be determined from

$$V_{CE} = V_C - V_E$$

and

$$V_C = V_{CE} + V_E$$

or

$$V_C = V_{CC} - I_C R_C$$

The voltage at the base with respect to ground can be determined from

$$V_B = V_{CC} - I_B R_B$$

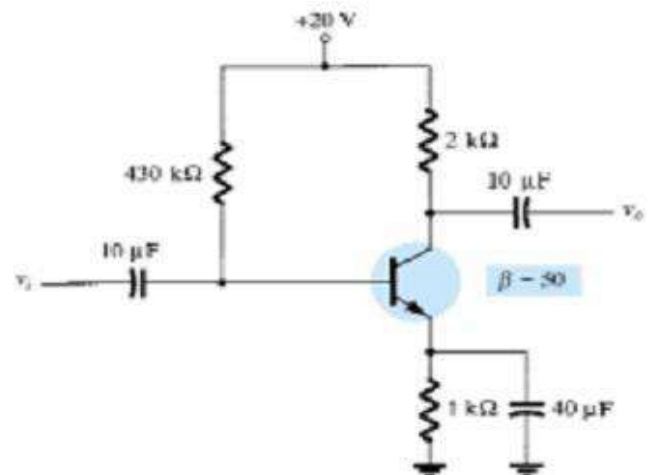
or

$$V_B = V_{BE} + V_E$$

Example:

For the emitter bias network of Fig.

- (a) I_B .
- (b) I_C .
- (c) V_{CE} .
- (d) V_C .
- (e) V_E .
- (f) V_B .
- (g) V_{BC} .



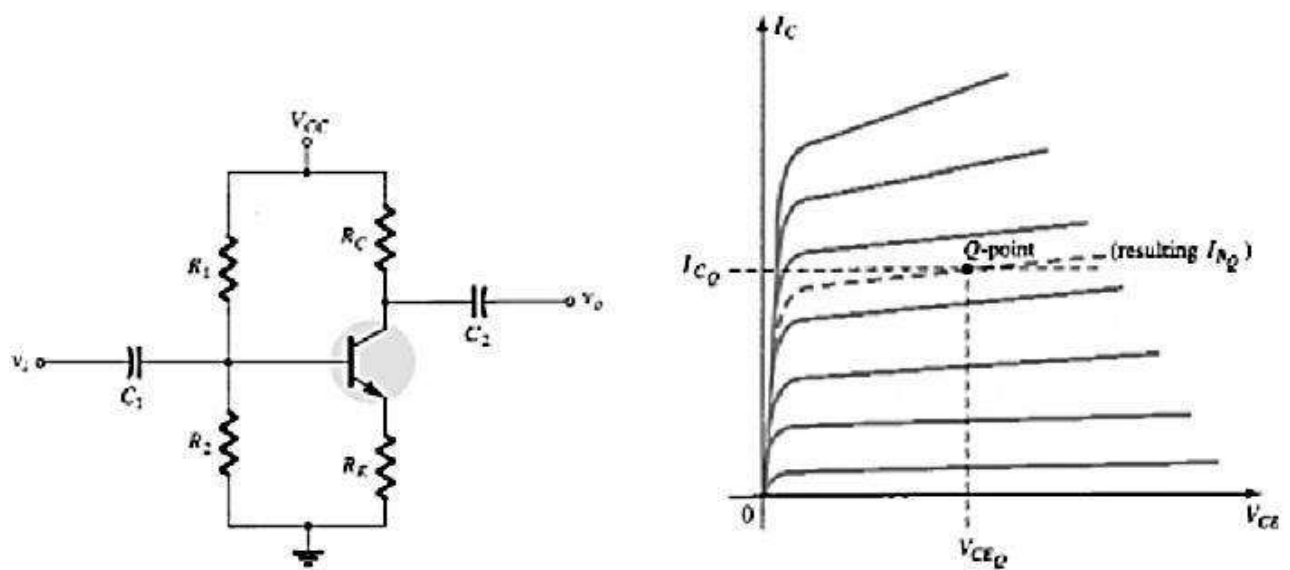
$$\begin{aligned}
\text{(a)} \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{430 \text{ k}\Omega + (51)(1 \text{ k}\Omega)} \\
&= \frac{19.3 \text{ V}}{481 \text{ k}\Omega} = \mathbf{40.1 \text{ }\mu\text{A}} \\
\text{(b)} \quad I_C &= \beta I_B \\
&= (50)(40.1 \text{ }\mu\text{A}) \\
&\cong \mathbf{2.01 \text{ mA}} \\
\text{(c)} \quad V_{CE} &= V_{CC} - I_C(R_C + R_E) \\
&= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega + 1 \text{ k}\Omega) = 20 \text{ V} - 6.03 \text{ V} \\
&= \mathbf{13.97 \text{ V}} \\
\text{(d)} \quad V_C &= V_{CC} - I_C R_C \\
&= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega) = 20 \text{ V} - 4.02 \text{ V} \\
&= \mathbf{15.98 \text{ V}} \\
\text{(e)} \quad V_E &= V_C - V_{CE} \\
&= 15.98 \text{ V} - 13.97 \text{ V} \\
&= \mathbf{2.01 \text{ V}} \\
\text{or } V_E &= I_E R_E \cong I_C R_E \\
&= (2.01 \text{ mA})(1 \text{ k}\Omega) \\
&= \mathbf{2.01 \text{ V}} \\
\text{(f)} \quad V_B &= V_{BE} + V_E \\
&= 0.7 \text{ V} + 2.01 \text{ V} \\
&= \mathbf{2.71 \text{ V}} \\
\text{(g)} \quad V_{BC} &= V_B - V_C \\
&= 2.71 \text{ V} - 15.98 \text{ V} \\
&= \mathbf{-13.27 \text{ V}} \quad (\text{reverse-biased as required})
\end{aligned}$$

The collector saturation level or maximum collector current for an emitter-bias de-sign can be determined

$$I_{Csat} = \frac{V_{CC}}{R_C + R_E}$$

2- Voltage-Divider Bias:

There are two methods that can be applied to analyze the voltage-divider configuration.



Exact Analysis:

The Thévenin equivalent network for the network to the left of the base terminal can then be found in the following manner:

$$R_{Th} = R_1 \parallel R_2$$

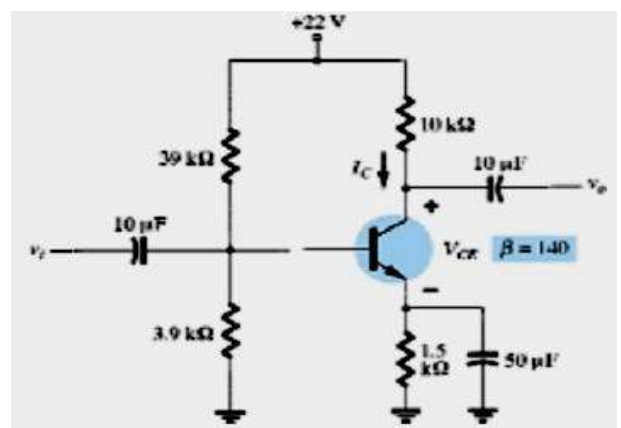
$$E_{Th} = V_{R_2} = \frac{R_2 V_{CC}}{R_1 + R_2}$$

$$E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

Example: Determine the dc bias voltage V_{CE} and the current I_C for the voltage-divider configuration of Fig.

$$\begin{aligned} R_{Th} &= R_1 \parallel R_2 \\ &= \frac{(39 \text{ k}\Omega)(3.9 \text{ k}\Omega)}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 3.55 \text{ k}\Omega \\ E_{Th} &= \frac{R_2 V_{CC}}{R_1 + R_2} \\ &= \frac{(3.9 \text{ k}\Omega)(22 \text{ V})}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 2 \text{ V} \end{aligned}$$



$$\begin{aligned}
 I_B &= \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \\
 &= \frac{2 \text{ V} - 0.7 \text{ V}}{3.55 \text{ k}\Omega + (141)(1.5 \text{ k}\Omega)} = \frac{1.3 \text{ V}}{3.55 \text{ k}\Omega + 211.5 \text{ k}\Omega} \\
 &= 6.05 \text{ }\mu\text{A} \\
 I_C &= \beta I_B \\
 &= (140)(6.05 \text{ }\mu\text{A}) \\
 &= \mathbf{0.85 \text{ mA}} \\
 V_{CE} &= V_{CC} - I_C(R_C + R_E) \\
 &= 22 \text{ V} - (0.85 \text{ mA})(10 \text{ k}\Omega + 1.5 \text{ k}\Omega) \\
 &= 22 \text{ V} - 9.78 \text{ V} \\
 &= \mathbf{12.22 \text{ V}}
 \end{aligned}$$

Approximate Analysis:

The resistance R_i is the equivalent resistance between base and ground for the transistor with an emitter resistor R_E .

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$$

The condition that will define whether the approximate approach can be applied will be the following:

$$\beta R_E \geq 10 R_2$$

$$V_E = V_B - V_{BE}$$

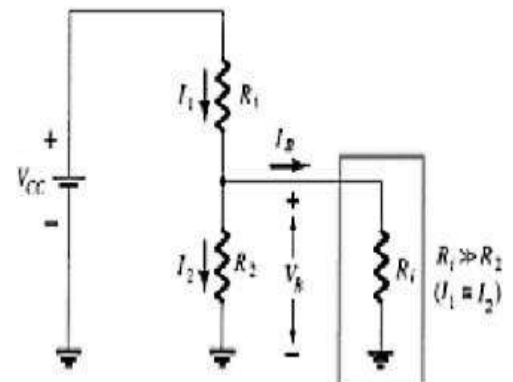
$$I_E = \frac{V_E}{R_E}$$

$$I_{CQ} \cong I_E$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

but since $I_E \cong I_C$,

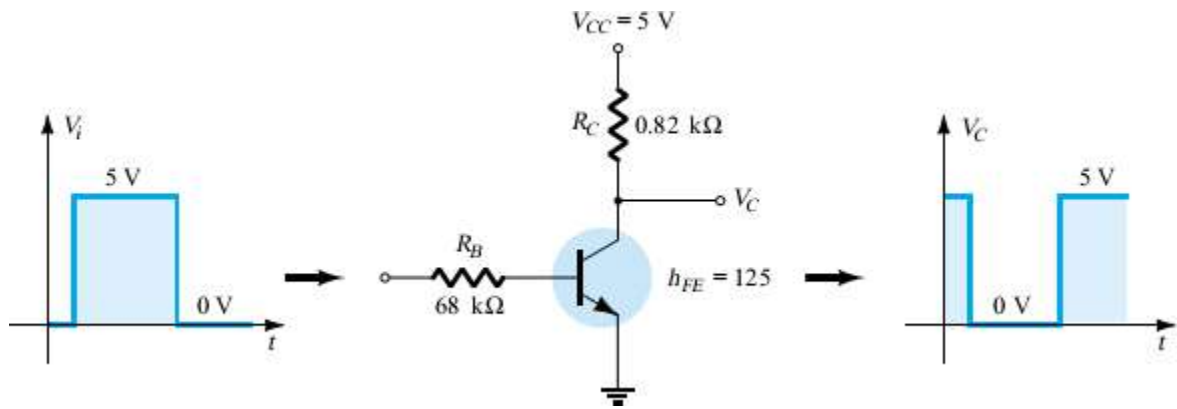
$$V_{CEQ} = V_{CC} - I_C(R_C + R_E)$$



H.W: Repeat the analysis of previous example using the approximate technique, and compare solutions for I_{CQ} and V_{CEQ}

3-Transistor Switching Networks:

The application of transistors is not limited solely to the amplification of signals. Through proper design it can be used as a switch for computer and control applications.



Proper design for the inversion process requires that the operating point switch from cutoff to saturation along the load line.

The saturation level for the collector current is defined by

$$I_{Csat} = \frac{V_{CC}}{R_C}$$

Clampers

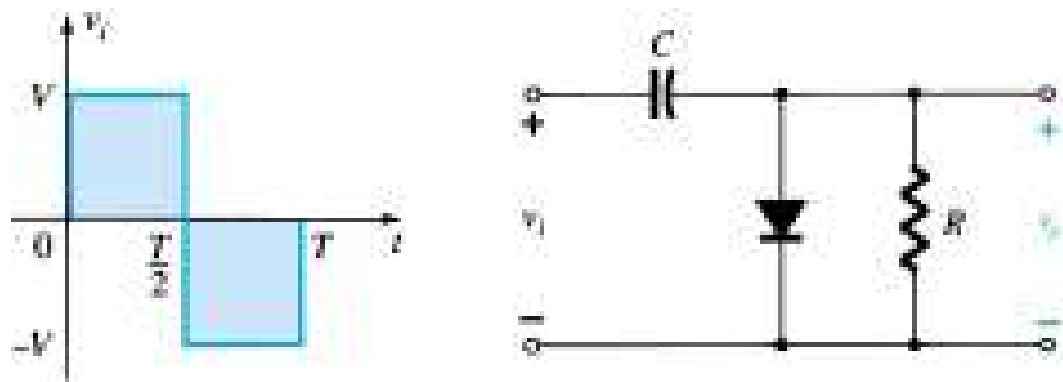


Fig. 12.1 Clamper.

The clamping network is one that will clamp a signal to a different dc level. It must have a capacitor a diode and resistance but it can also employ dc supply to introduce additional shift.

The magnitude of R and C must be chosen because the time constant τ ,

$$\tau = RC$$

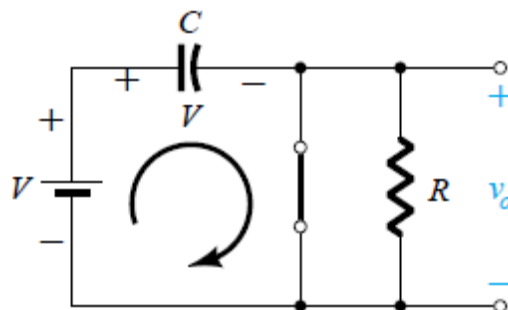


Fig. 12.2 not Diode "on" and the capacitor charging to V volts.

And the capacitor will fully charge or discharge in 5τ , τ must be large enough to ensure that the voltage across capacitor does not discharge significantly during the interval the diode is off.

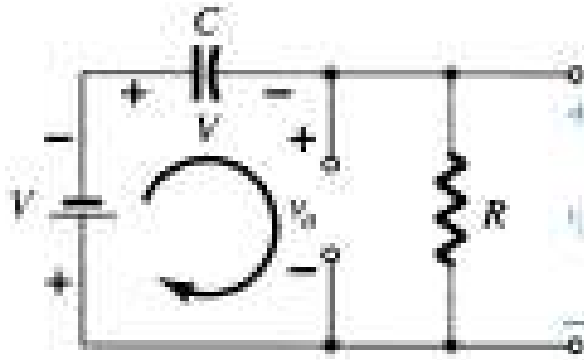


Fig.12.3 Determining v_o with the diode "off."

During the positive half cycle the network will appear as shown with the diode in on state, τ is very small that the capacitor will charge very quickly $v_o=0$.

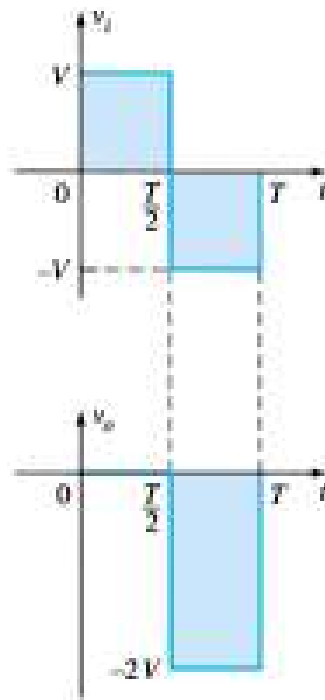
During the negative half cycle the network will appear as shown, diode off, τ is large, the capacitor holds during the period

$$-V - V - v_o = 0$$

$$v_o = -2V$$

The following steps may be helpful to analysis clampers.

- Start analysis by the part of input signal that will forward bias the diode
- During the period assume that the capacitor will charge to voltage level determined by the network (v_c , v_o)
- Assume that during the diode **OFF** the capacitor will hold to its established voltage level.
- Keep in mind the general rule that the total swing of the output must match the swing of input signal.

Fig. 12.4 Sketching v_o

Example

Determine v_o

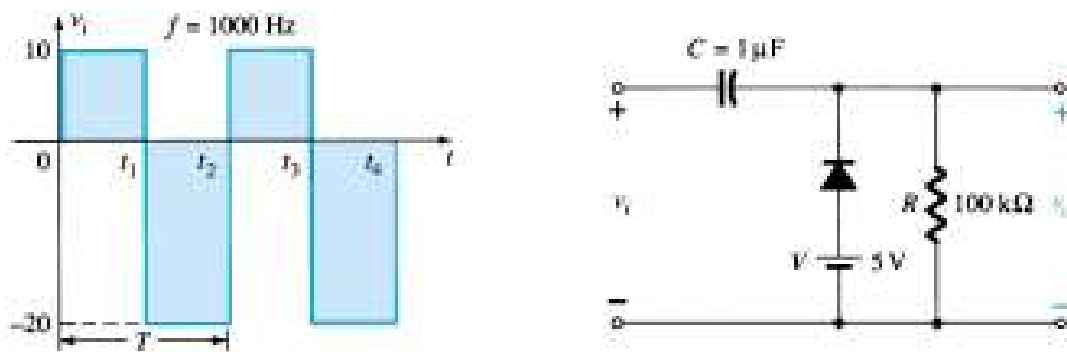


Fig 12.5 Applied signal and network for Example 12.1.

Solution

$$T = \frac{1}{f} = \frac{1}{1000} = 10^{-3} \text{ s} \quad \text{Half time-period} = \frac{T}{2} = 0.5 \times 10^{-3} \text{ s}$$

-Begin with forward bias, 2nd half Period, the network will appear as shown

$$v_o = 5V$$

$$20 + 5 - v_C = 0$$

$$v_C = 25V$$

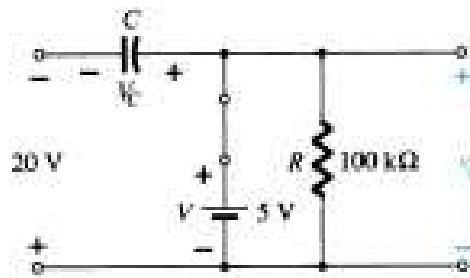


Fig 12.6 Determining v_o and v_C with the diode in the "on" state.

-During the positive half cycle Diode reverse bias

$$10 + 25 - v_o = 0$$

$$v_o = 35V$$

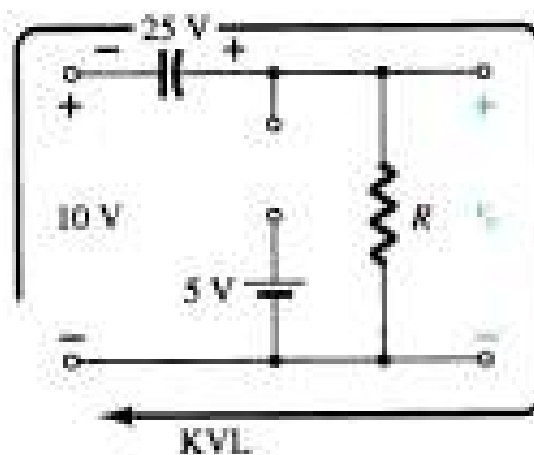


Fig 12.7 Determining v_o with the diode in the "off" state

The output swing of 30V match the swing of input signal.

Time constant of the discharge network

$$\tau = RC = (100k)(0.1\mu) = 0.01s = 10ms$$

The total discharge times

$$5\tau = 50ms$$

$$\text{Half time-period} = T/2 = 0.5 \times 10^{-3}s$$

It's a good approximation that the capacitor will hold its voltage during the discharge period.

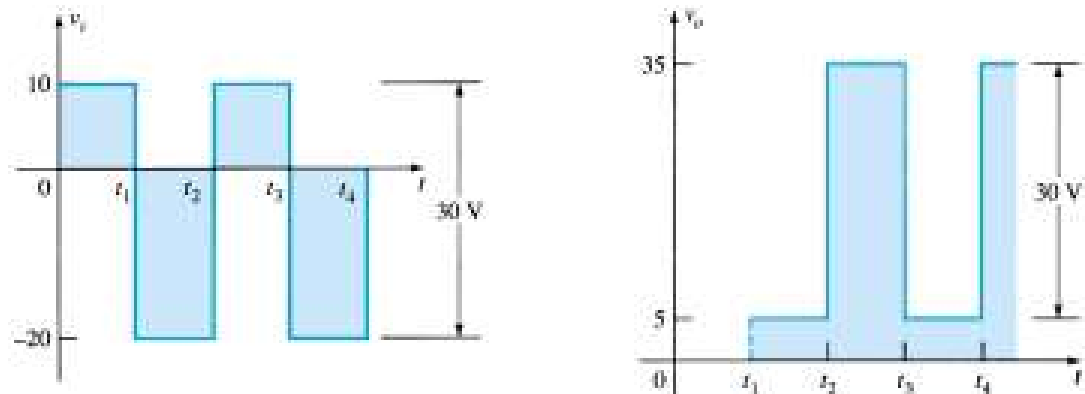


Fig 12.8 v_i and v_o for the clamper

Example

Repeat example using a silicon diode

Solution

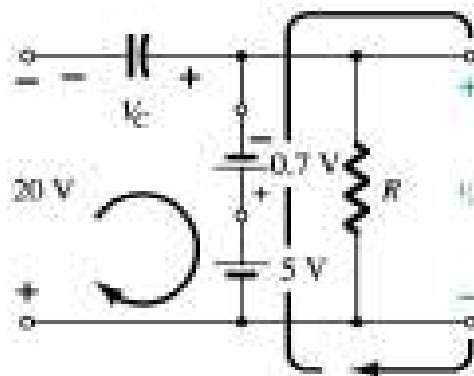


Fig 12.9 Determining v_o and V_c with the diode in the "on" state.

For short circuit state

$$5 - 0.7 - v_o = 0$$

$$v_o = 4.3V$$

for input section

$$-20 - v_c + 0.7 - 5 = 0 \quad v_c = 24.3V$$

-For open circuit state Diode reverse bias

$$10 - 24.3 - v_o = 0$$

$$v_o = 34.3V$$

The output swing of 30V match the swing of input signal.

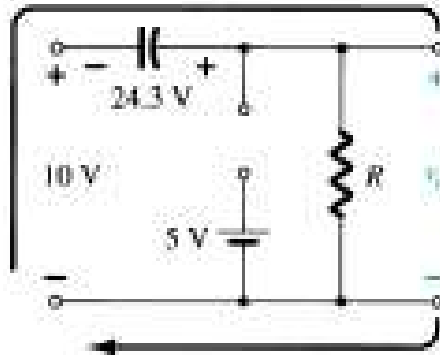


Fig 12.10 Determining v_o with the diode in the “off” state.

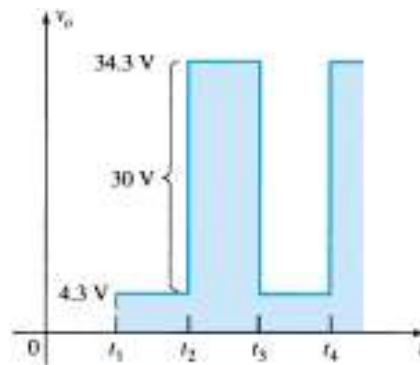


Fig 12.11 v_o for the clamper

Note: -

For sinusoidal signal one approach to the analysis of clamping networks is to replace the sinusoidal signal by a square wave of the same peak value.

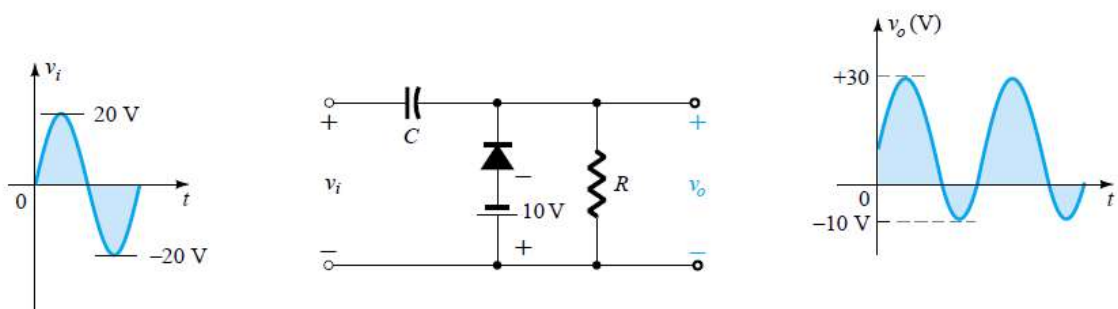
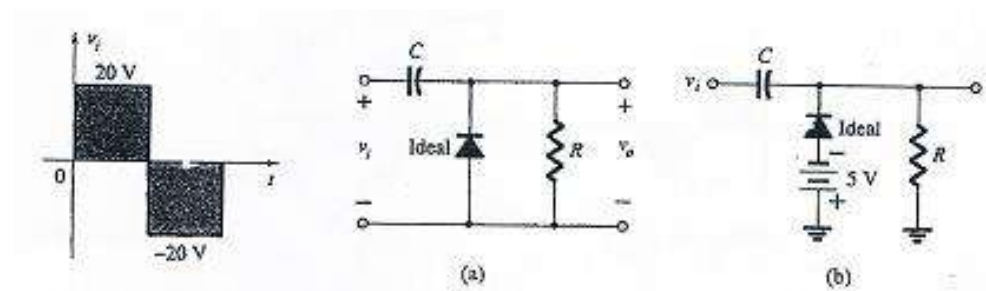


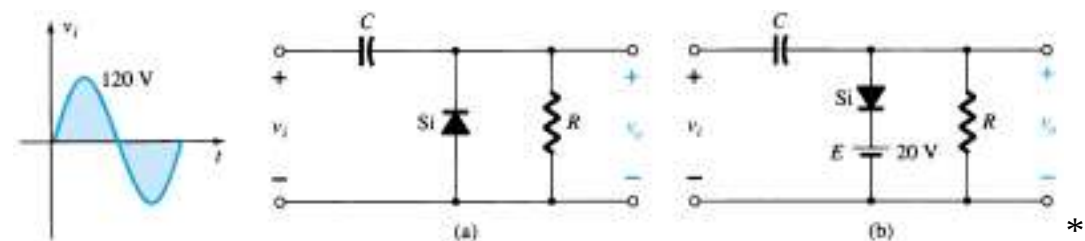
Fig 12.12 Clamping network with a sinusoidal input.

Problems

Q1: Sketch and determine v_o for configurations shown.

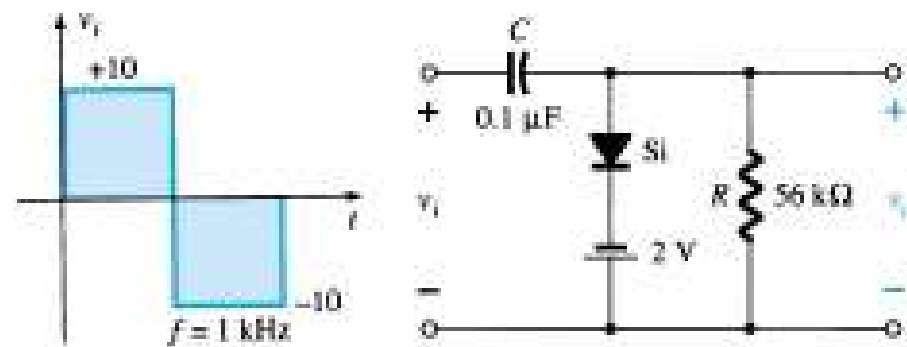


Q2: Sketch v_o for each network of Figure for the input shown. Would it be a good approximation to consider the diode to be ideal for both configurations? Why?



Q3. For the network of Figure:

- Calculate τ .
- Compare τ to half the period of the applied signal.
- Sketch v_o .



- CONSTRUCTION AND CHARACTERISTICS OF JFETS:

The basic construction of the n -channel JFET is shown in Fig. 2. Note that the major part of the structure is the n -type material that forms the channel between the embedded layers of p -type material. The top of the n -type channel is connected through an ohmic contact to a terminal referred to as the *drain* (D), while the lower end of the same material is connected through an ohmic contact to a terminal referred to as the *source* (S). The two p -type materials are connected together and to the *gate* (G) terminal. In essence, therefore, the drain and source are connected to the ends of the n -type channel and the gate to the two layers of p -type material. In the absence of any applied potentials the JFET has two p - n junctions under no-bias conditions.

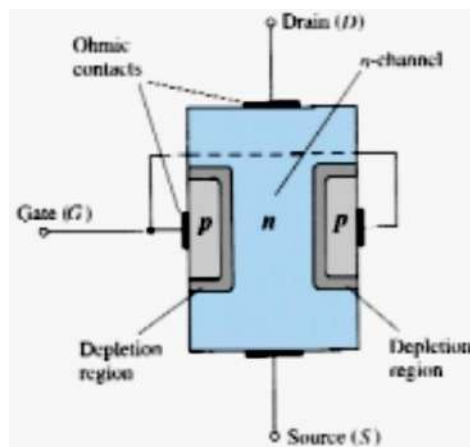


Figure 2

In Fig. 3, a positive voltage V_{DS} has been applied across the channel and the gate has been connected directly to the source to establish the condition $V_{GS} = 0$ V. As the voltage V_{DS} is increased from 0 to a few volts, the current will increase as determined by Ohm's law and the plot of I_D versus V_{DS} will appear. If V_{DS} is increased to a level where it appears that the two depletion regions would “touch”, a condition referred to as *pinch-off* will result.

I_{DSS} is the maximum drain current for a JFET and is defined by the conditions $V_{GS} = 0$ V and $V_{DS} > |V_p|$.

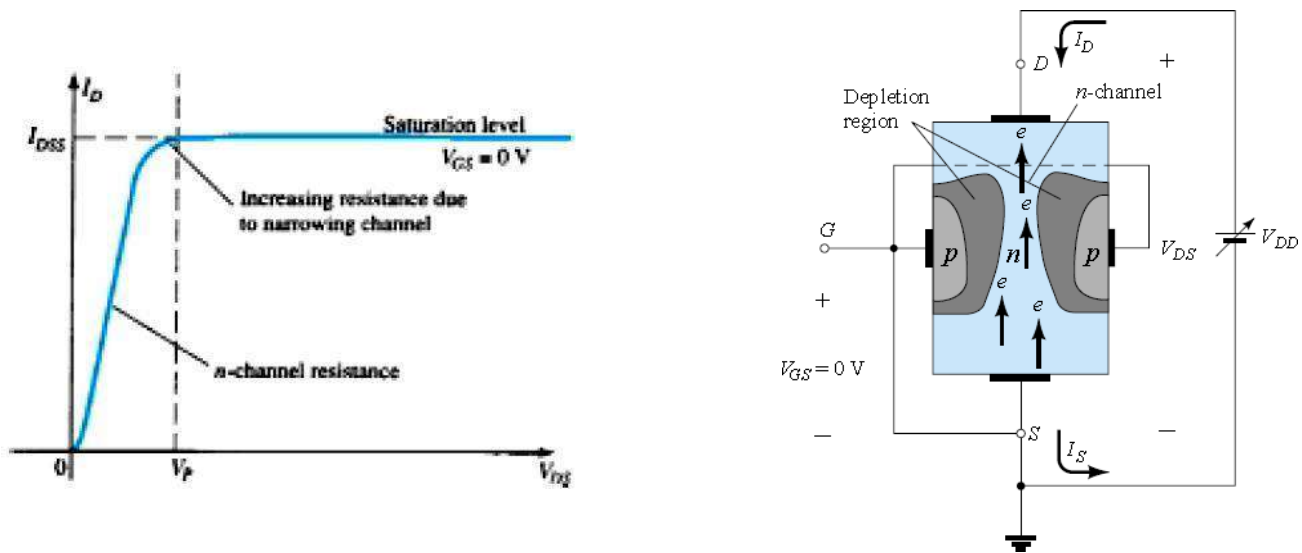


Figure 3

The voltage from gate to source, denoted V_{GS} , is the controlling voltage of the JFET, curves of I_D versus V_{DS} for various levels of V_{GS} can be developed for the JFET. Note also on Fig. 4 how the pinch-off voltage continues to drop in a parabolic manner as V_{GS} becomes more and more negative. Eventually, V_{GS} when $V_{GS} = V_p$ will be sufficiently negative to establish a saturation level that is essentially 0 mA, and for all practical purposes the device has been “turned off.”

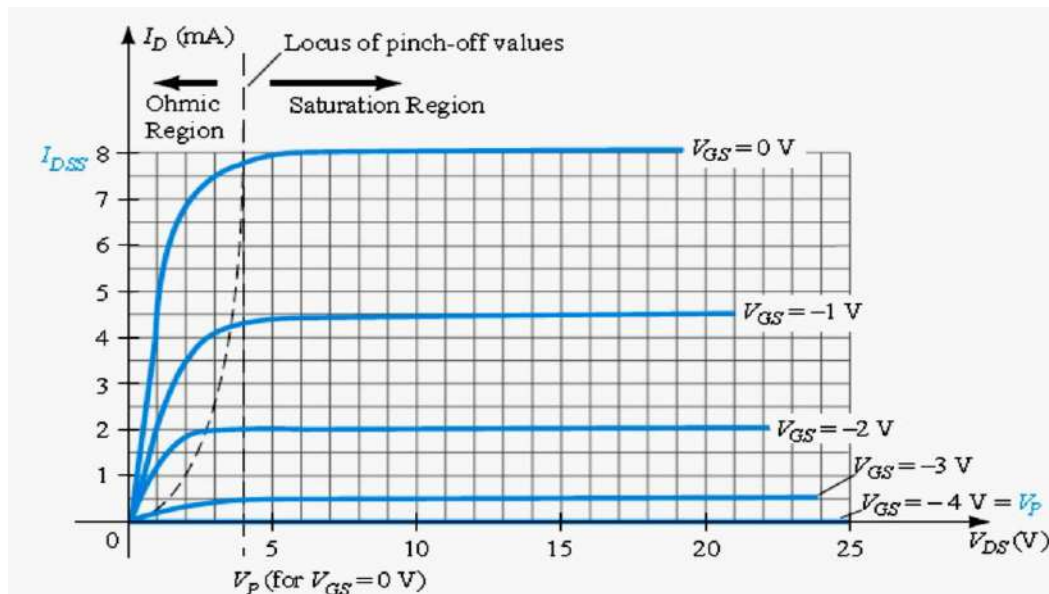


Figure 4: n-Channel JFET characteristics with $I_{DSS} = 8$ mA and $V_p = -4$ V.

The level of V_{GS} that results in $I_D = 0$ mA is defined by $V_{GS} = V_p$, with V_p being a negative voltage for n -channel devices and a positive voltage for p -channel JFETs.

- Voltage-Controlled Resistor:

The region to the left of the pinch-off locus of Fig. 4 is referred to as the *ohmic* or *voltage-controlled resistance region*. In this region the JFET can actually be employed as a variable resistor (possibly for an automatic gain control system) whose resistance is controlled by the applied gate-to-source voltage

$$r_d = \frac{r_o}{(1 - V_{GS}/V_p)^2}$$

where r_o is the resistance with $V_{GS} = 0$ V and r_d the resistance at a particular level of V_{GS} . For example, if n -channel JFET with r_o equal to 10 k Ω ($V_{GS} = 0$ V), $V_p = -6$ V, will result in 40 k Ω at $V_{GS} = -3$ V.

- p-Channel Devices:

The p -channel JFET is constructed in exactly the same manner as the n -channel device of Fig. 3, but with a reversal of the p - and n -type materials as shown in Fig. 5.

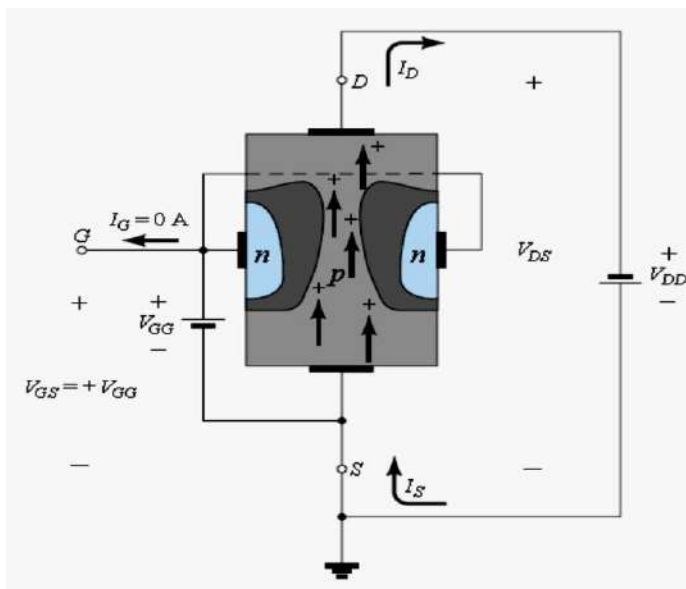


Figure 5

The defined current directions are reversed, as are the actual polarities for the voltages V_{GS} and V_{DS} . For the p -channel device, the channel will be constricted by increasing positive voltages from gate to source and the double-subscript notation for V_{DS} will result in negative voltages for V_{DS} on the characteristics of Fig. 6.

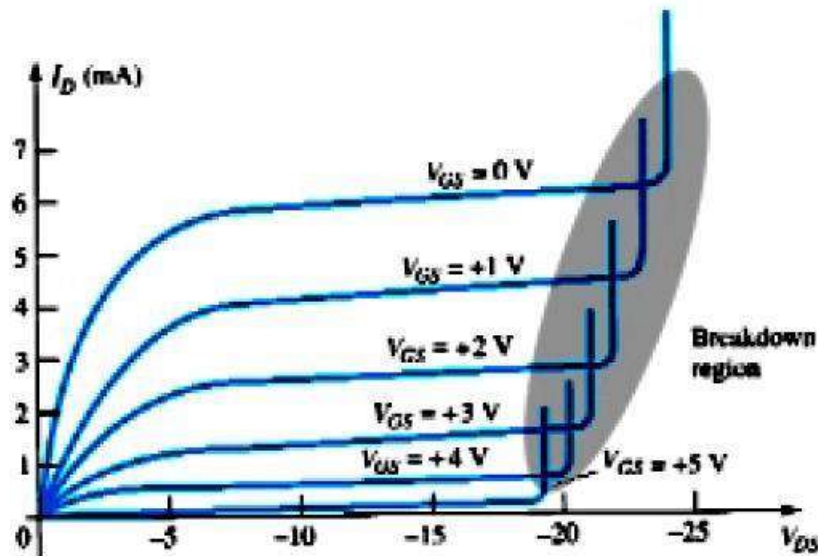
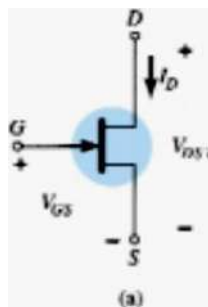
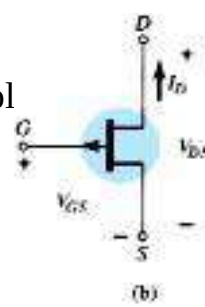


Figure 6: p -Channel JFET characteristics with $I_{DSS} = 6$ mA and $V_p = +6$ V.

Figure 7: JFET



symbol (a) n -channel; (b) p -channel.



- TRANSFER CHARACTERISTICS:

The relationship between I_D and V_{GS} is defined by *Shockley's equation*:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

constants
control variable

The squared term of the equation will result in a nonlinear relationship between I_D and V_{GS} , producing a curve that grows exponentially with decreasing magnitudes of V_{GS} .

The transfer characteristics defined by Shockley's equation are unaffected by the network in which the device is employed.

By using basic algebra we can obtain an equation for the resulting level of V_{GS} for a given level of I_D .

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right)$$

In Fig.8 two graphs are provided, with the vertical scaling in milliamperes for each graph. One is a plot of I_D versus V_{DS} , while the other is I_D versus V_{GS} .

In review:

When $V_{GS} = 0$ V, $I_D = I_{DSS}$, and when $V_{GS} = V_P$, $I_D = 0$ mA.

The total device dissipation at 25°C (room temperature) is the maximum power the device can dissipate under normal operating conditions and is defined by

$$P_D = V_{DS} I_D$$

- IMPORTANT RELATIONSHIPS:

<i>JFET</i>		<i>BJT</i>
$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$	$\Leftrightarrow$	$I_C = \beta I_B$
$I_D = I_S$	$\Leftrightarrow$	$I_C \cong I_E$
$I_G \cong 0$ A	$\Leftrightarrow$	$V_{BE} \cong 0.7$ V

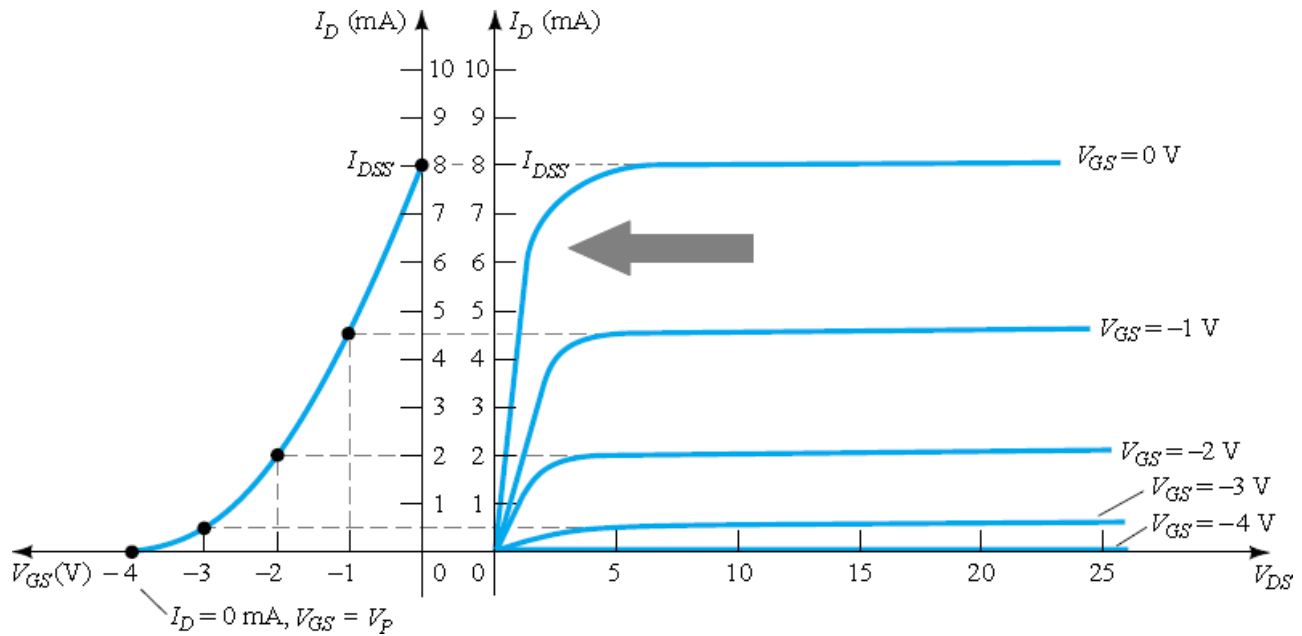


Figure 8

- FET Biasing:

The general relationships that can be applied to the dc analysis of all FET amplifiers are

$$I_G \approx 0 \text{ A} \quad \text{and} \quad I_D = I_S$$

The simplest of biasing arrangements for the n -channel JFET appears in Fig. 9. Referred to as the fixed-bias configuration, it is one of the few FET configurations that can be solved just as directly using either a mathematical or graphical approach.

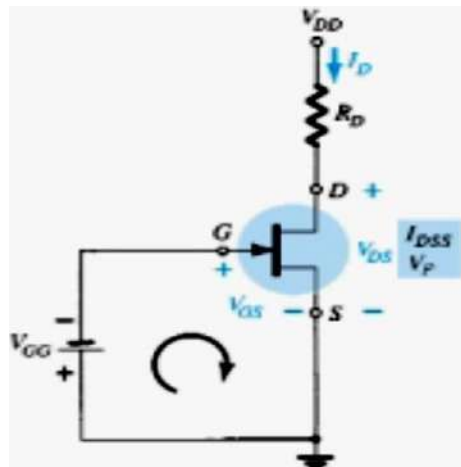


Figure 9



$$I_G = 0 \text{ A} \quad \text{and} \quad V_{RG} = I_G R_G = (0 \text{ A})R_G = 0 \text{ V}$$

Applying Kirchhoff's voltage law in the clockwise direction of the indicated loop of Fig.9 will result in

$$-V_{GG} - V_{GS} = 0 \quad \text{and} \quad V_{GS} = -V_{GG}$$

The drain-to-source voltage of the output section can be determined by applying Kirchhoff's voltage law as follows:

$$+V_{DS} + I_D R_D - V_{DD} = 0 \quad \text{and} \quad V_{DS} = V_{DD} - I_D R_D$$

For the configuration of Fig. 9,

$$V_S = 0$$

Using double-subscript notation:

$$V_{DS} = V_D - V_S \quad \text{or} \quad V_D = V_{DS} + V_S = V_{DS} + 0V \quad \text{and} \quad V_D = V_{DS}$$

$$\text{In addition,} \quad V_{GS} = V_G - V_S \quad \text{or} \quad V_G = V_{GS} + V_S = V_{GS} + 0V \quad \text{and} \quad V_G = V_{GS}$$

Example 1: Determine the following for the network of Fig. 10.

- (a) V_{GSQ} .
- (b) I_{DQ} .
- (c) V_{DS} .
- (d) V_D .
- (e) V_G .
- (f) V_S .

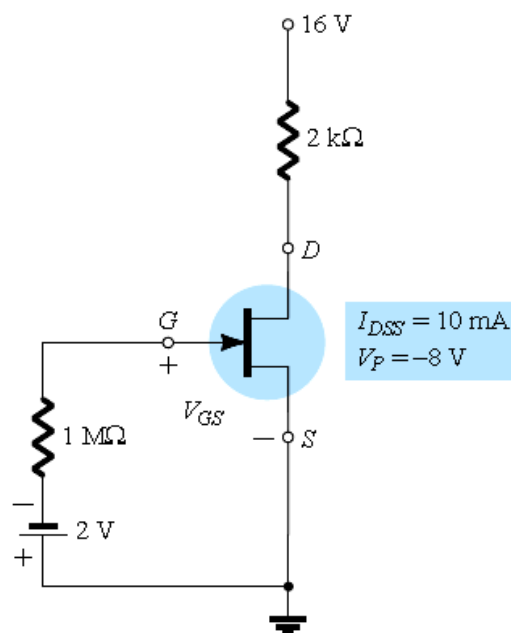


Figure 10

Mathematical Approach:

(a) $V_{GSQ} = -V_{GG} = -2 \text{ V}$

(b) $I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 10 \text{ mA} \left(1 - \frac{-2 \text{ V}}{-8 \text{ V}} \right)^2$
 $= 10 \text{ mA} (1 - 0.25)^2 = 10 \text{ mA} (0.75)^2 = 10 \text{ mA} (0.5625)$
 $= \mathbf{5.625 \text{ mA}}$

(c) $V_{DS} = V_{DD} - I_D R_D = 16 \text{ V} - (5.625 \text{ mA})(2 \text{ k}\Omega)$
 $= 16 \text{ V} - 11.25 \text{ V} = \mathbf{4.75 \text{ V}}$

(d) $V_D = V_{DS} = \mathbf{4.75 \text{ V}}$

(e) $V_G = V_{GS} = -2 \text{ V}$

(f) $V_S = 0 \text{ V}$

H.W: Repeat example 1 for the following figures:

